CONTAINERIZED SPARK ON KUBERNETES

William Benton

Red Hat, Inc.

@willb • willb@redhat.com

SPARK SUMMIT EUROPE 2016

BACKGROUND

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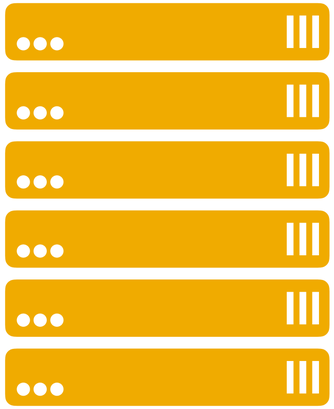

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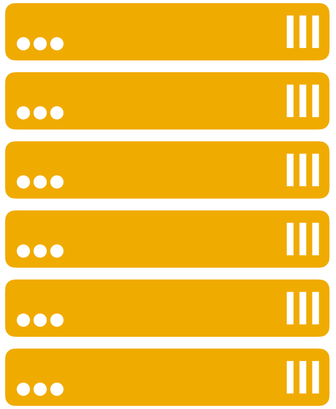

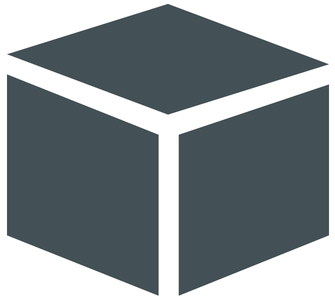

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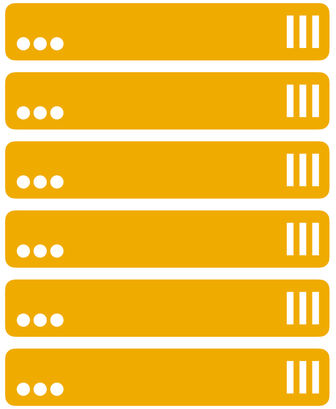

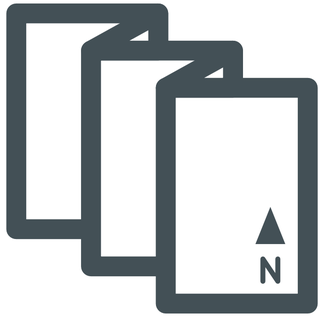

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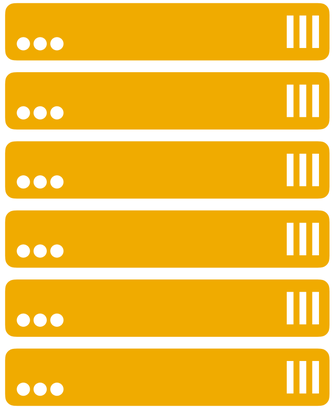

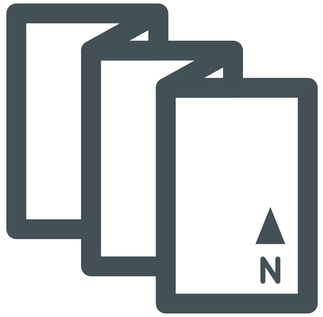

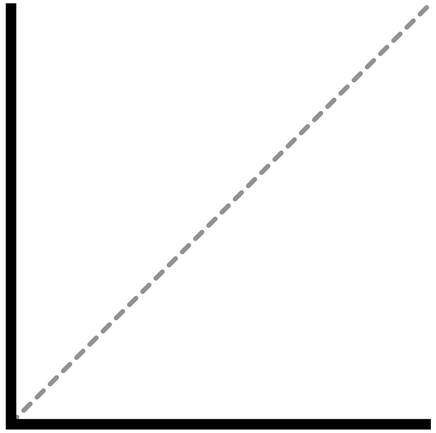

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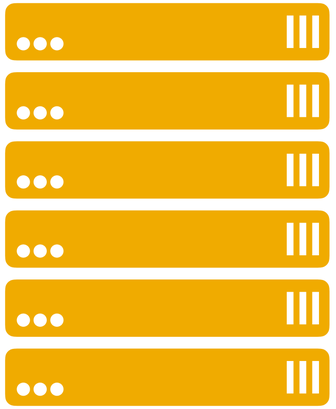

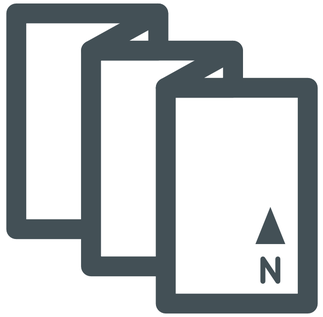

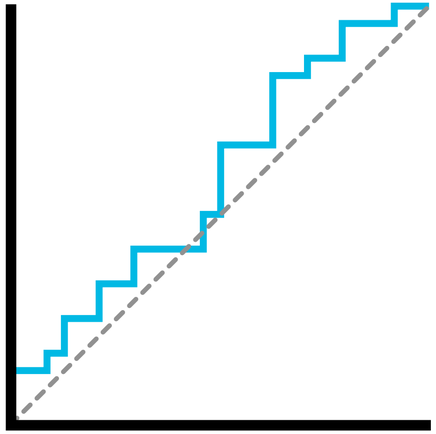

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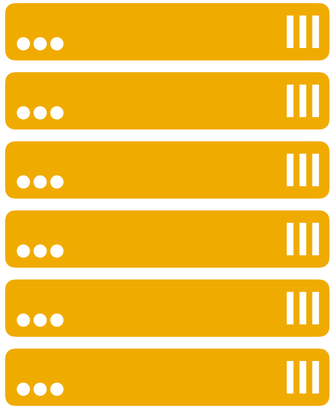

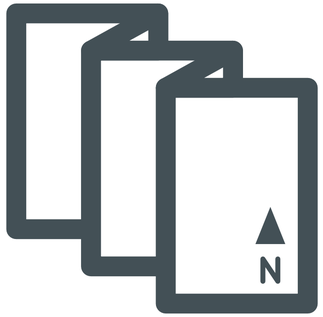

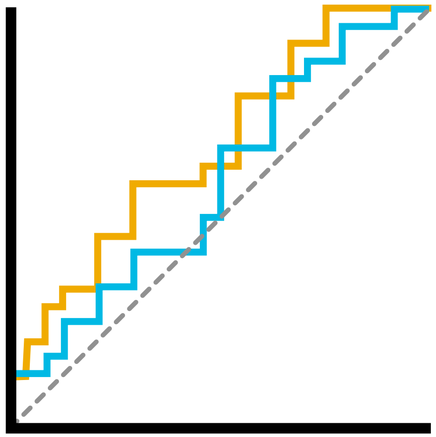

WHAT OUR SPARK CLUSTER LOOKED LIKE IN 2014

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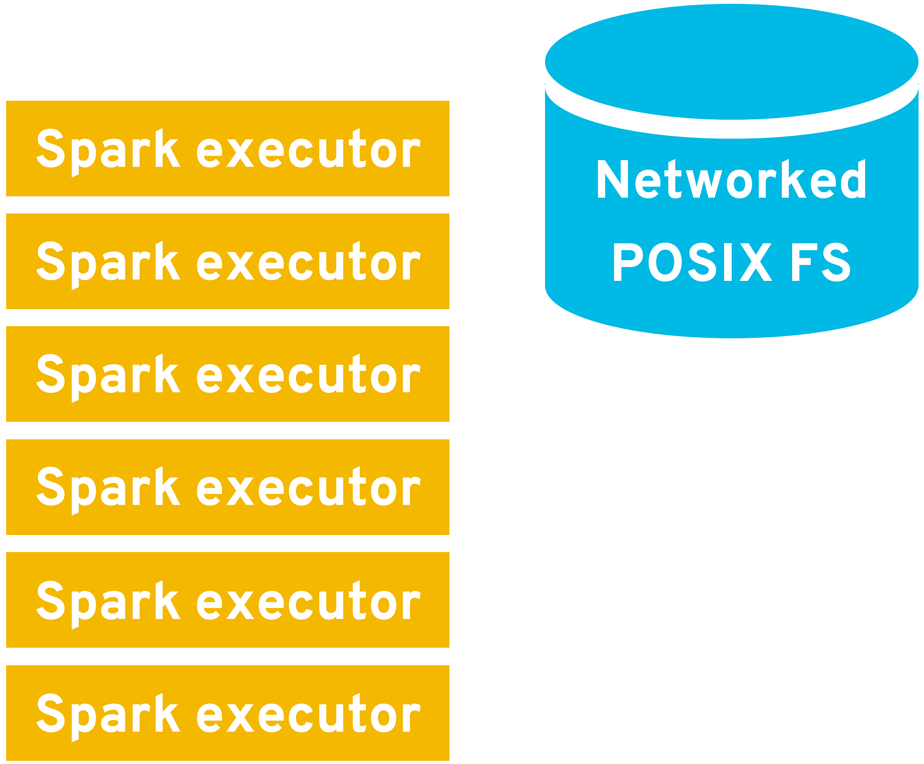

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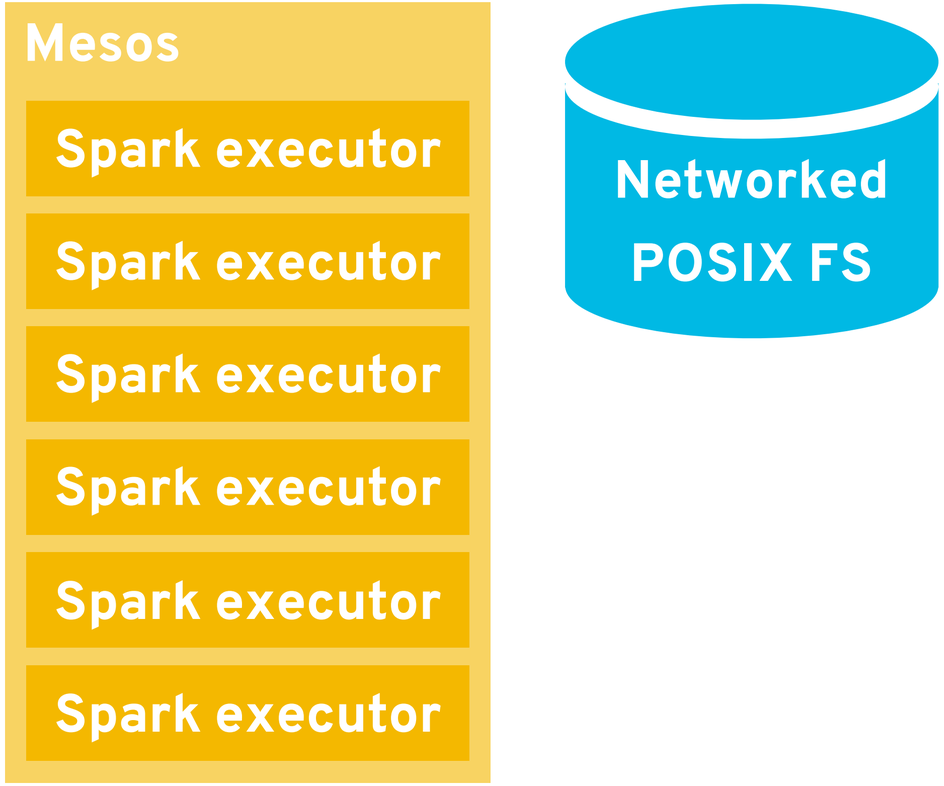

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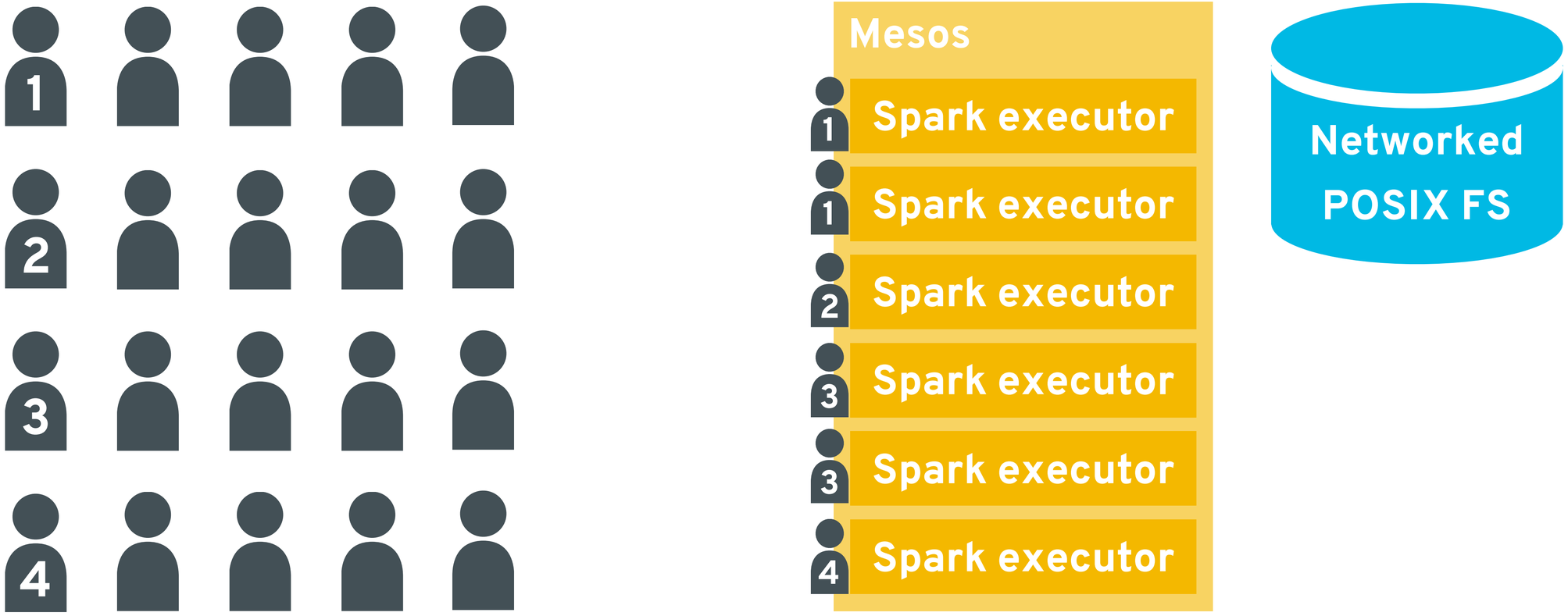

Analytics is no longer a separate workload.

Analytics is an essential component of modern data-driven applications.

OUR GOALS

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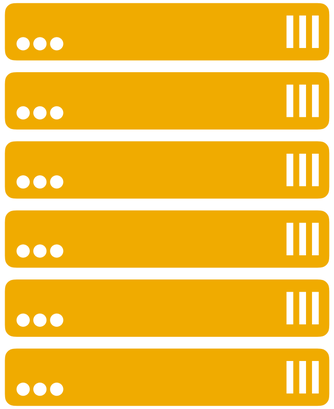

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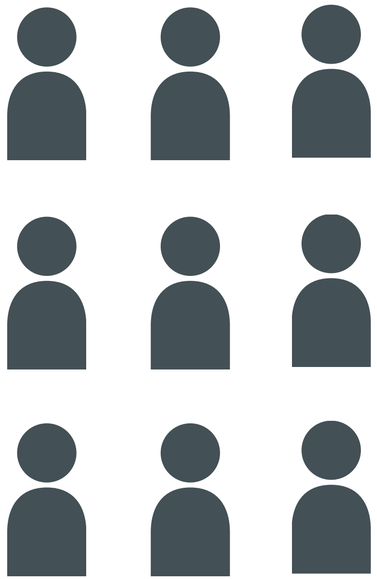

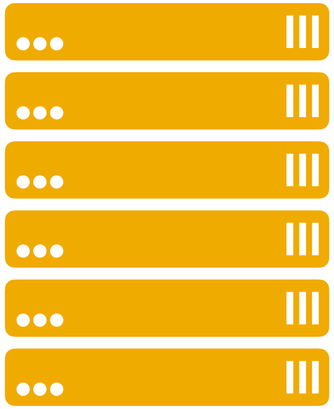

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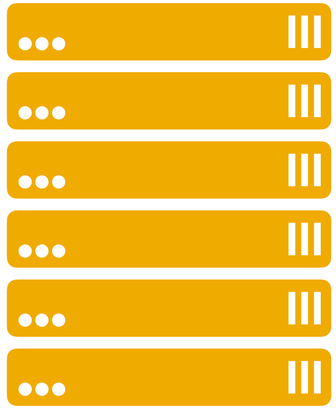

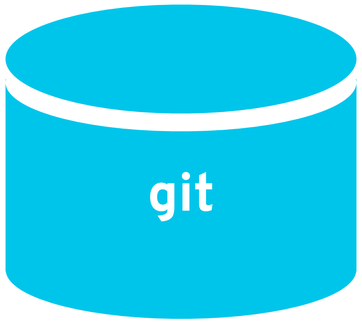

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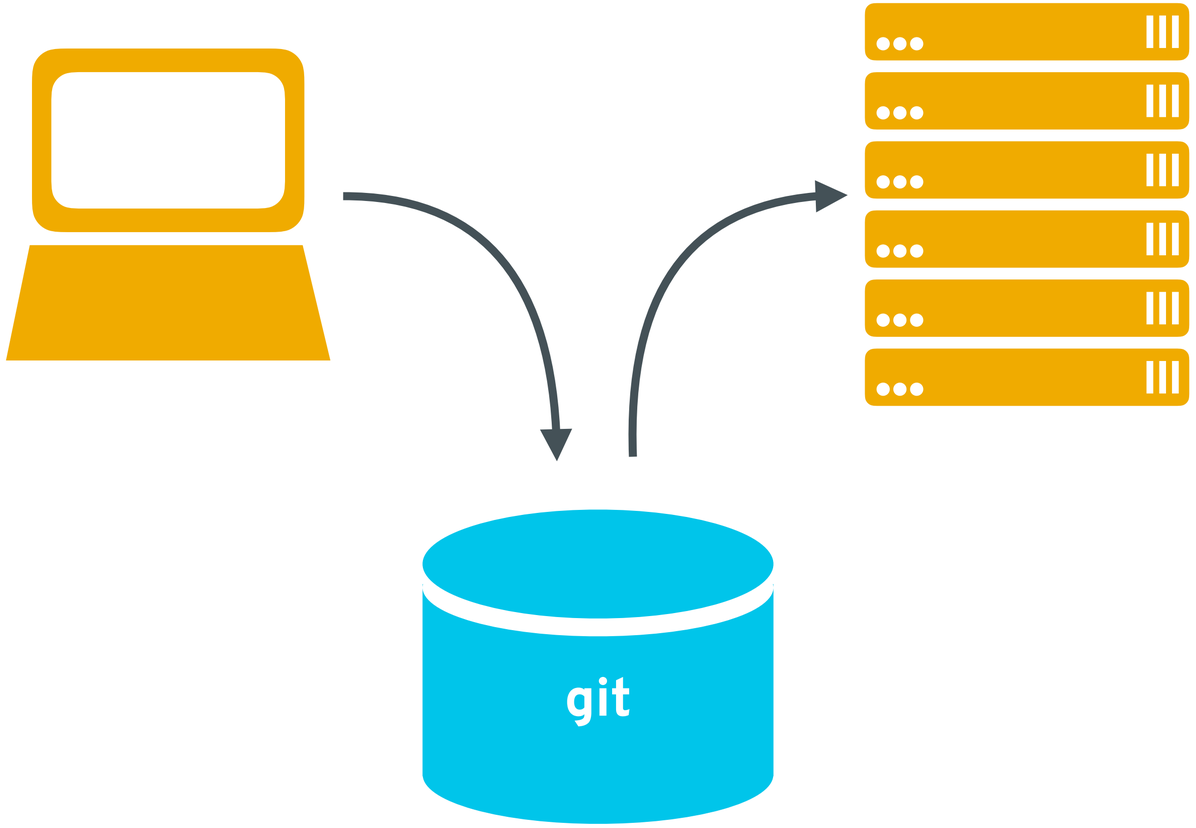

FORECAST

Motivating containerized microservices

Architectures for analytics and applications

Spark clusters in containers: practicalities and pitfalls

Play along at home

Future work

MOTIVATING MICROSERVICES

A **microservice** architecture employs **lightweight**, **modular**, and typically **stateless** components with **well-defined interfaces and contracts.**

BENEFITS OF MICROSERVICE ARCHITECTURES

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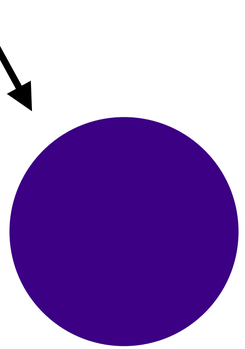
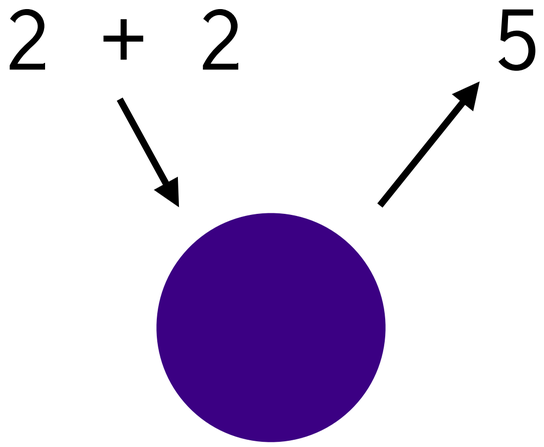


BENEFITS OF MICROSERVICE ARCHITECTURES



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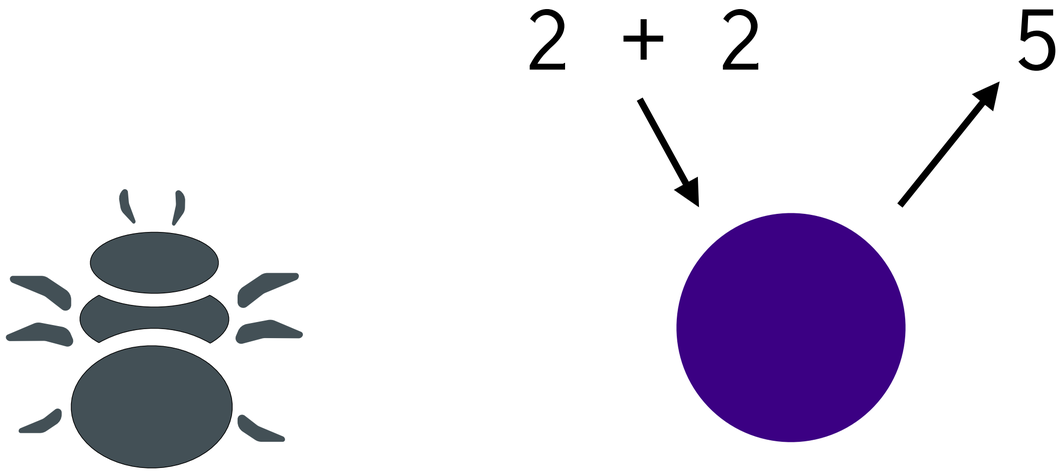

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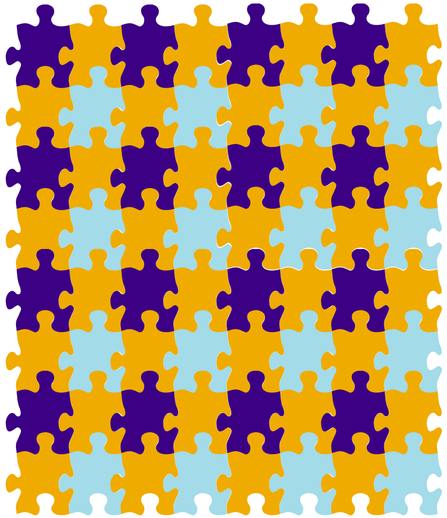

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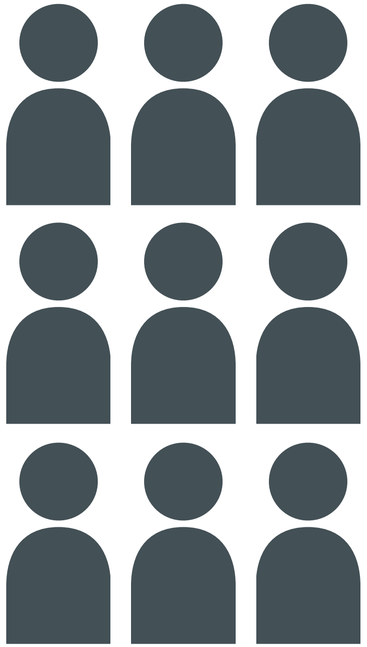

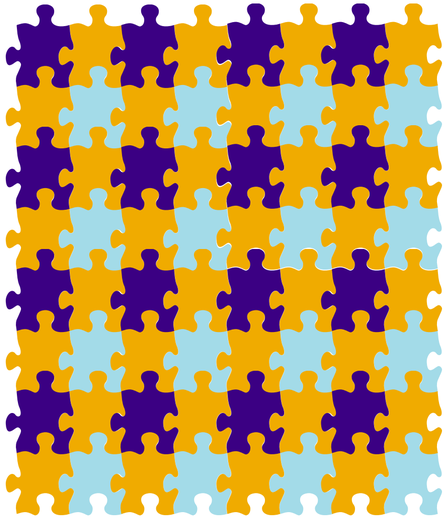

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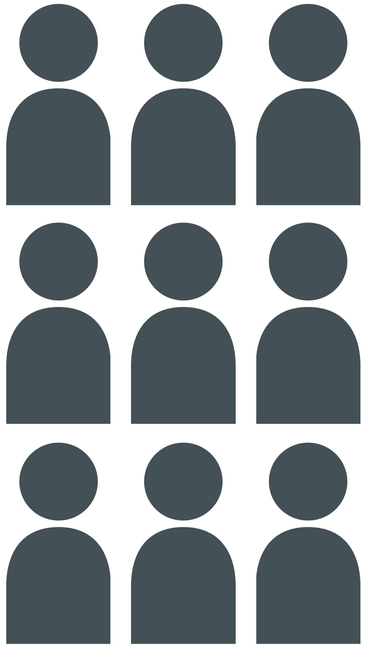

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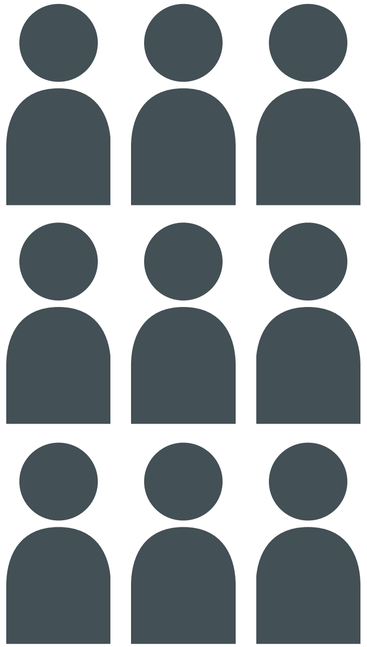

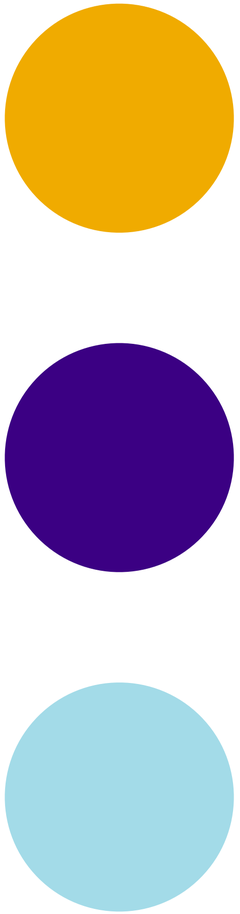

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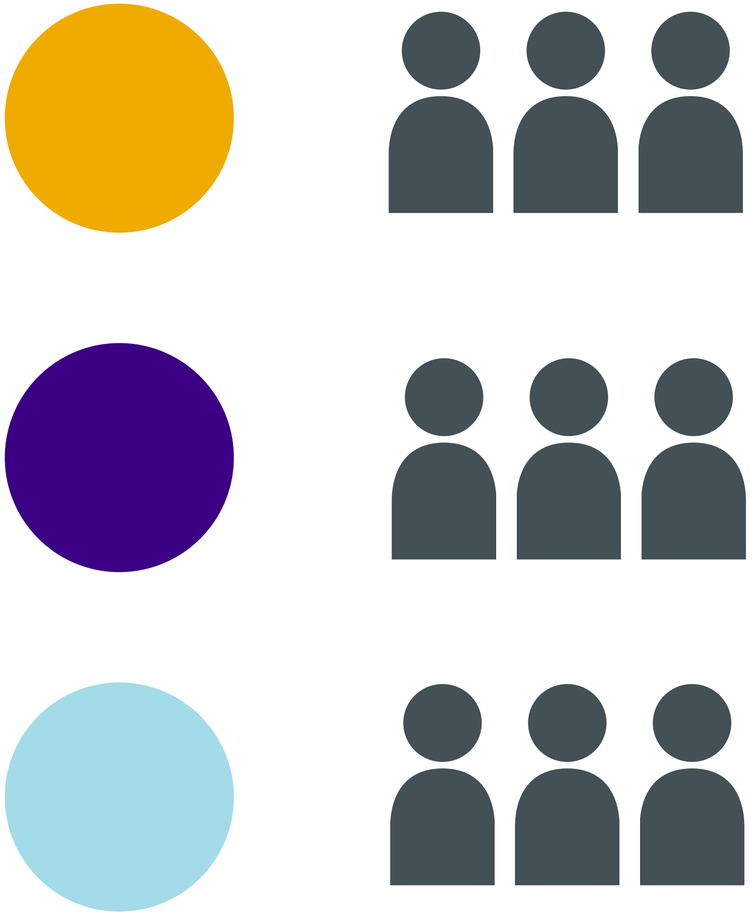
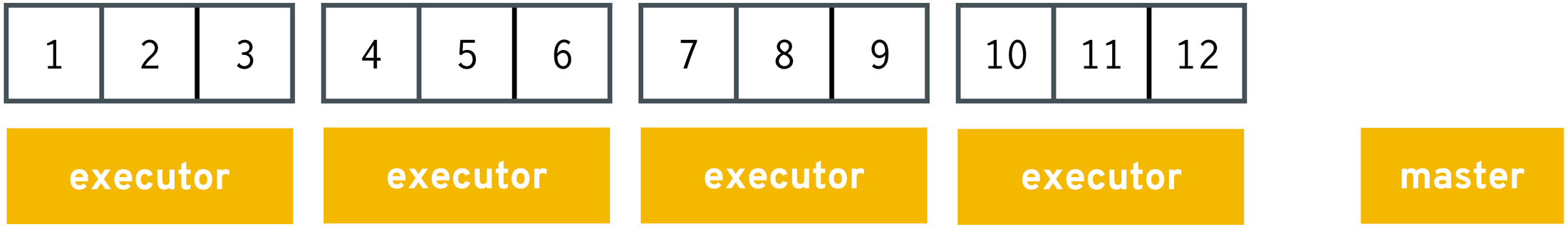


MICROSERVICES AND SPARK



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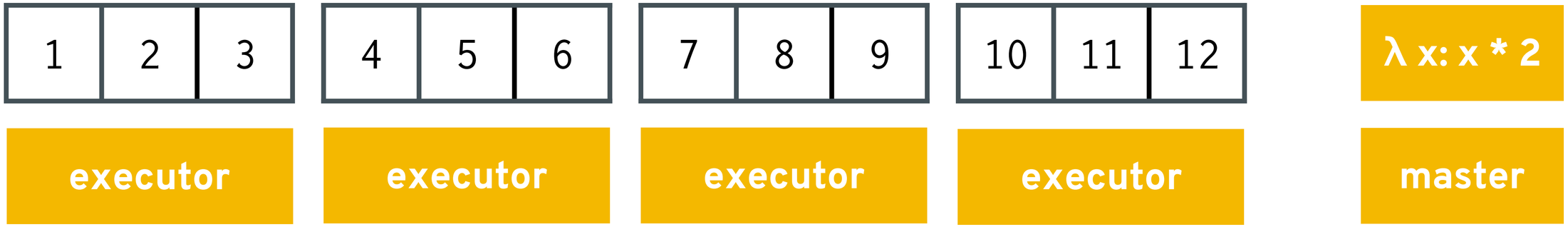

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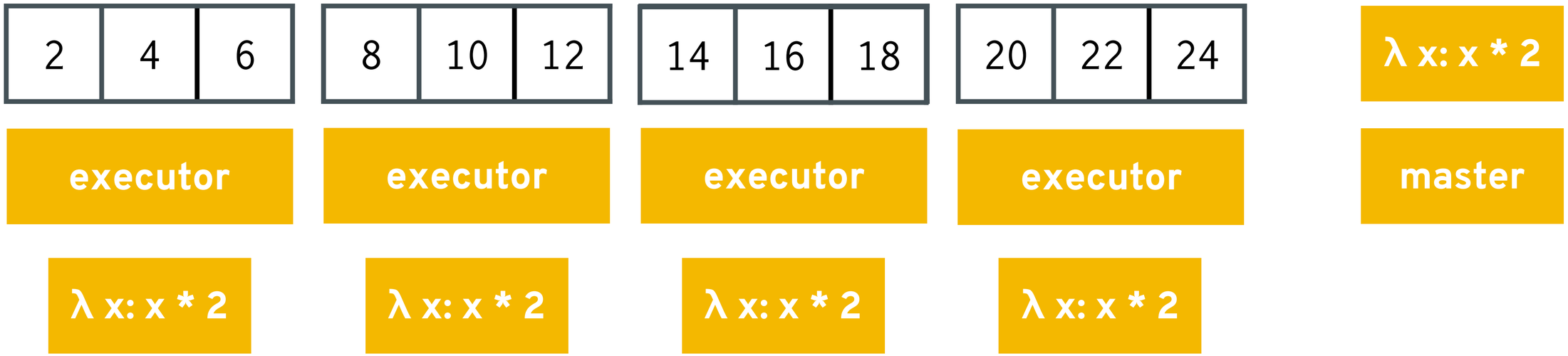

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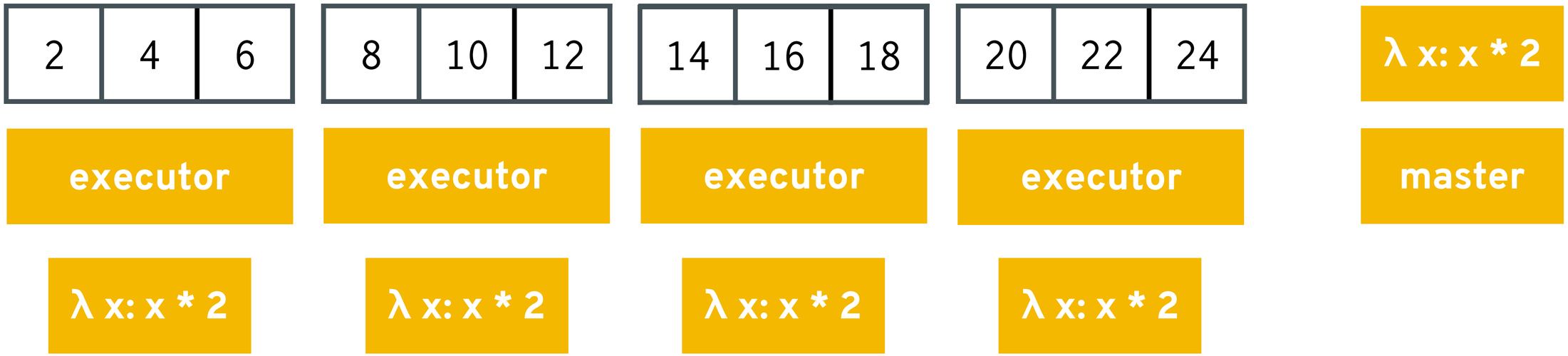

ARCHITECTURES FOR ANALYTICS AND APPLICATIONS

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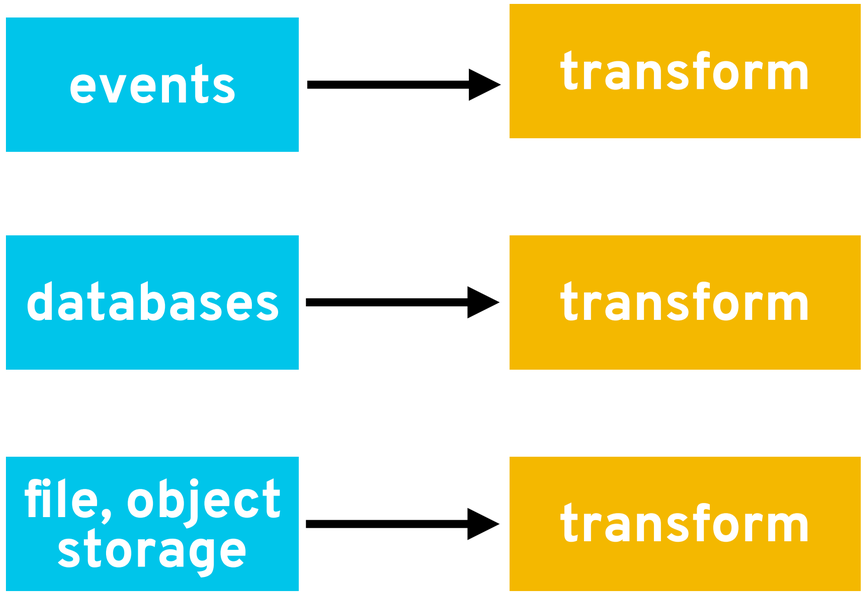

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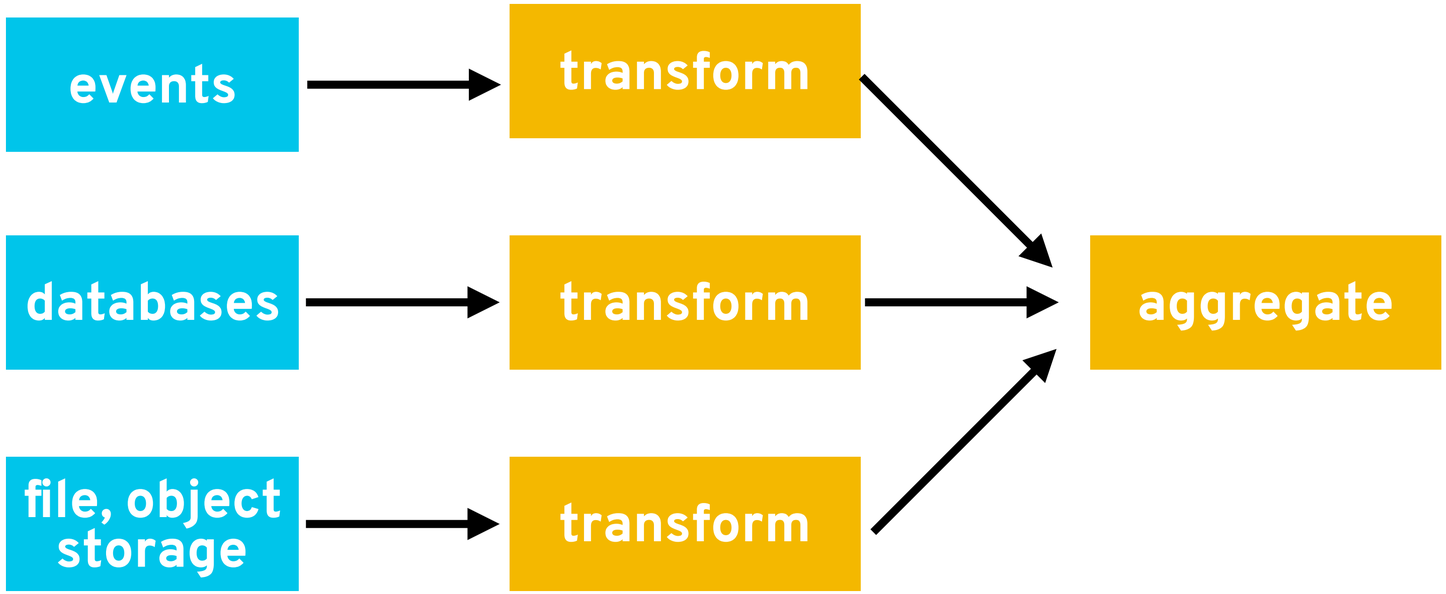

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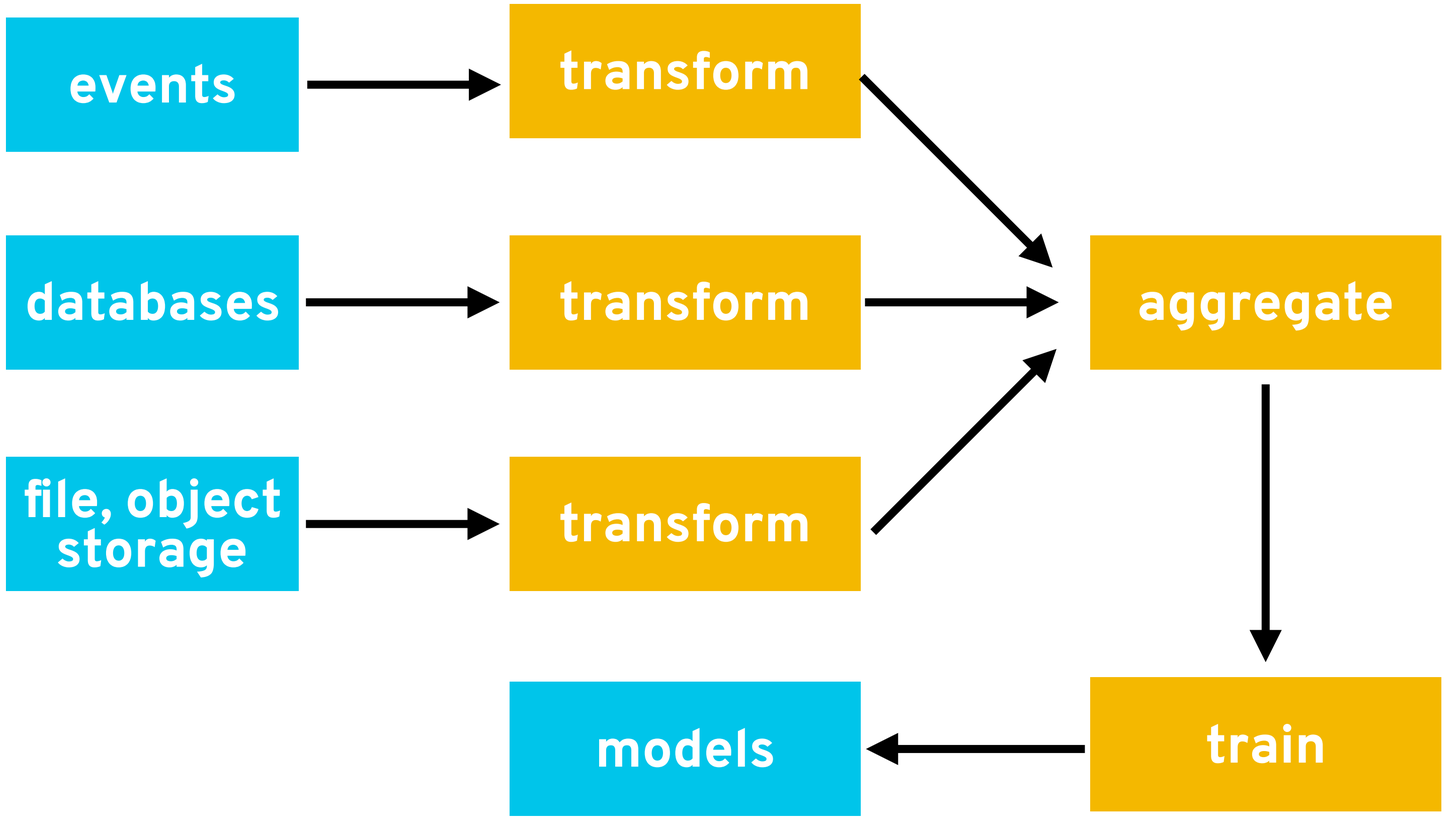

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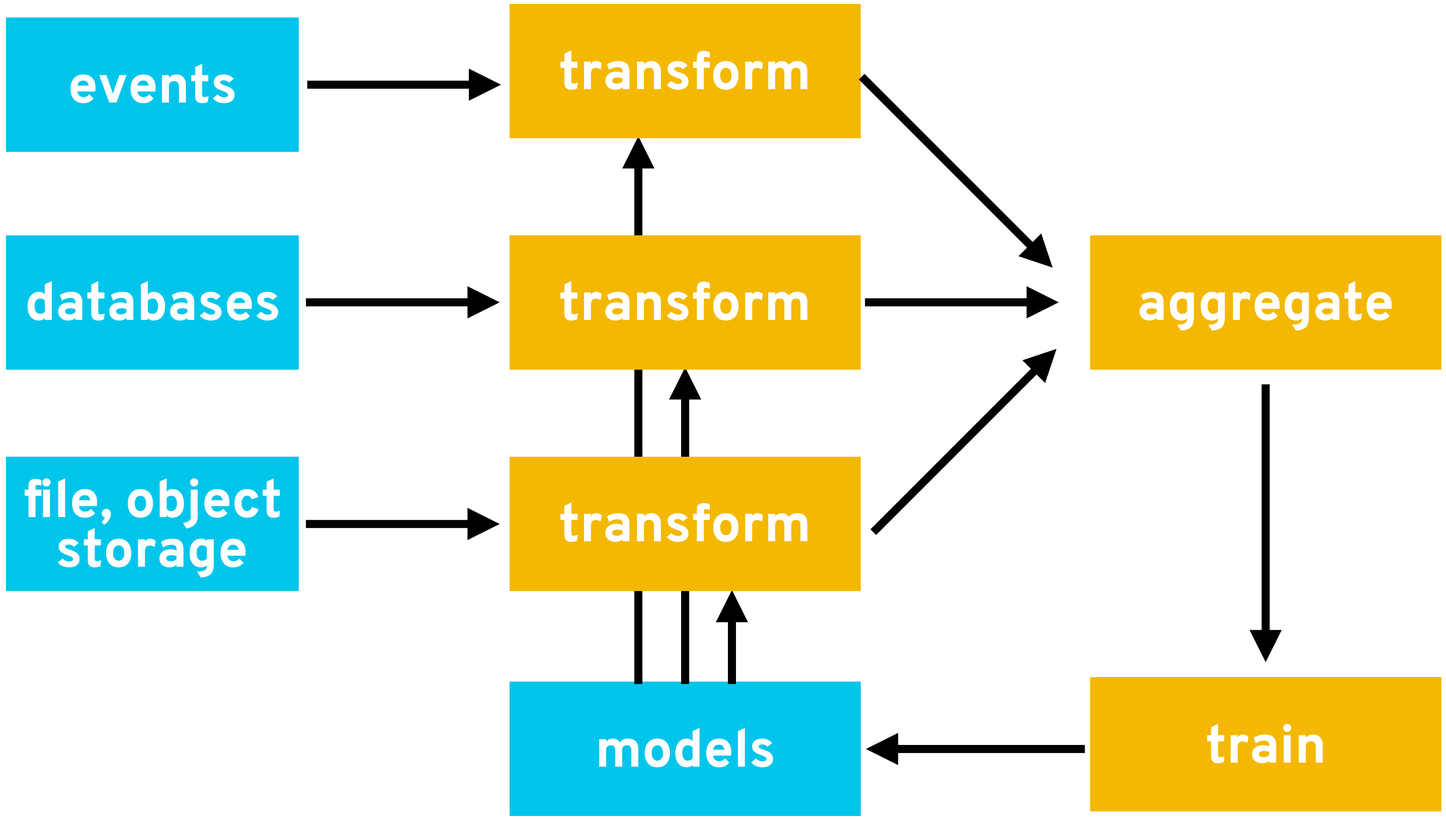

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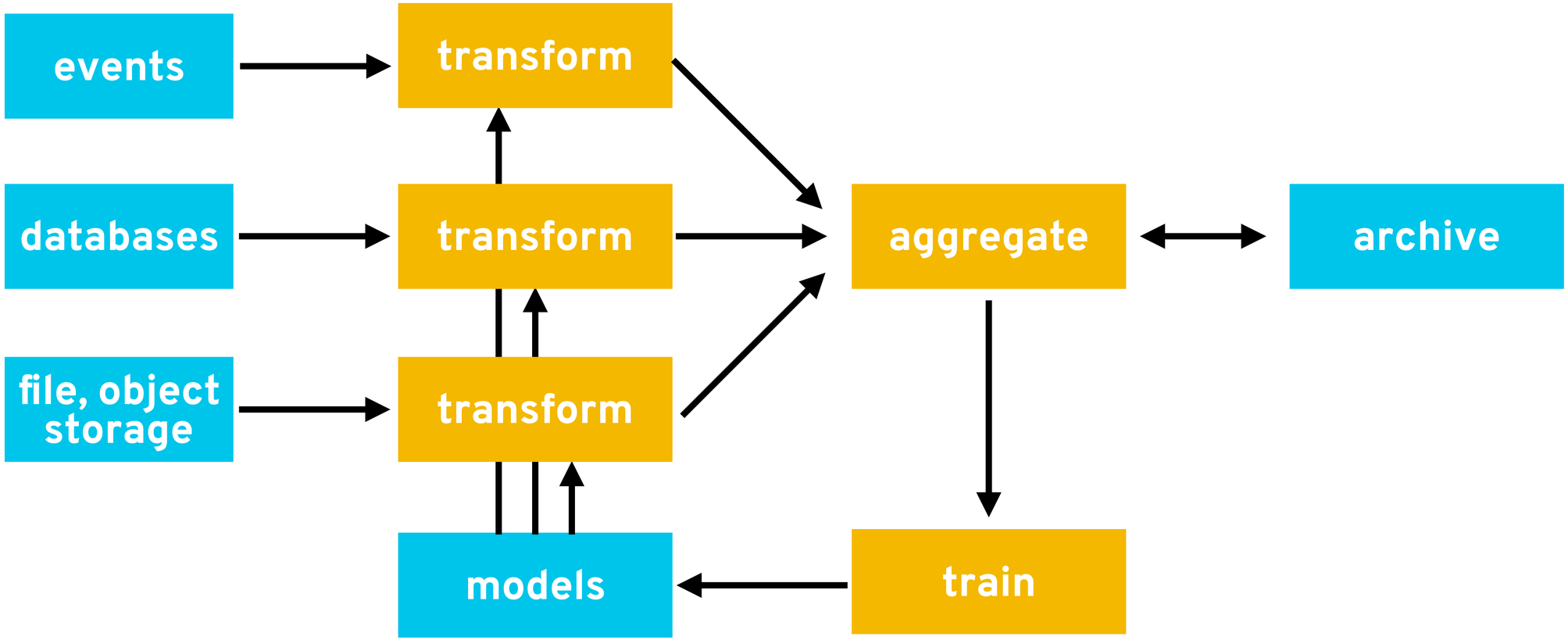

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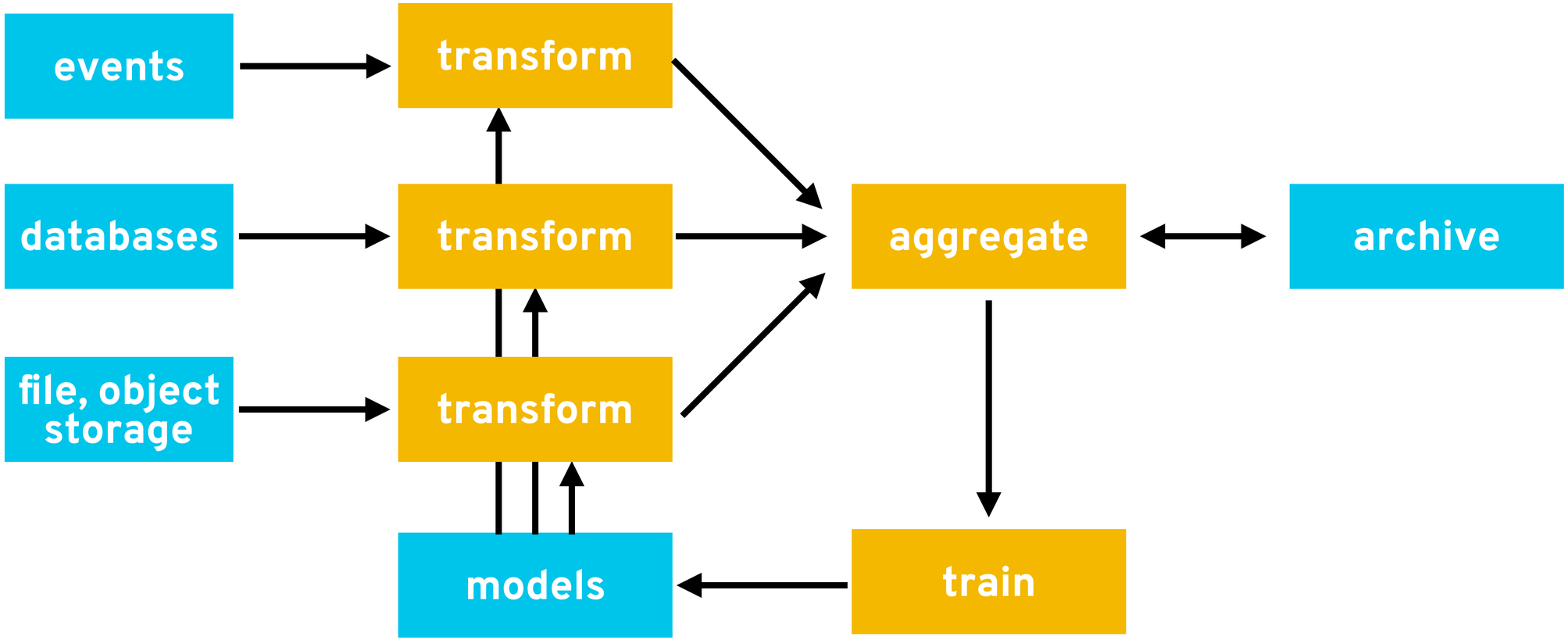

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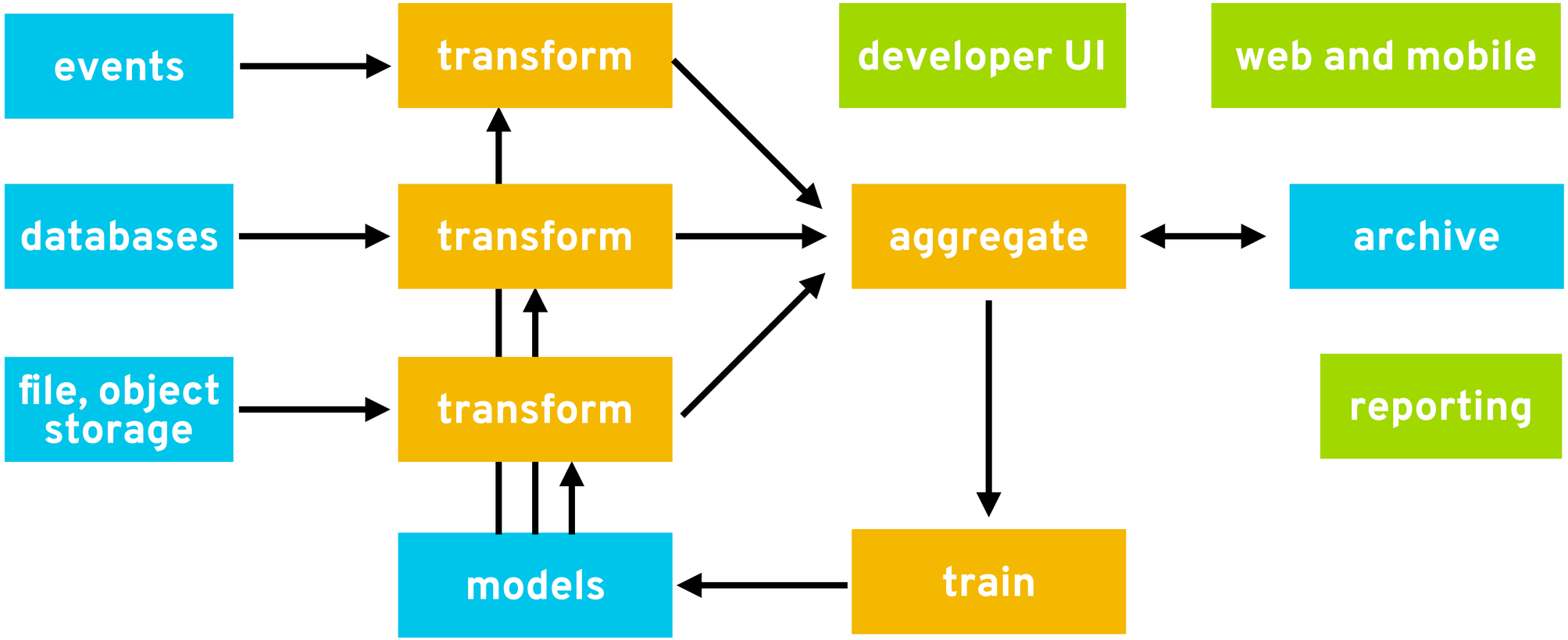

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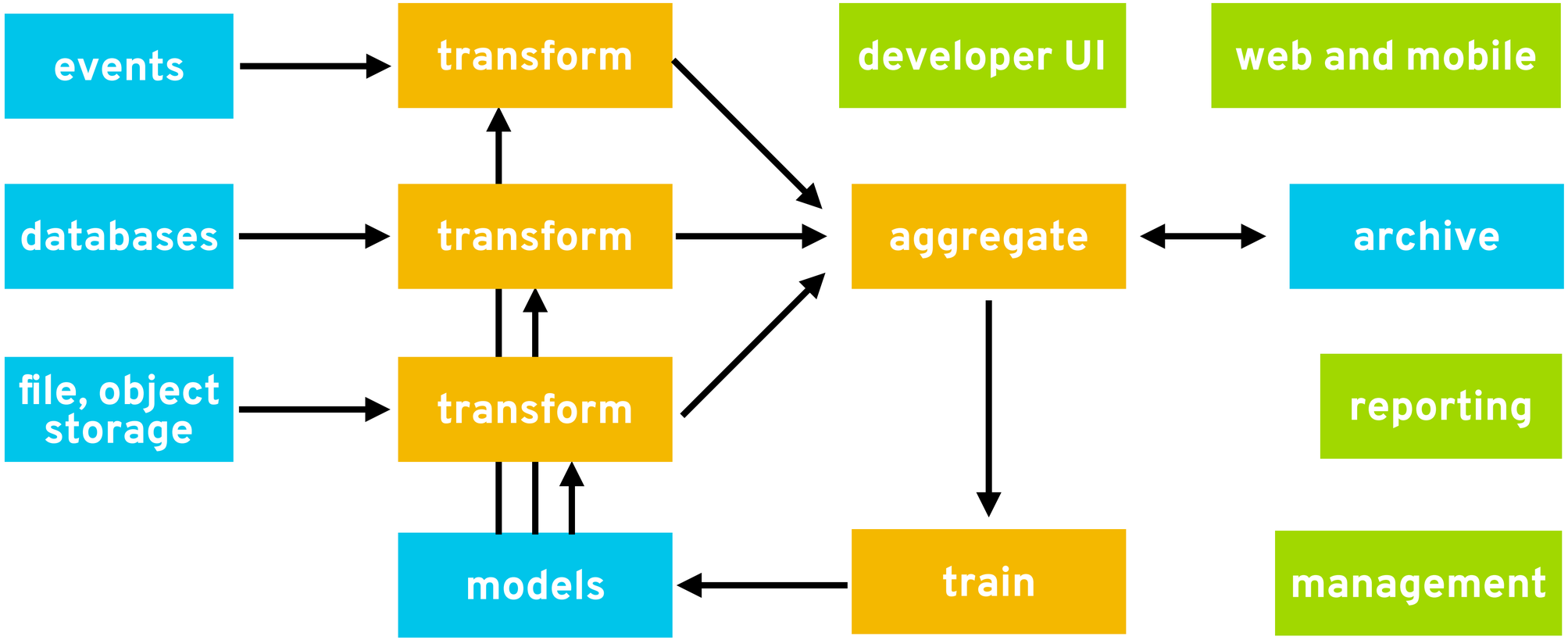

LEGACY ARCHITECTURES

CONVENTIONAL DATA WAREHOUSE

events

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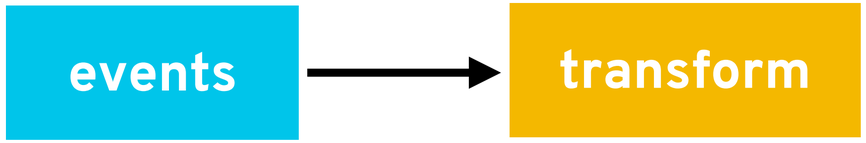

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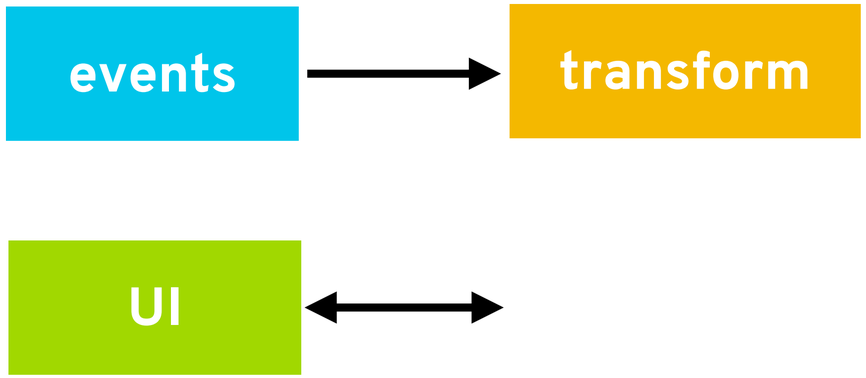

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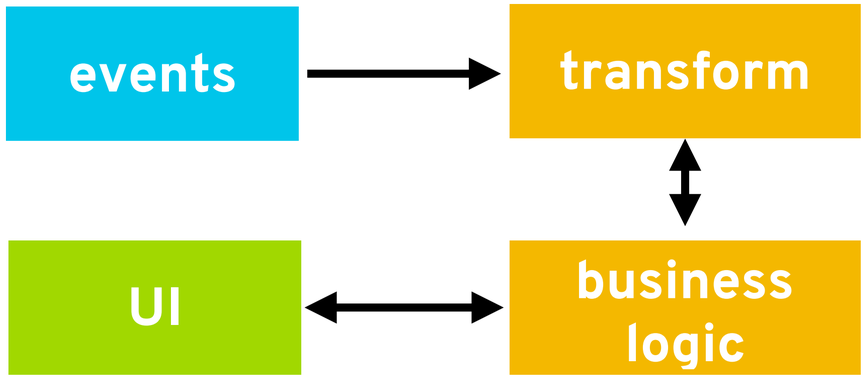

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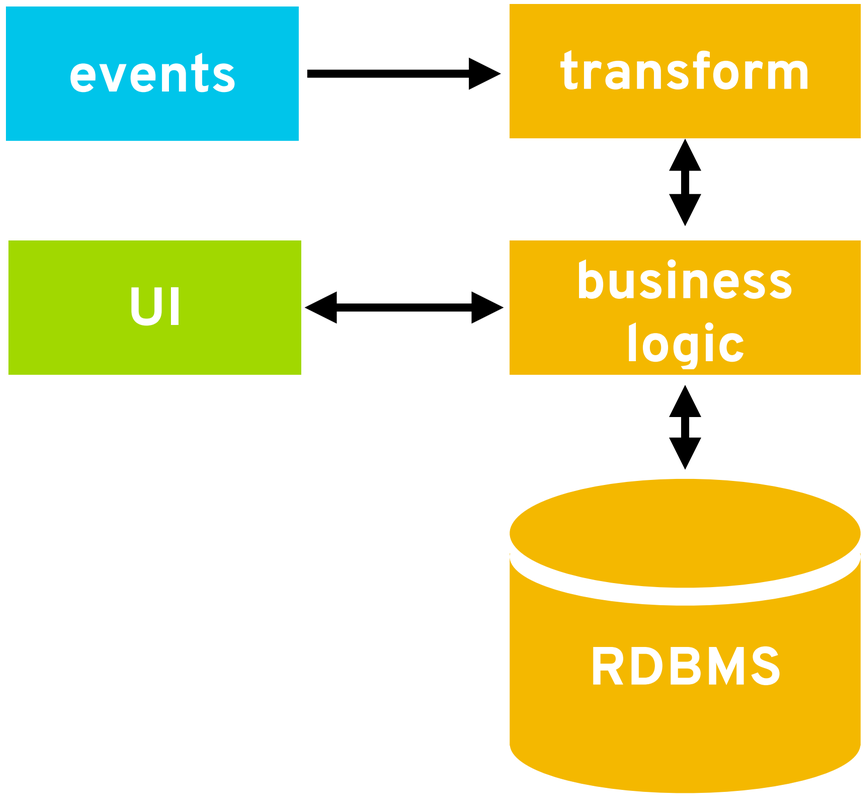

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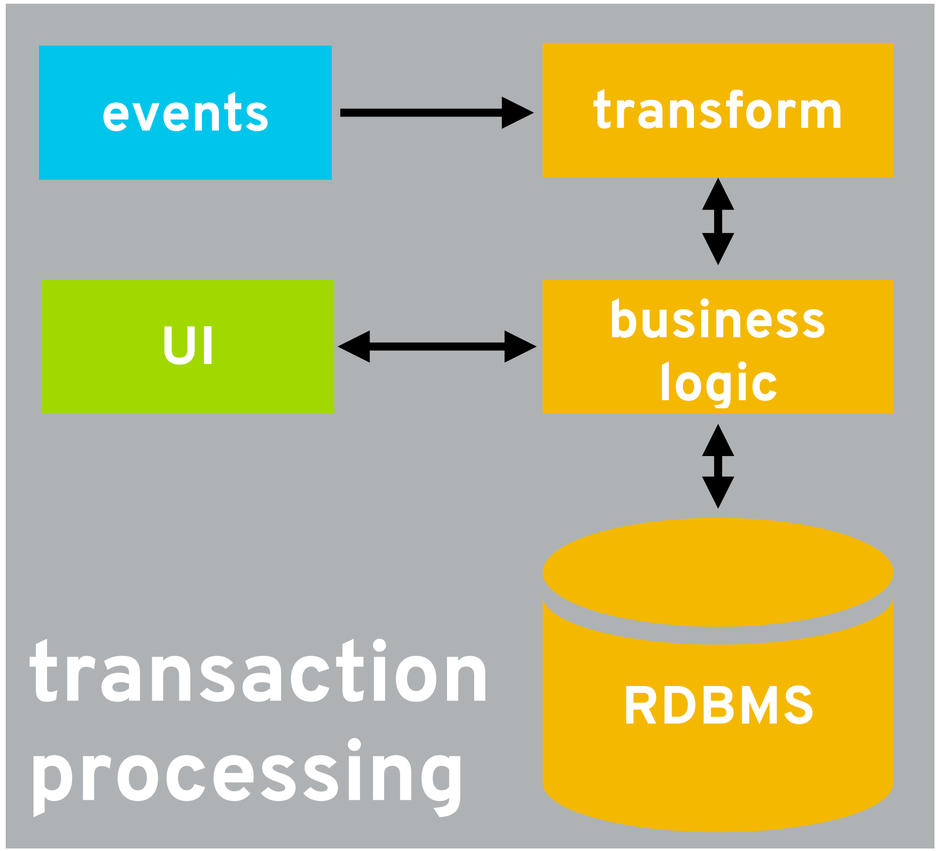

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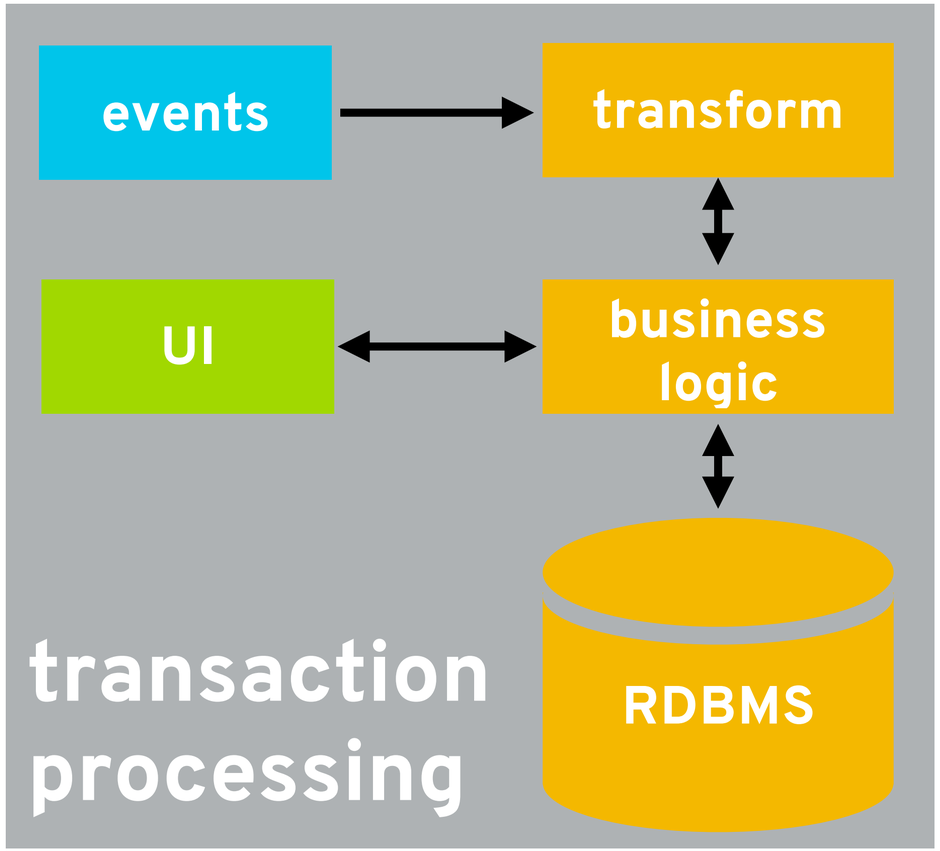

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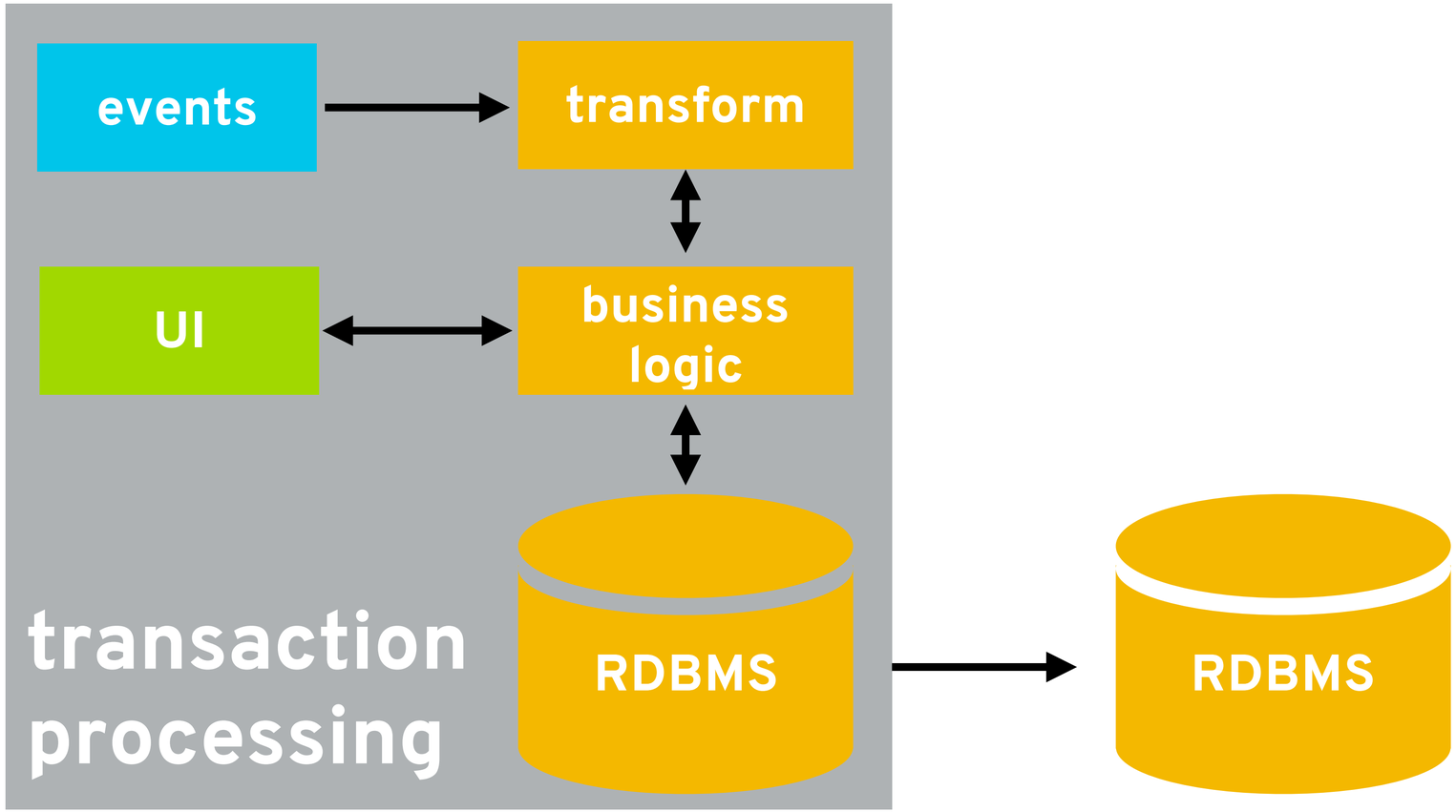

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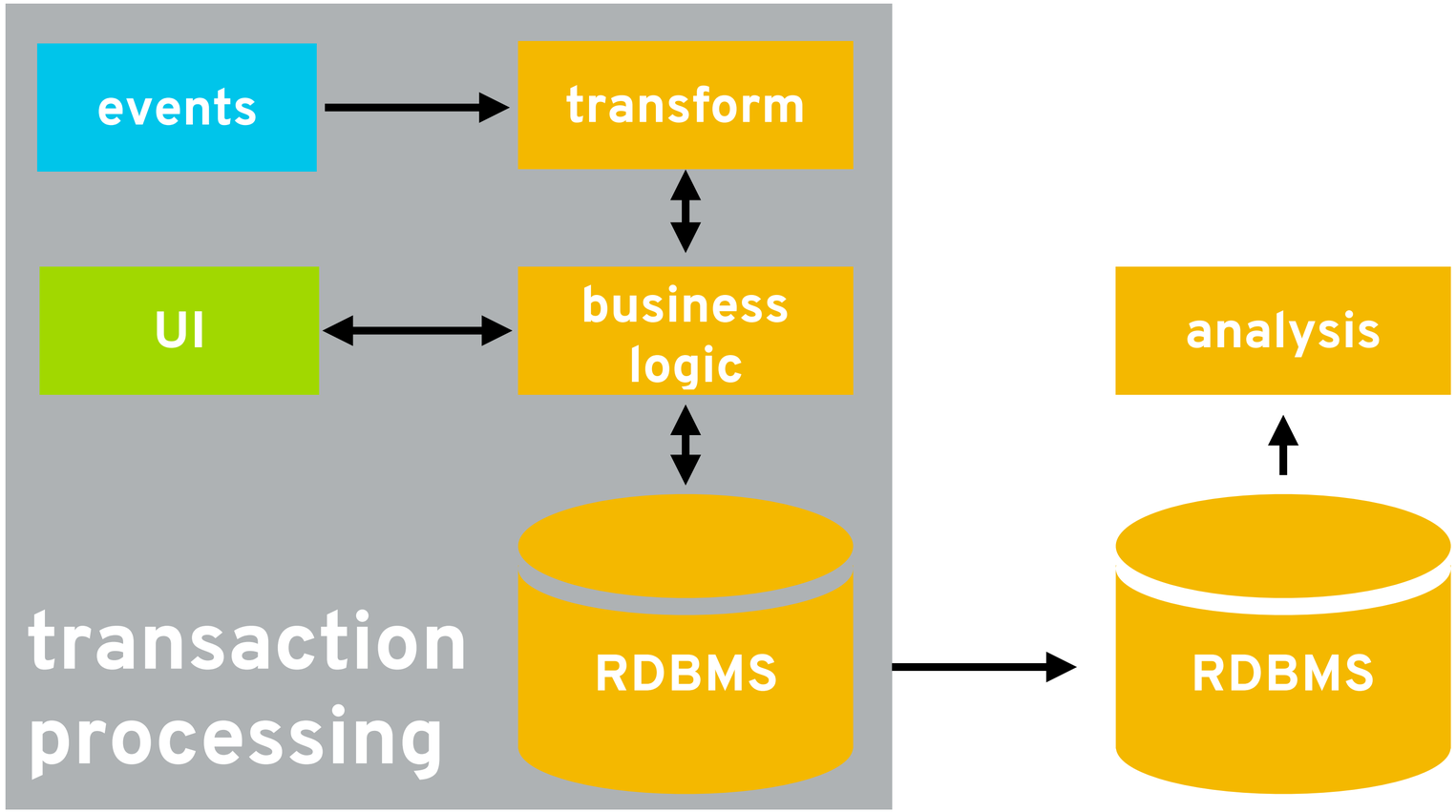

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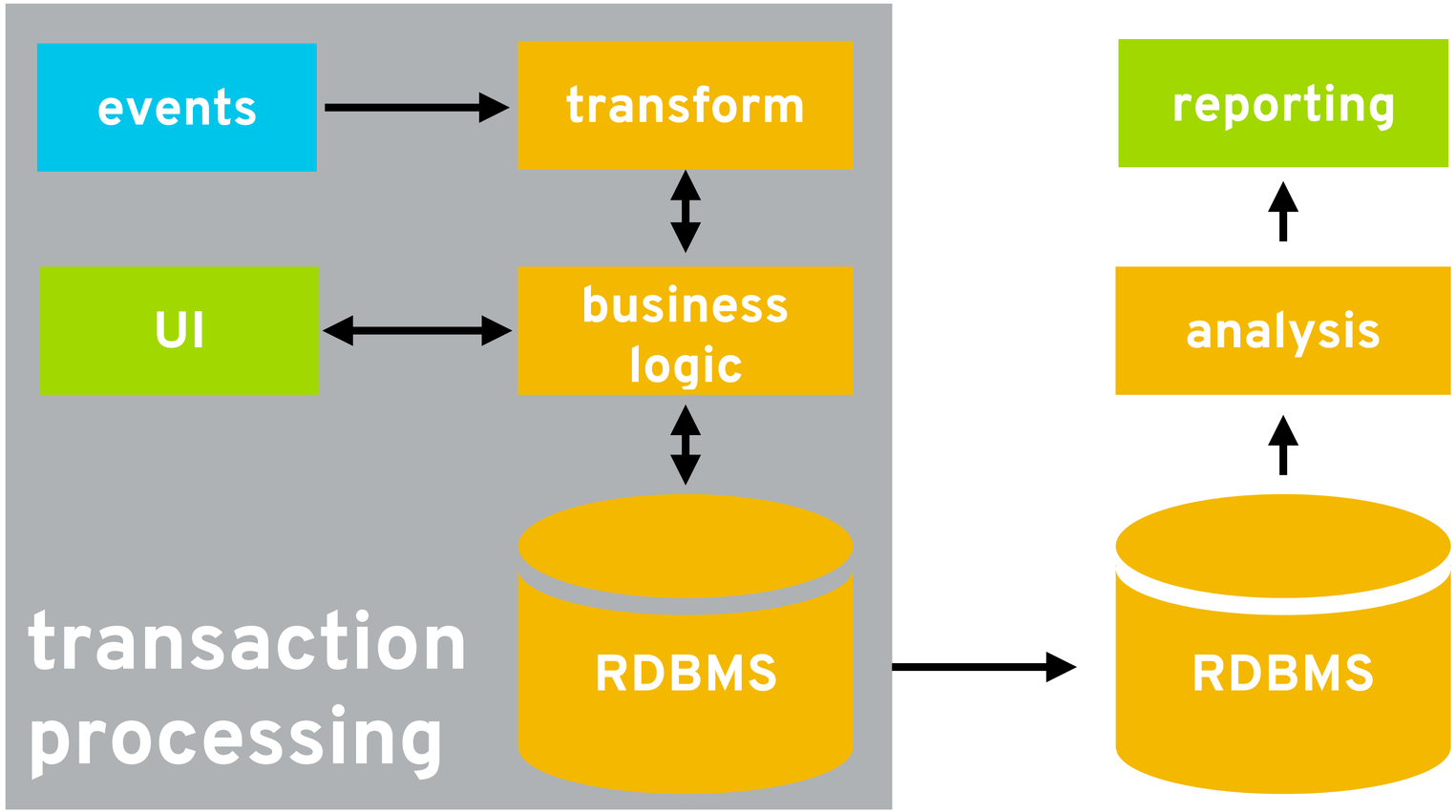

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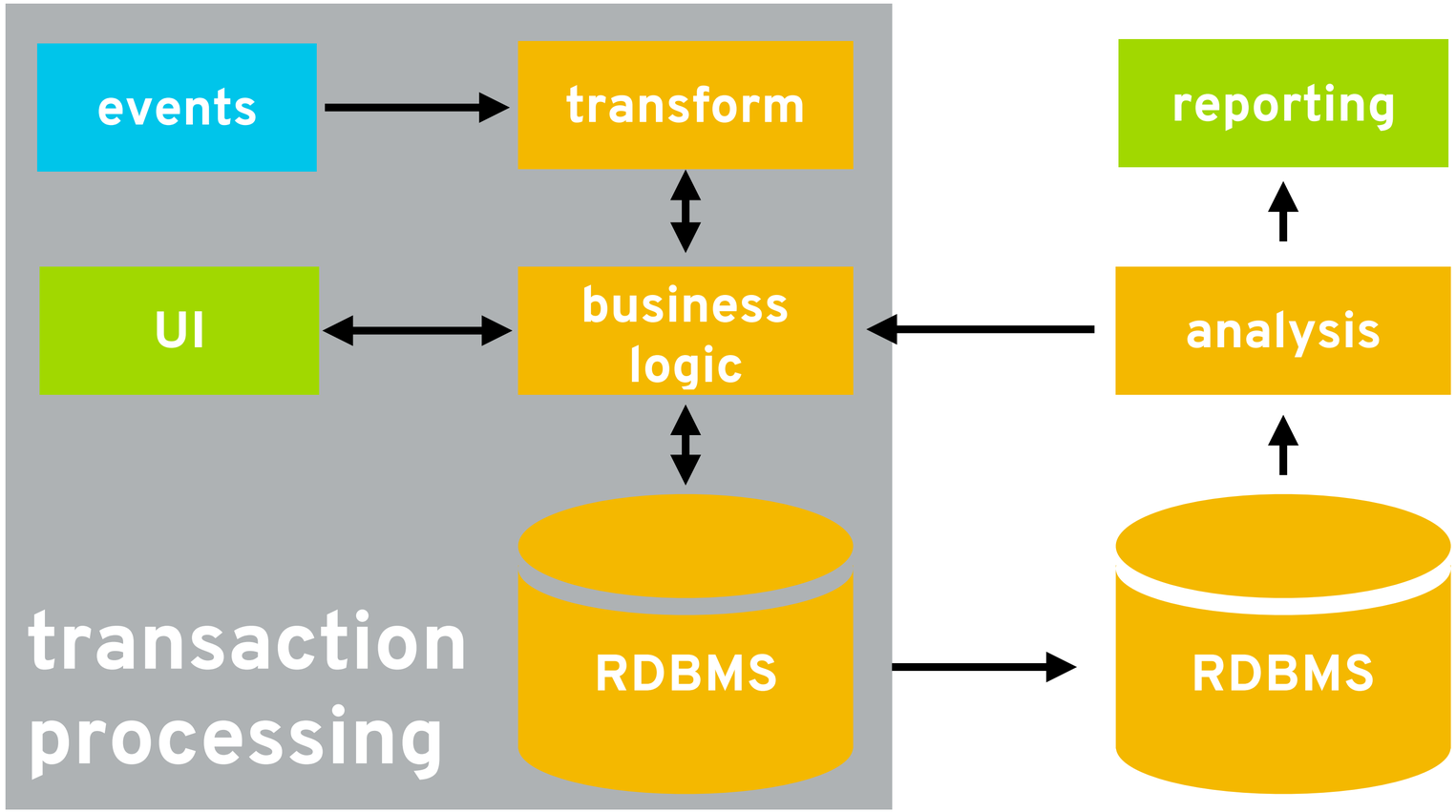

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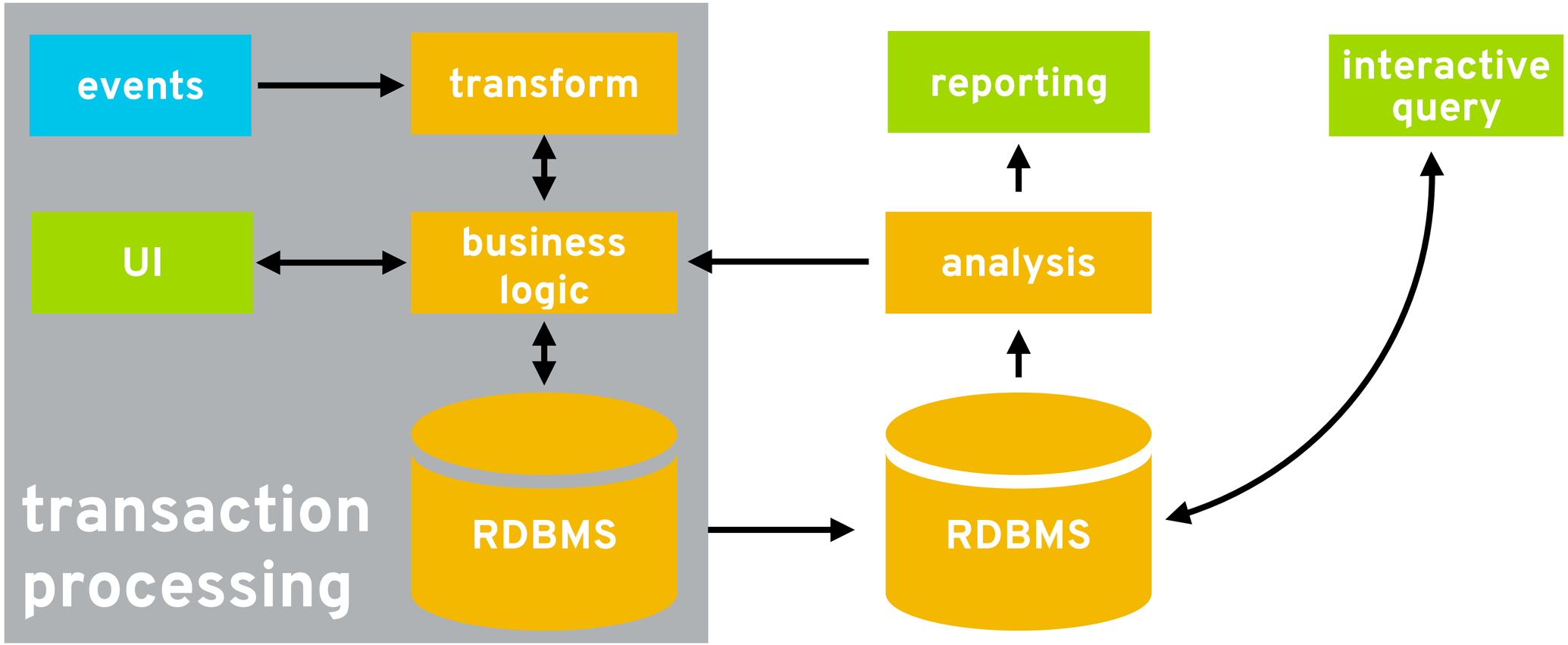

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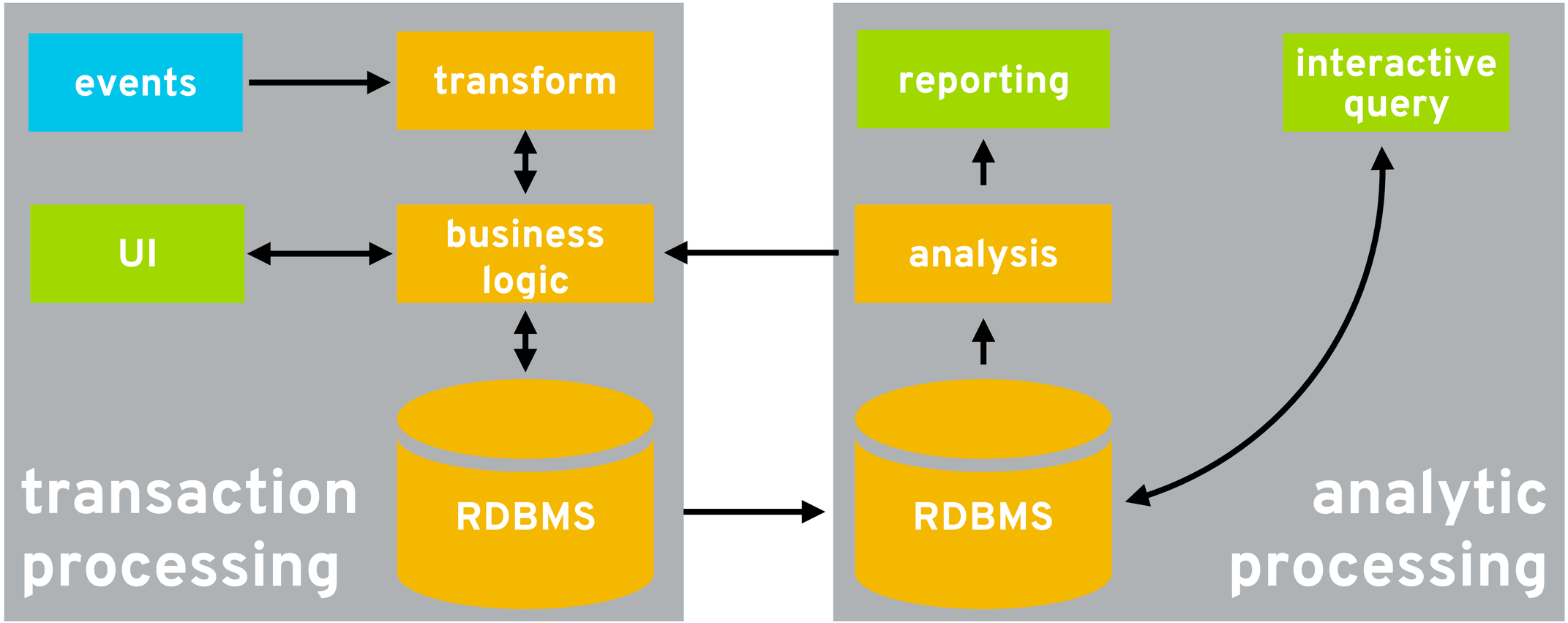
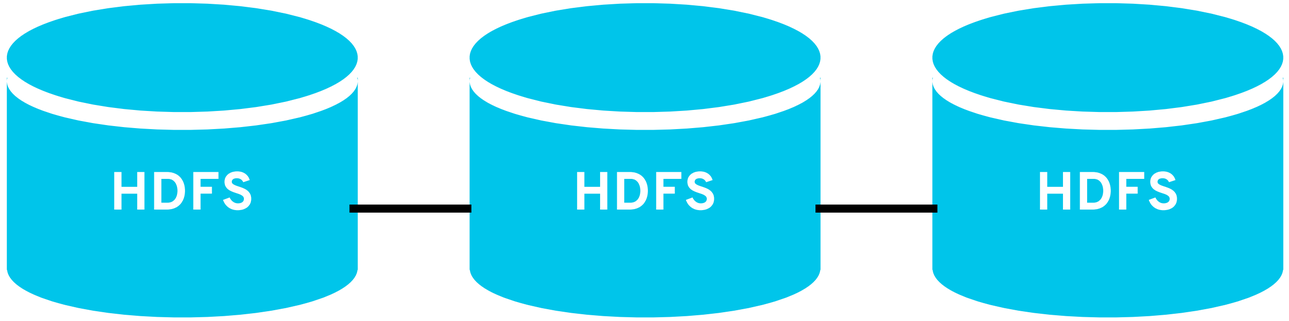


HADOOP-STYLE “DATA LAKE”



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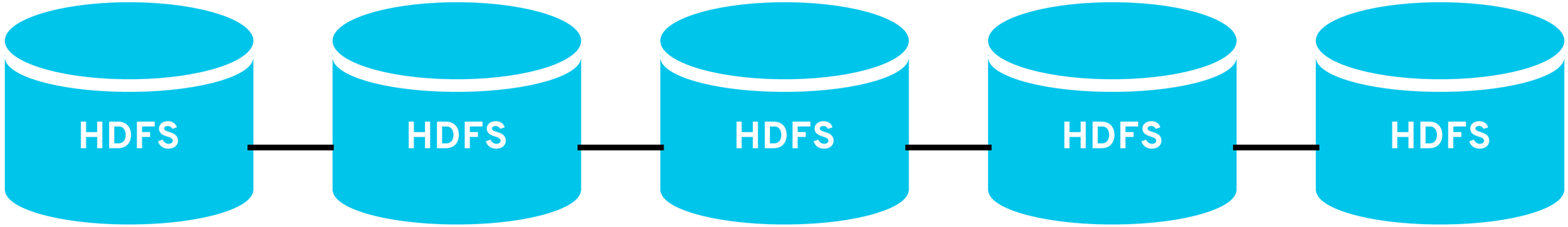

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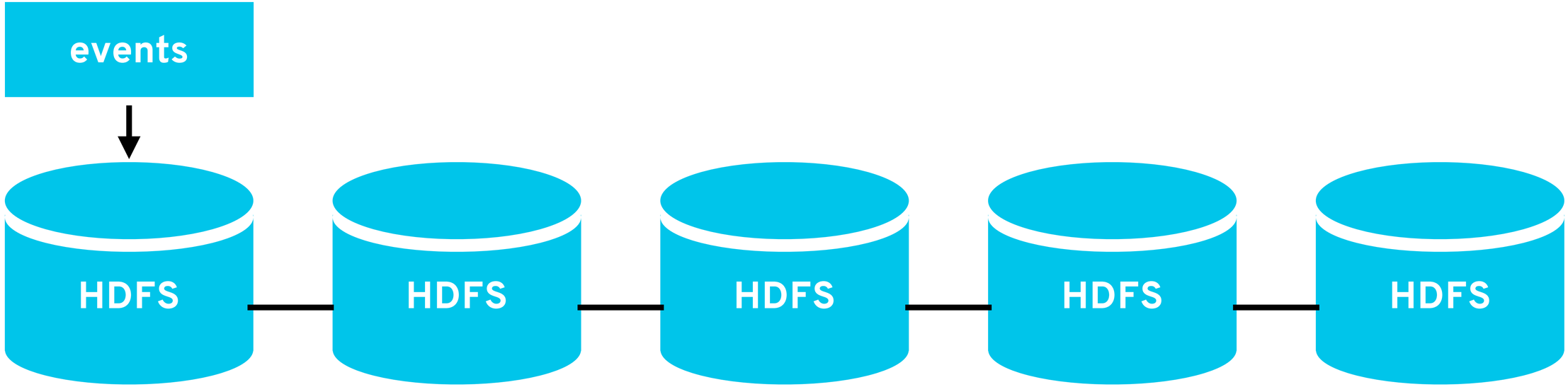

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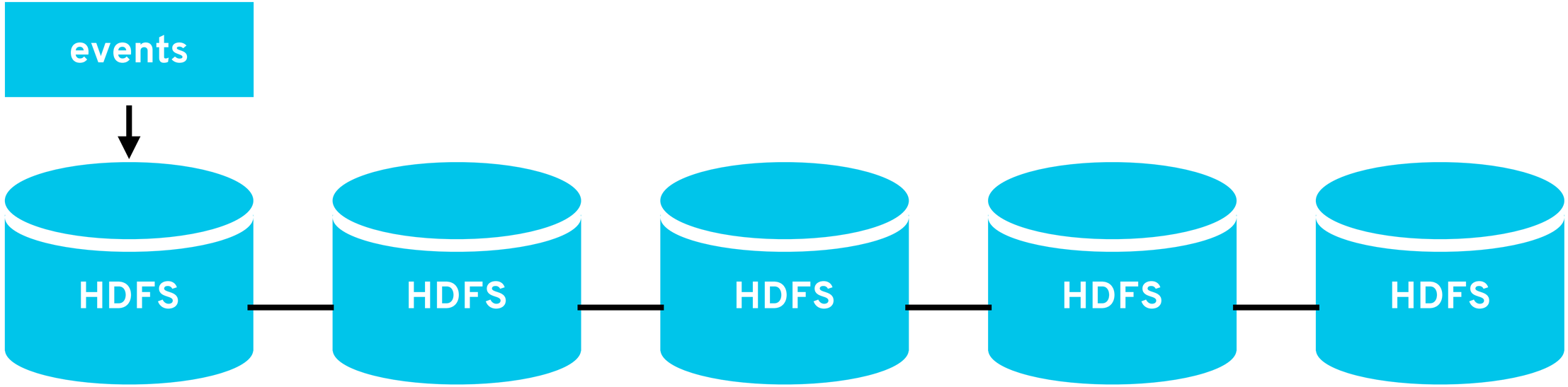

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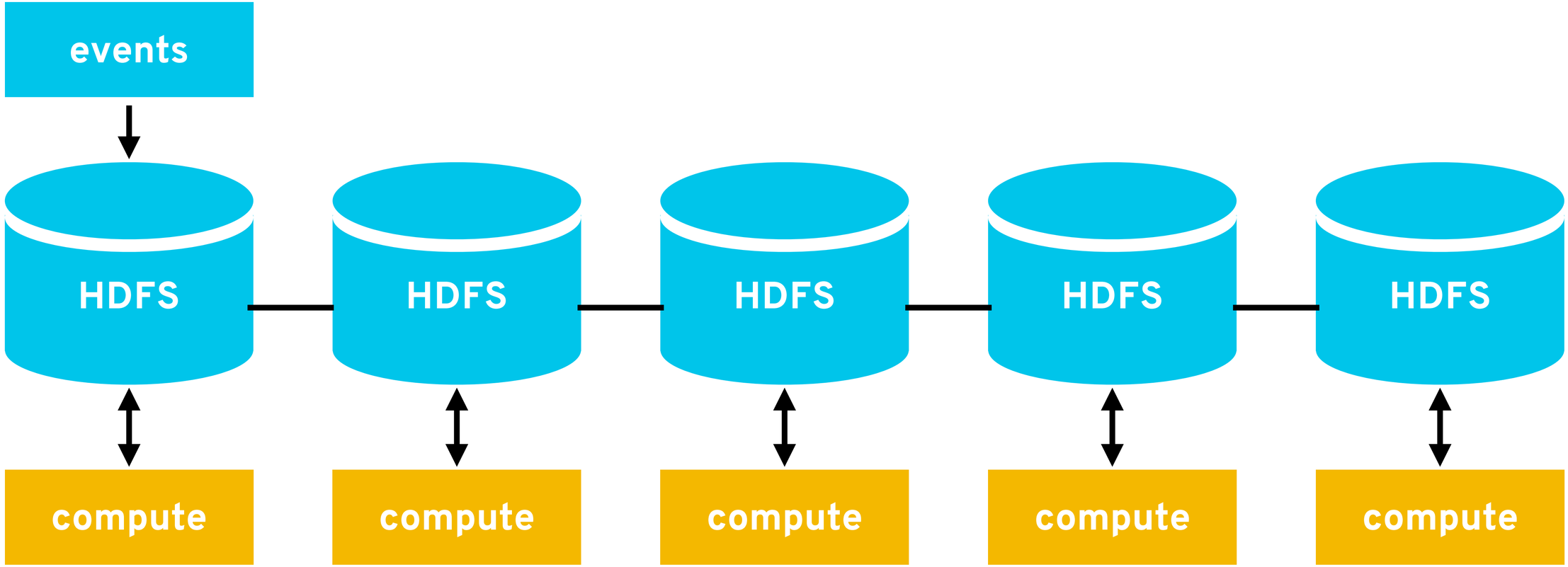
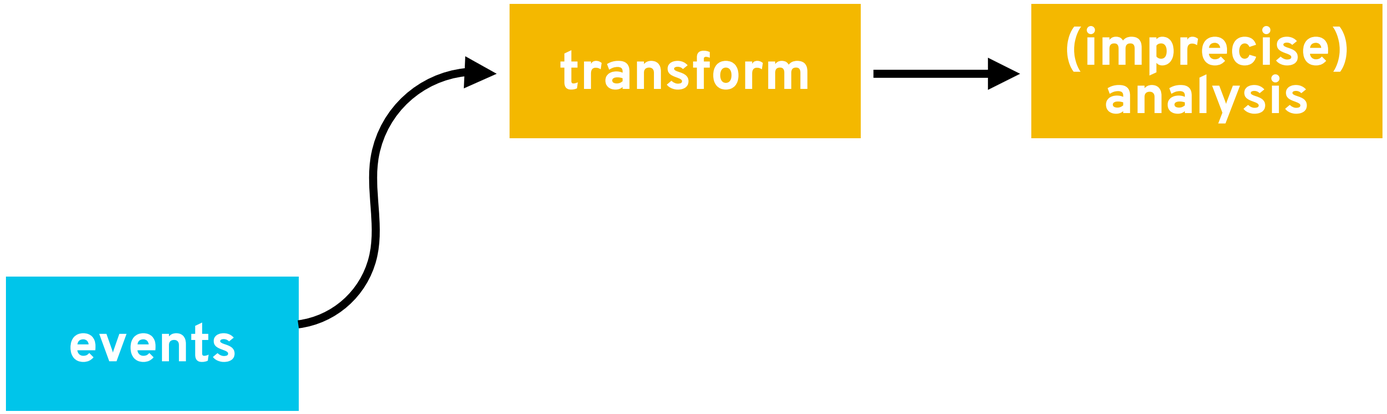
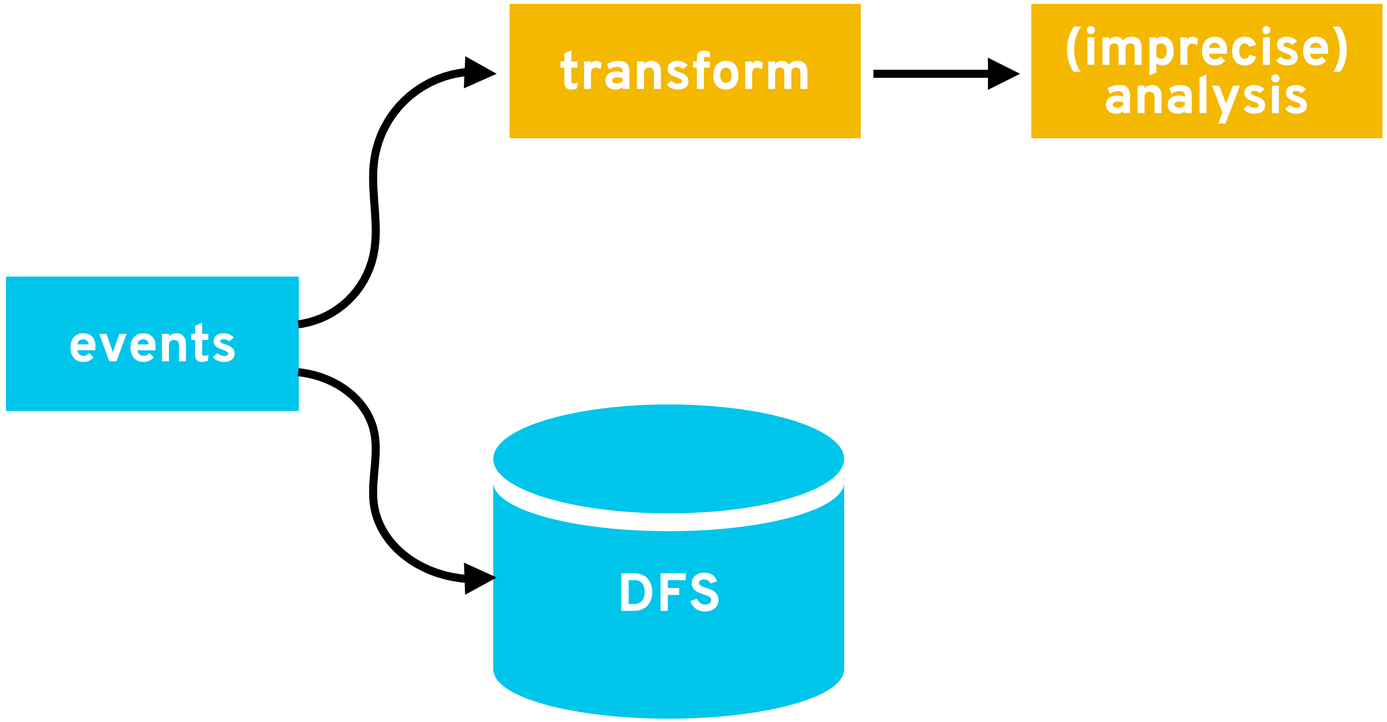


MODERN ARCHITECTURES

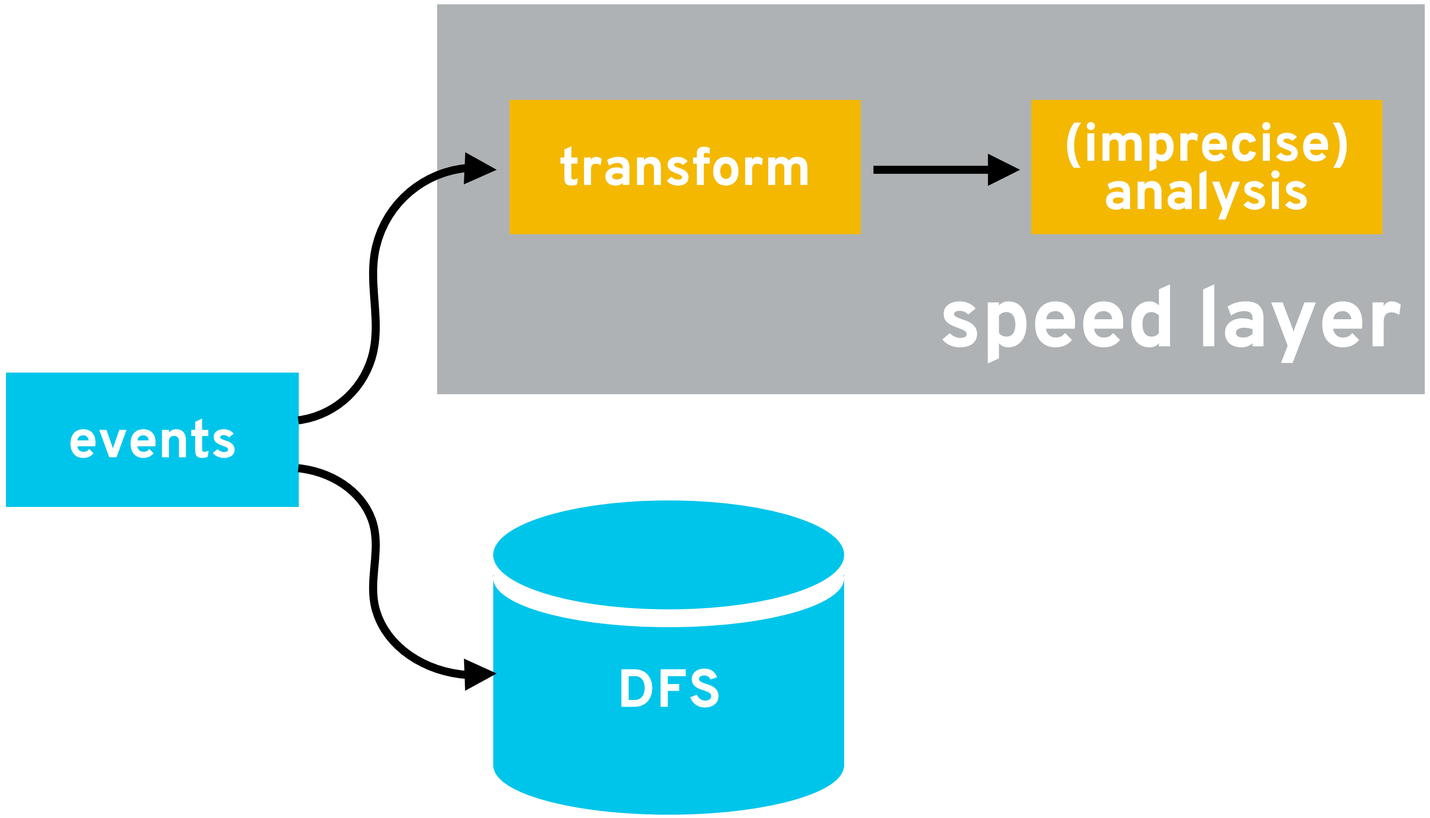
THE LAMBDA ARCHITECTURE



THE LAMBDA ARCHITECTURE



THE LAMBDA ARCHITECTURE



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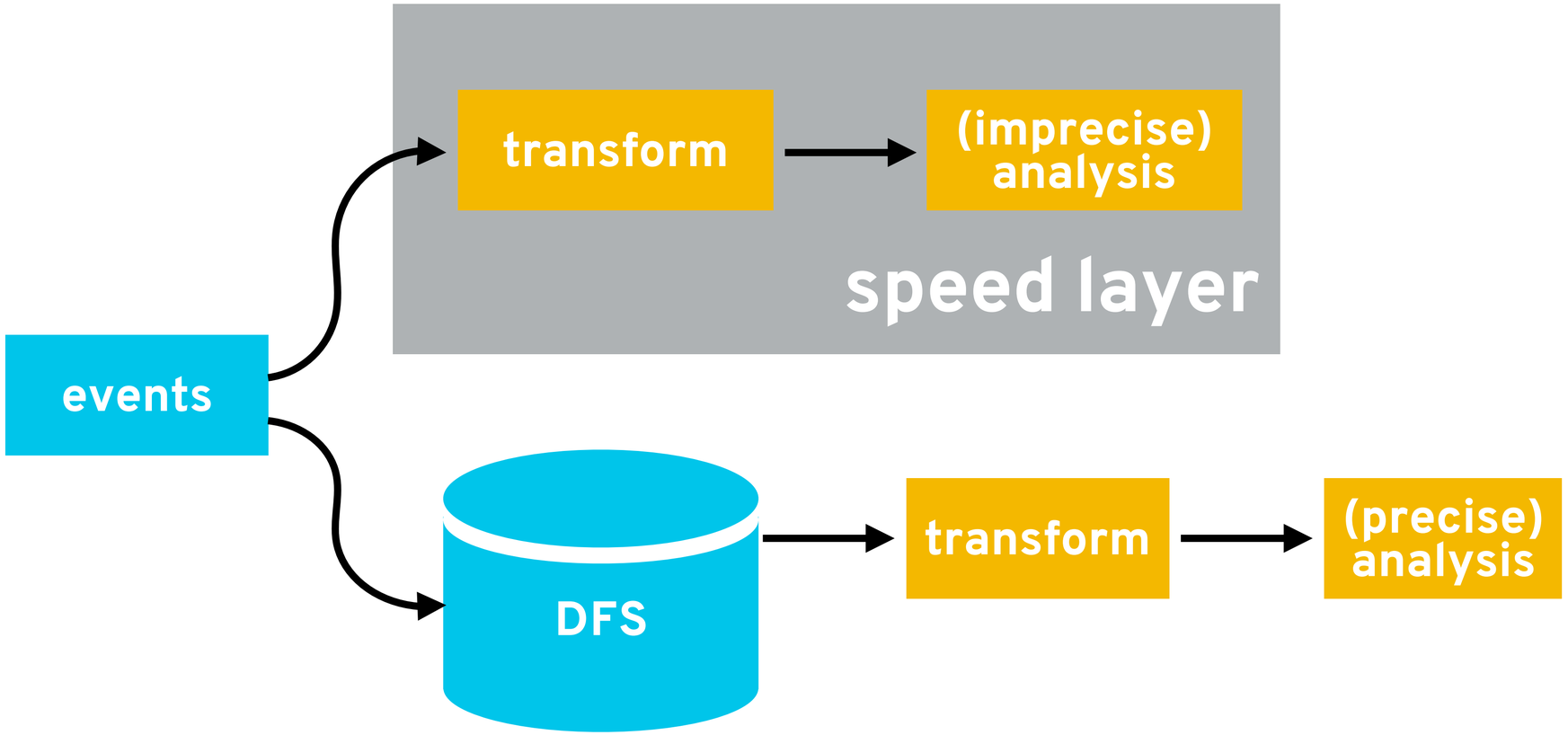

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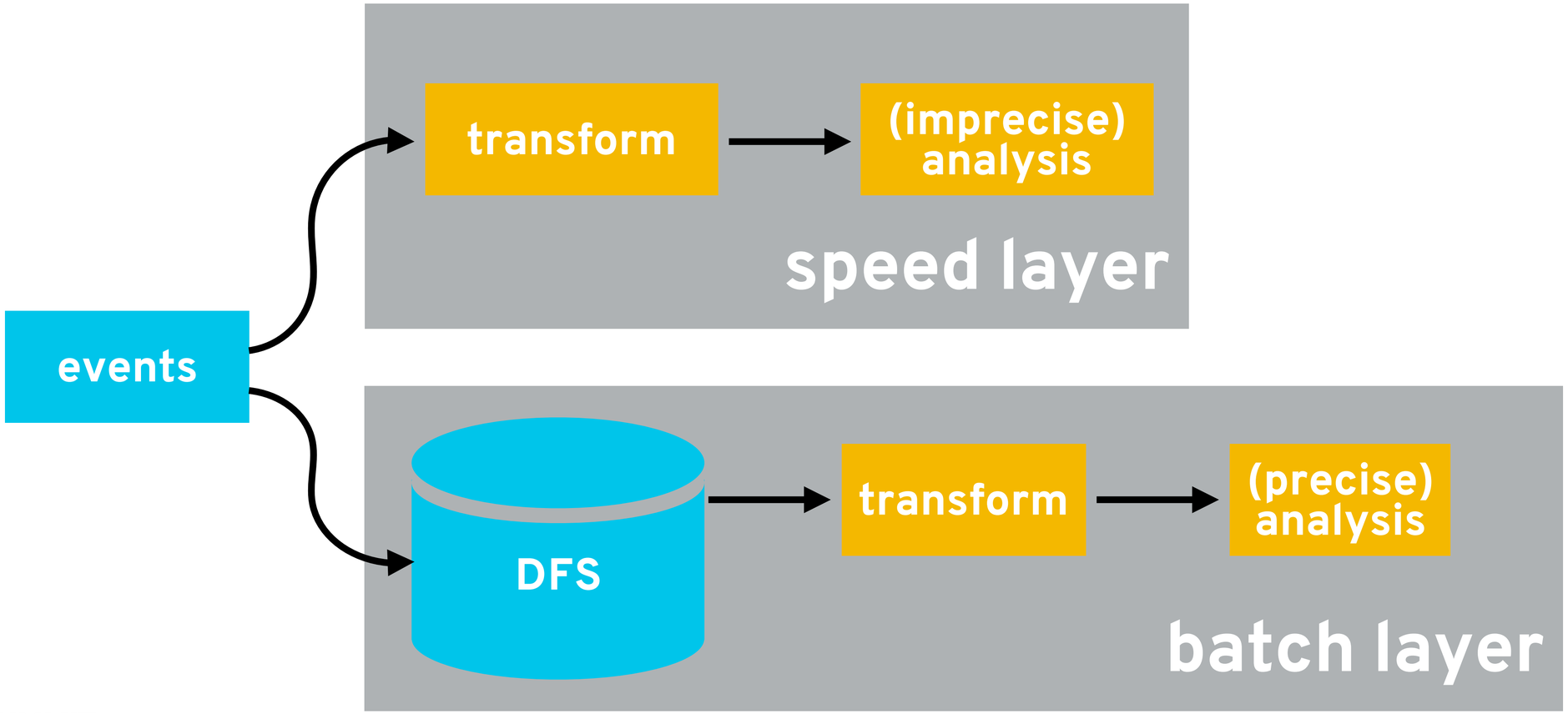

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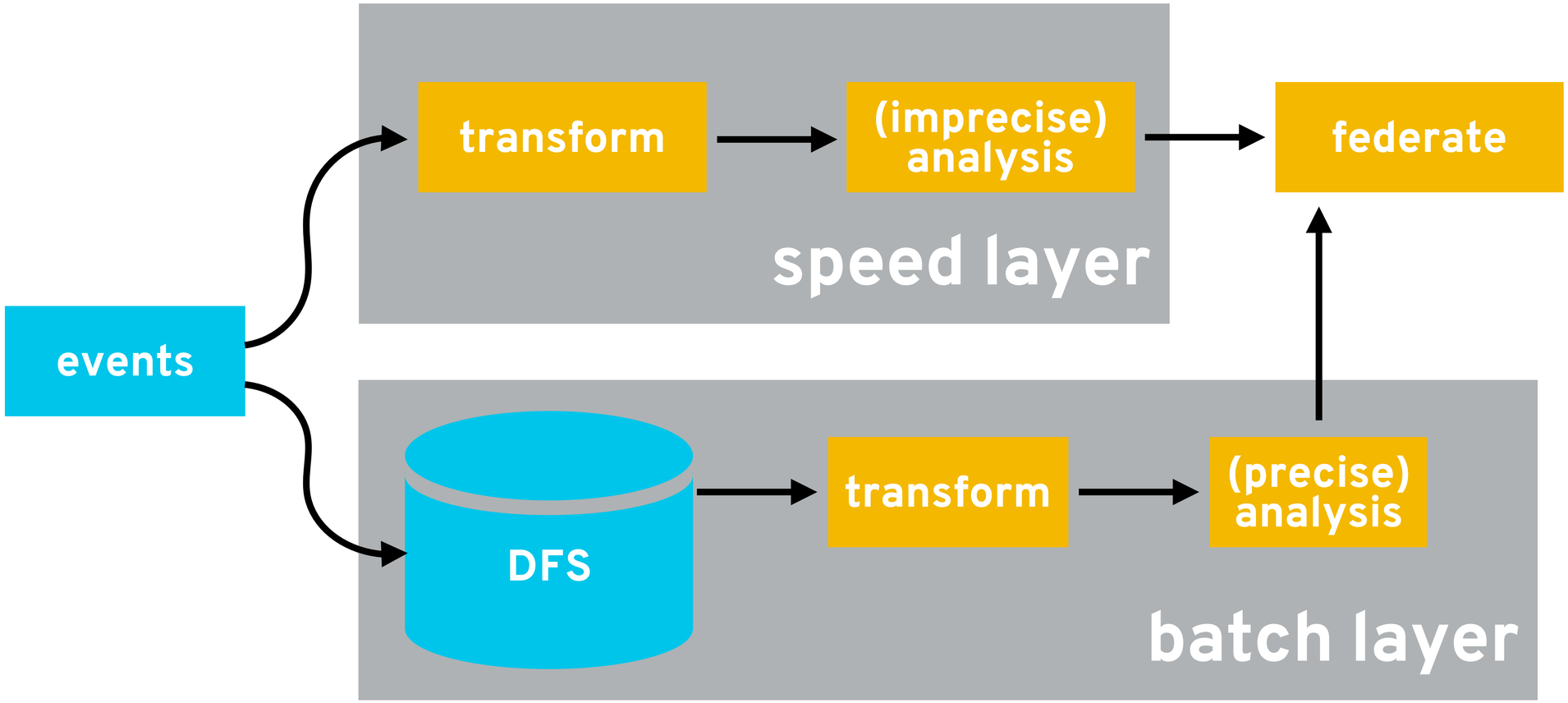

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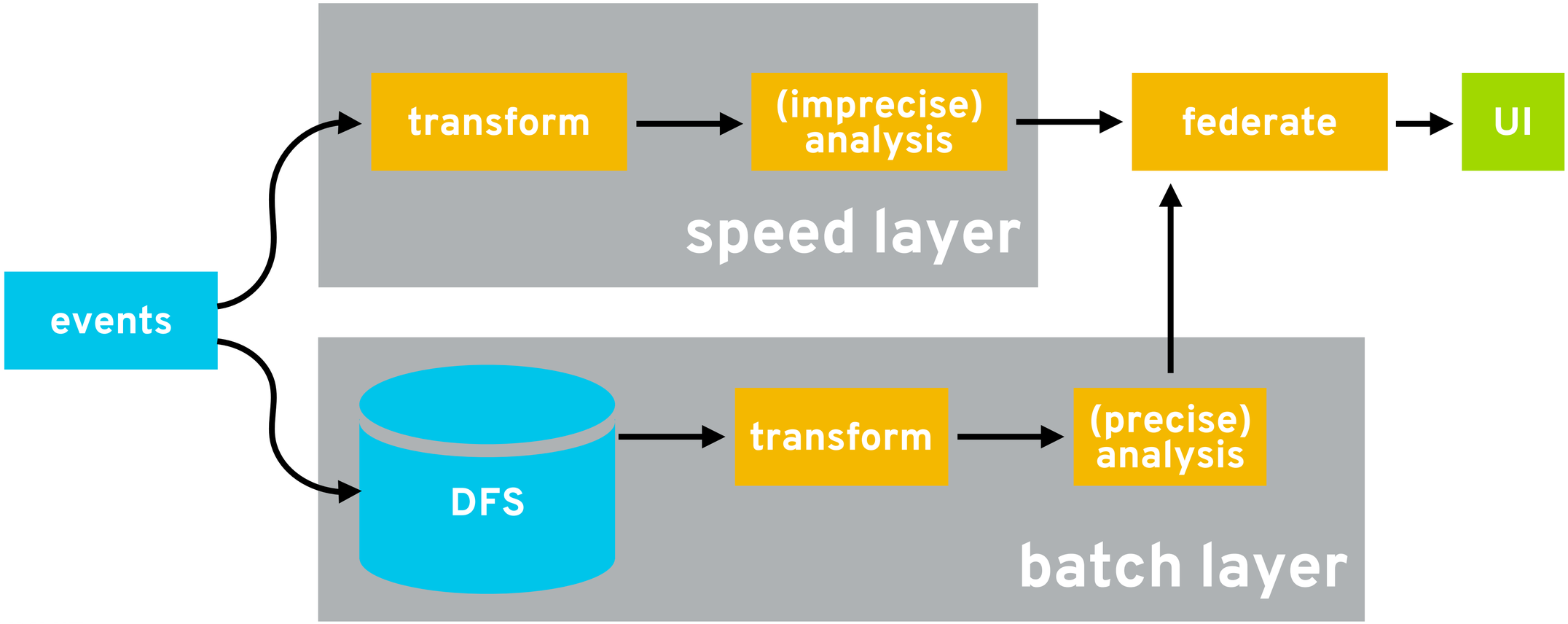

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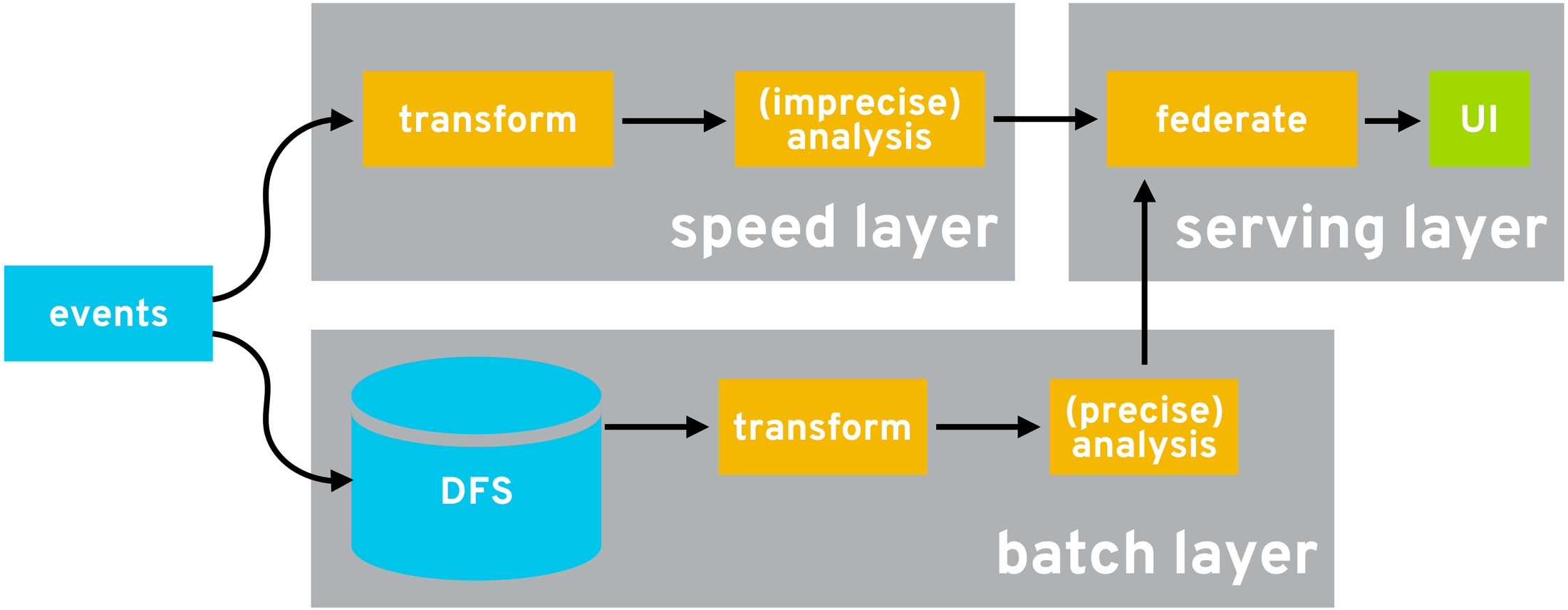

THE KAPPA ARCHITECTURE

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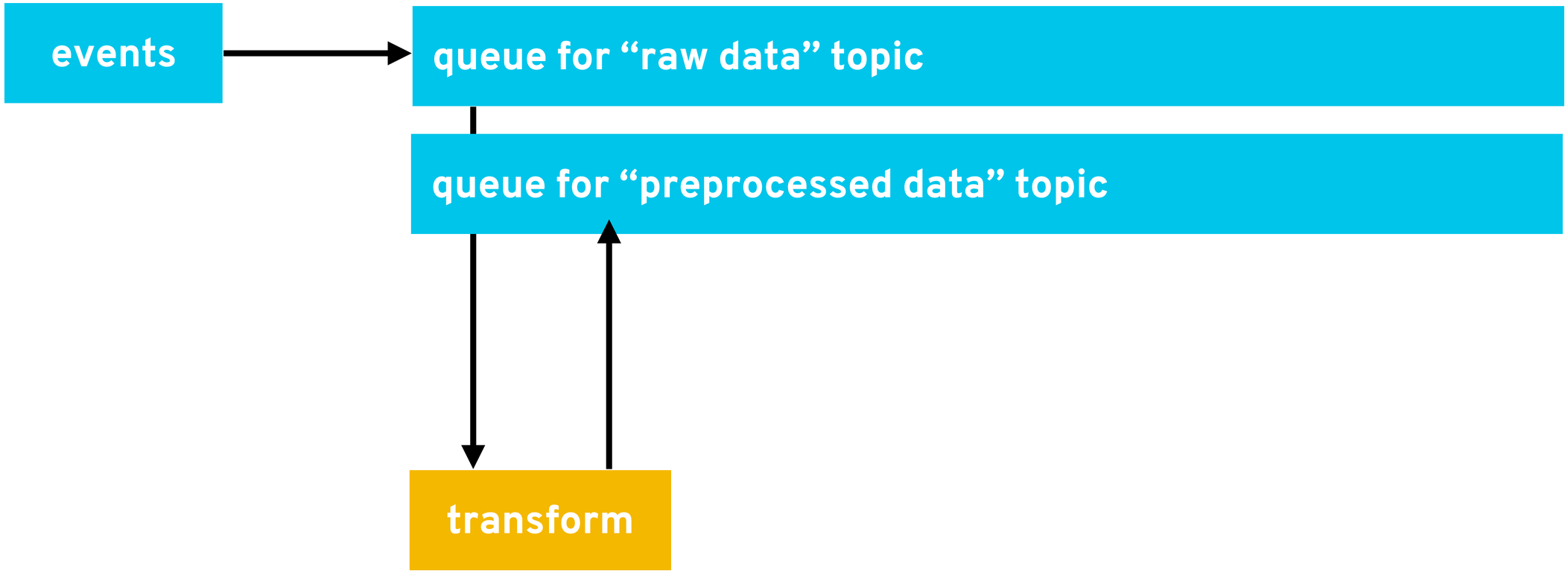

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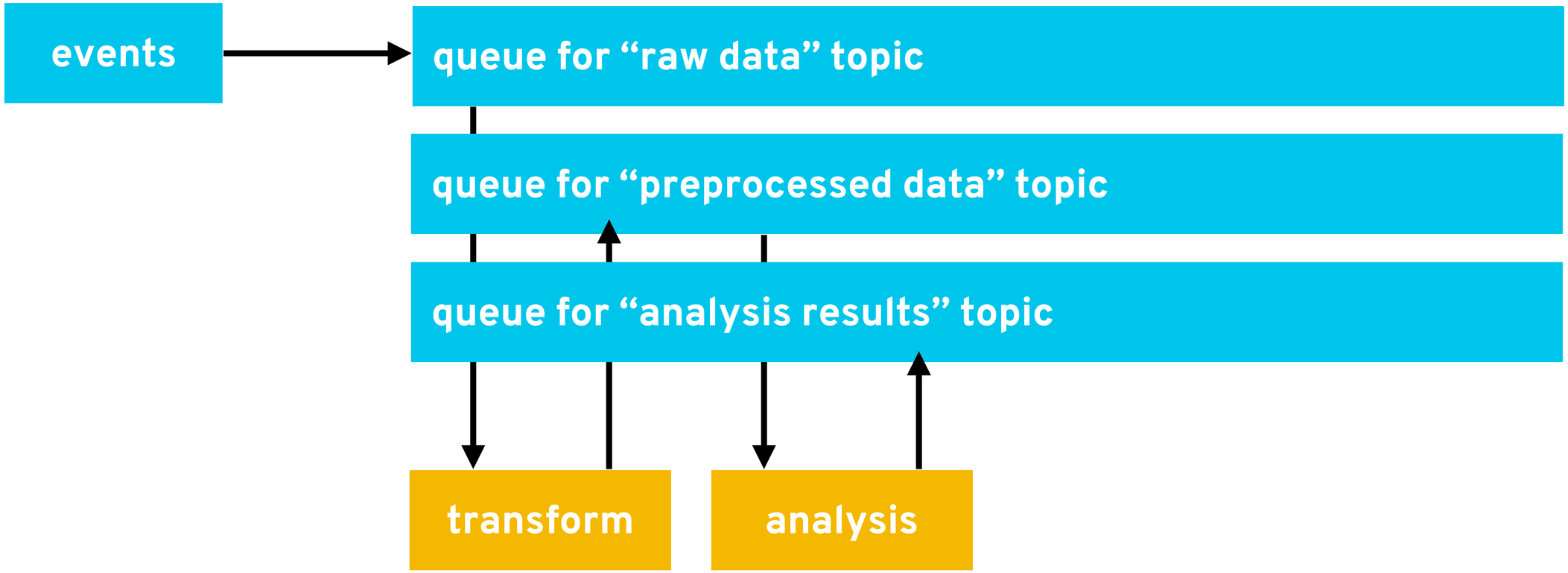
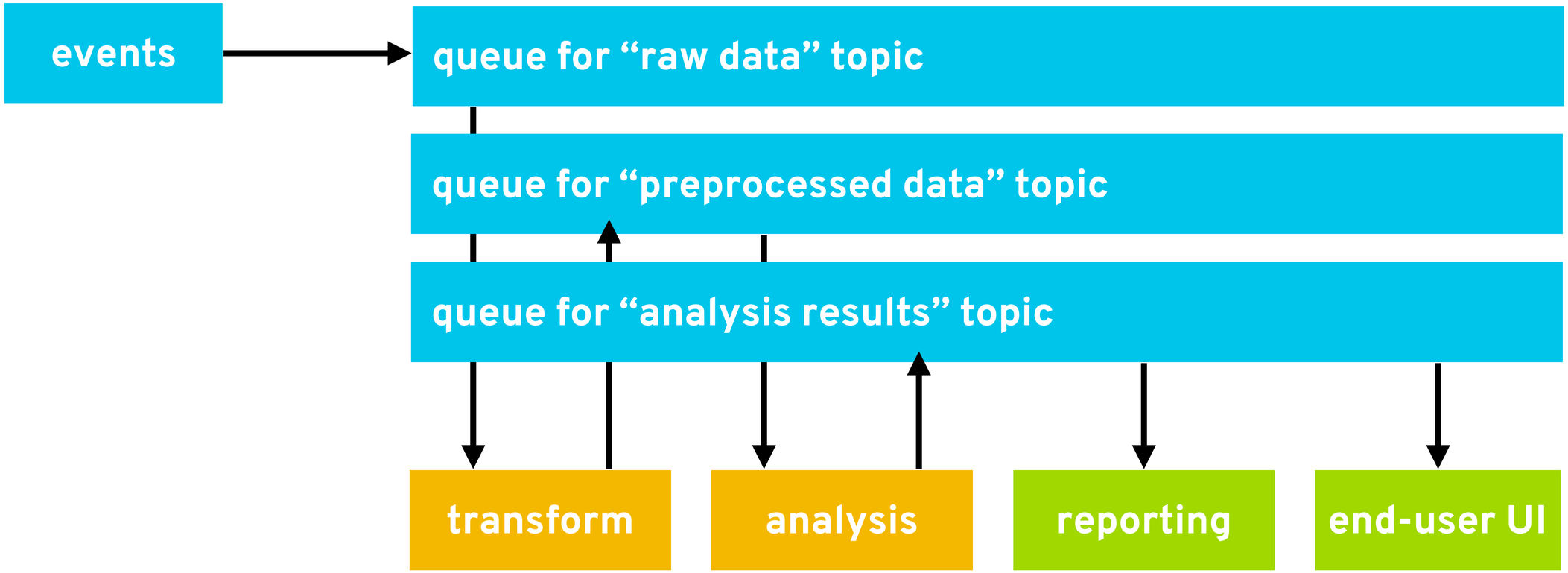
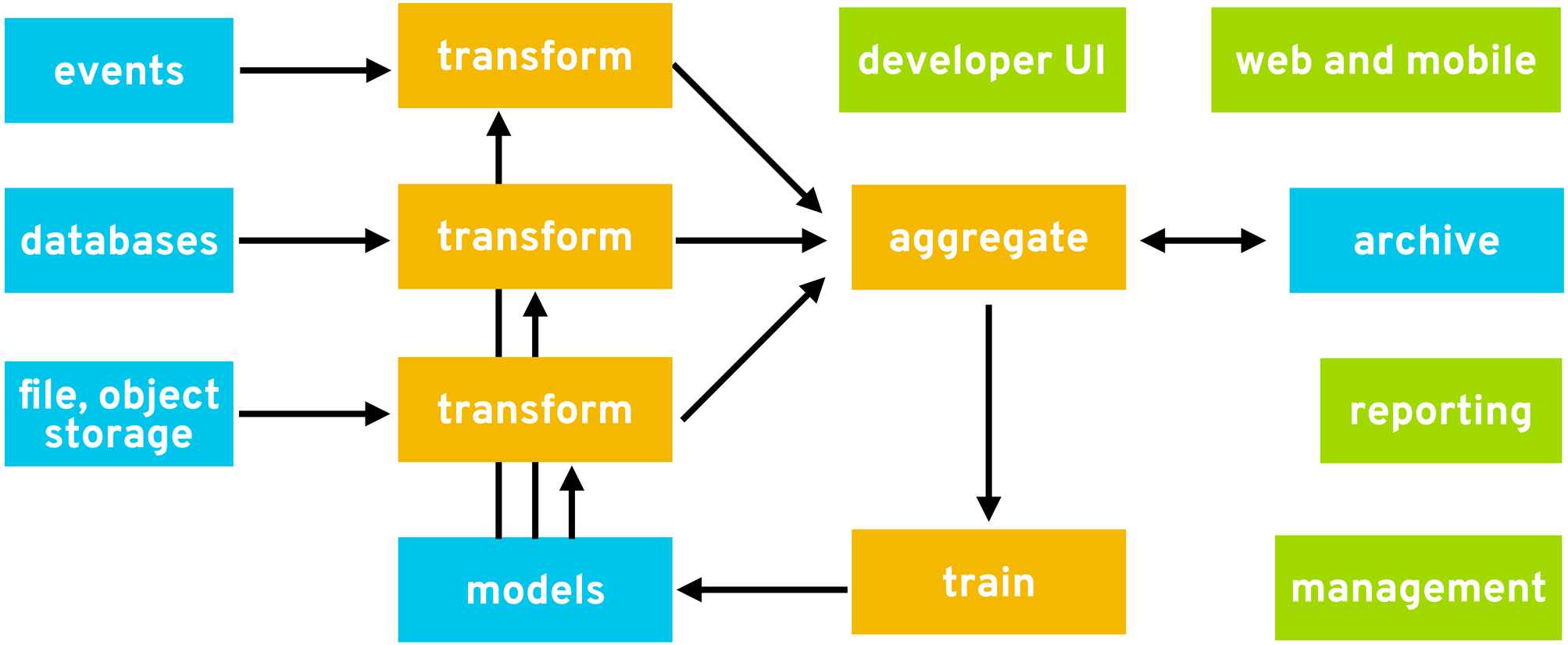


THE KAPPA ARCHITECTURE



DATA FEDERATION IN THE COMPUTE LAYER



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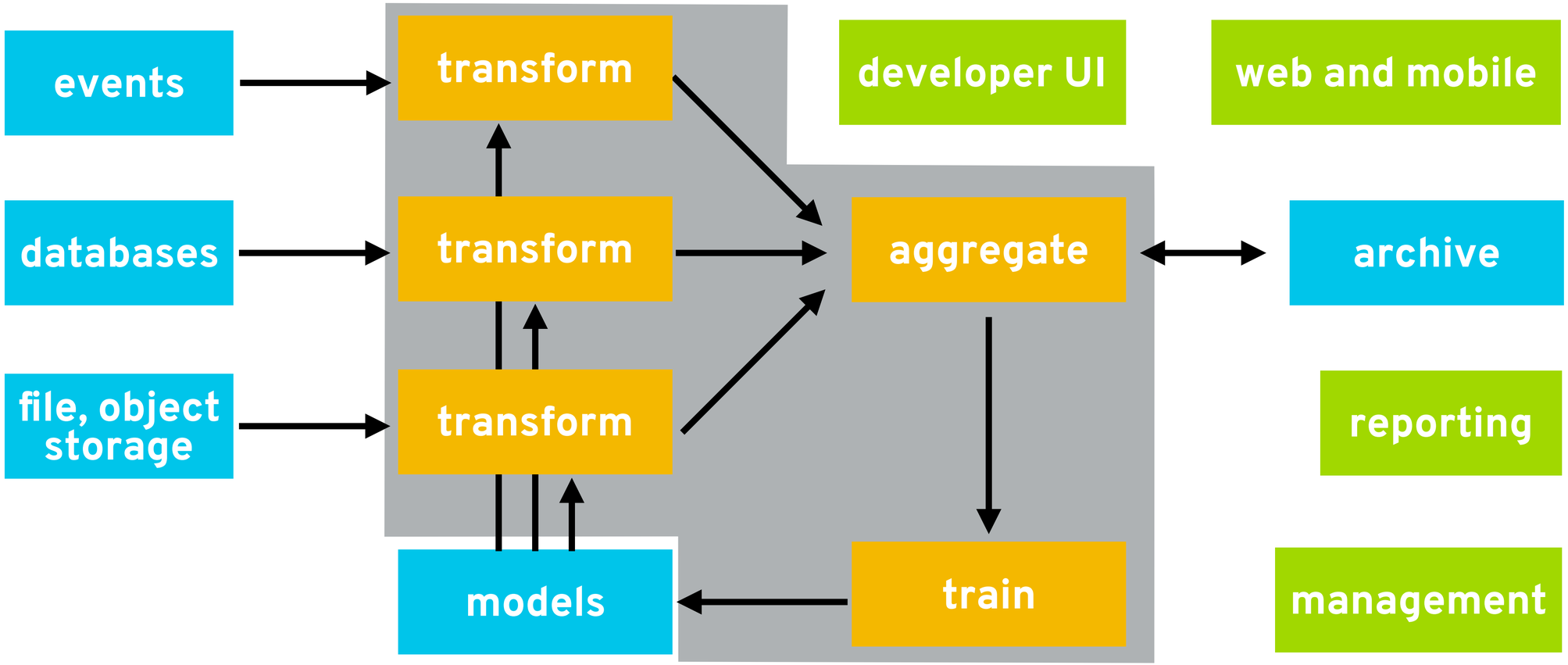
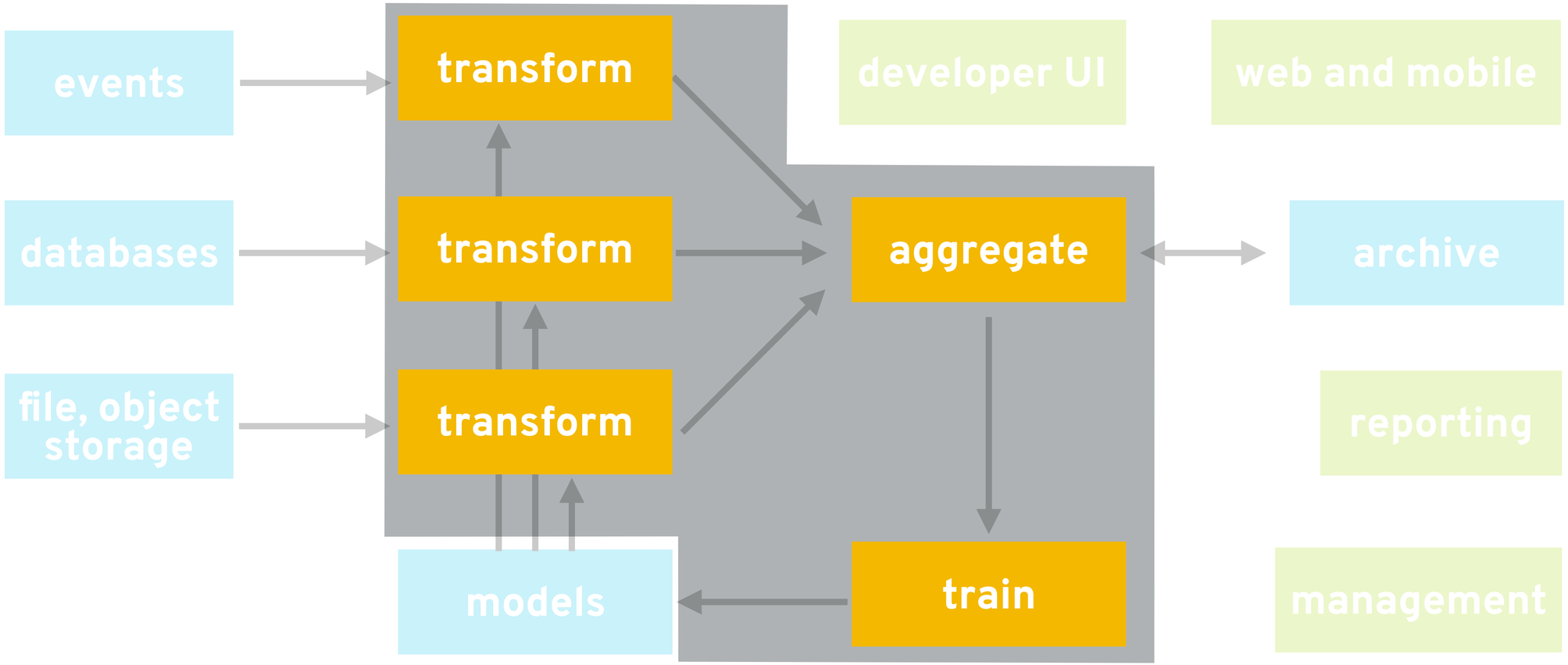
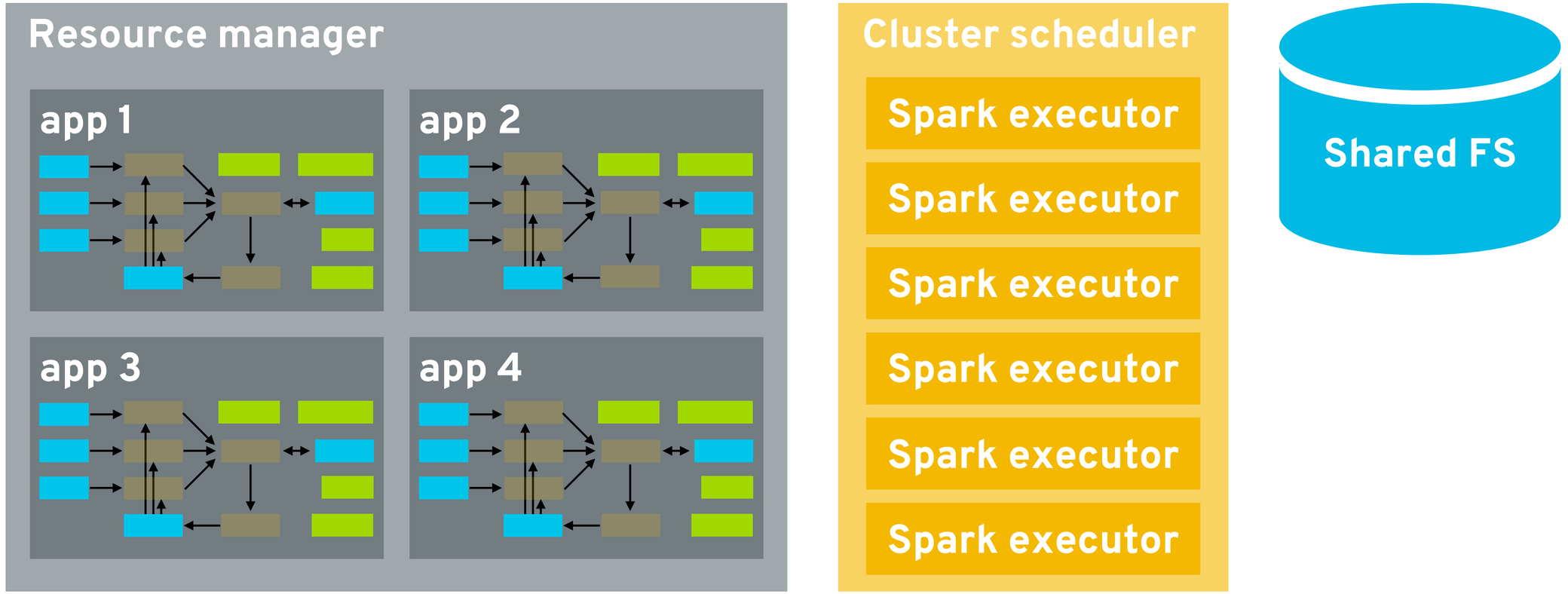


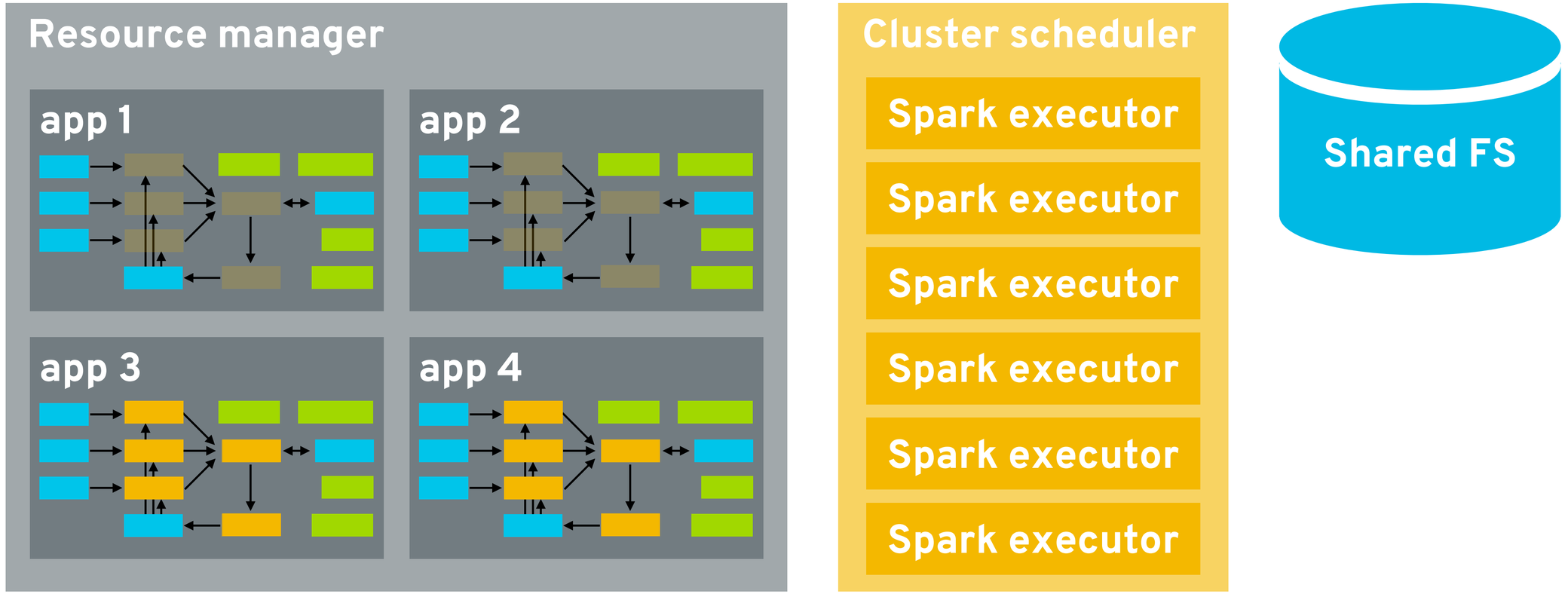
DATA FEDERATION IN THE COMPUTE LAYER



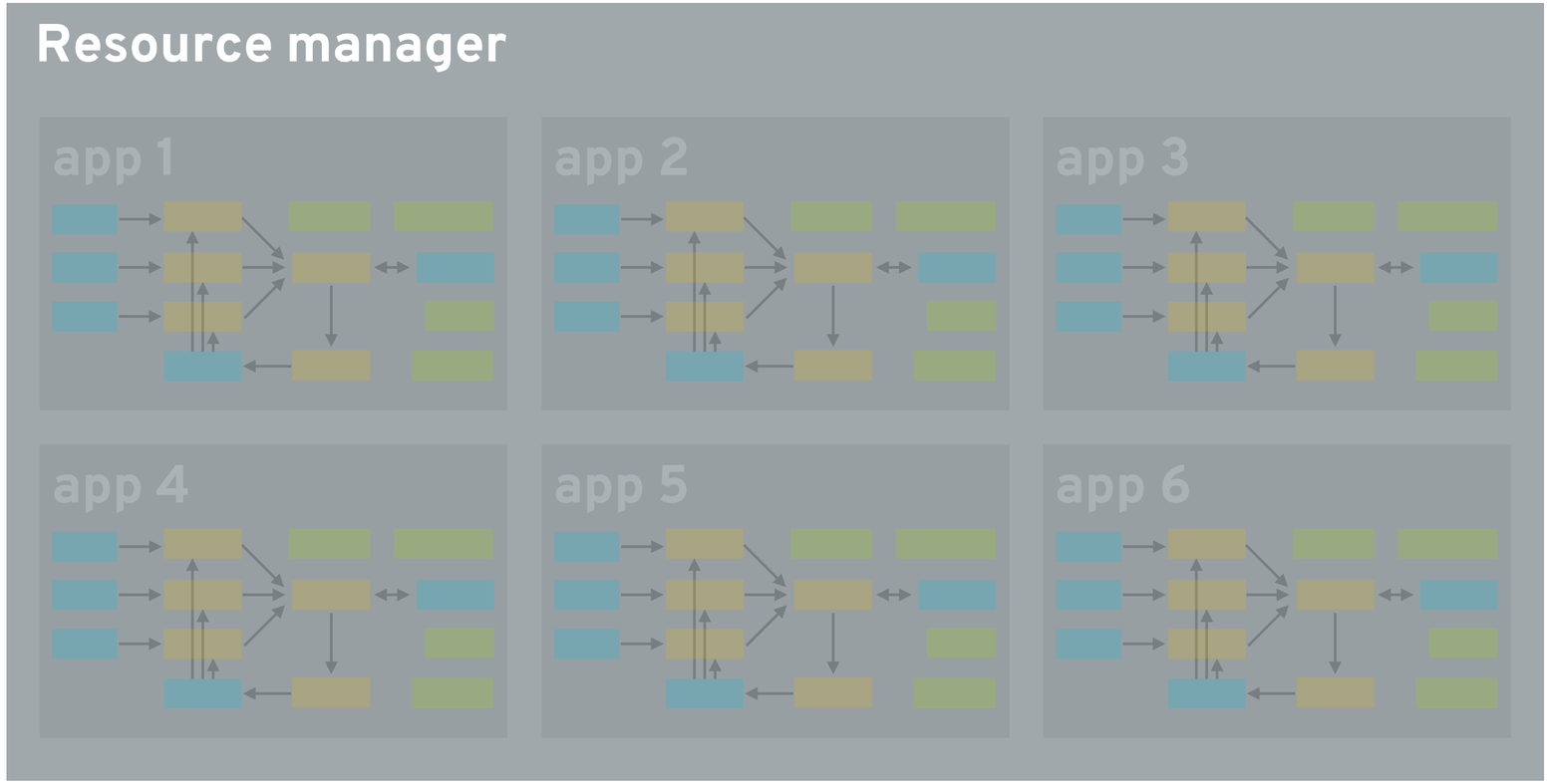
SIDEBAR: THE MONOLITHIC SPARK ANTIPATTERN



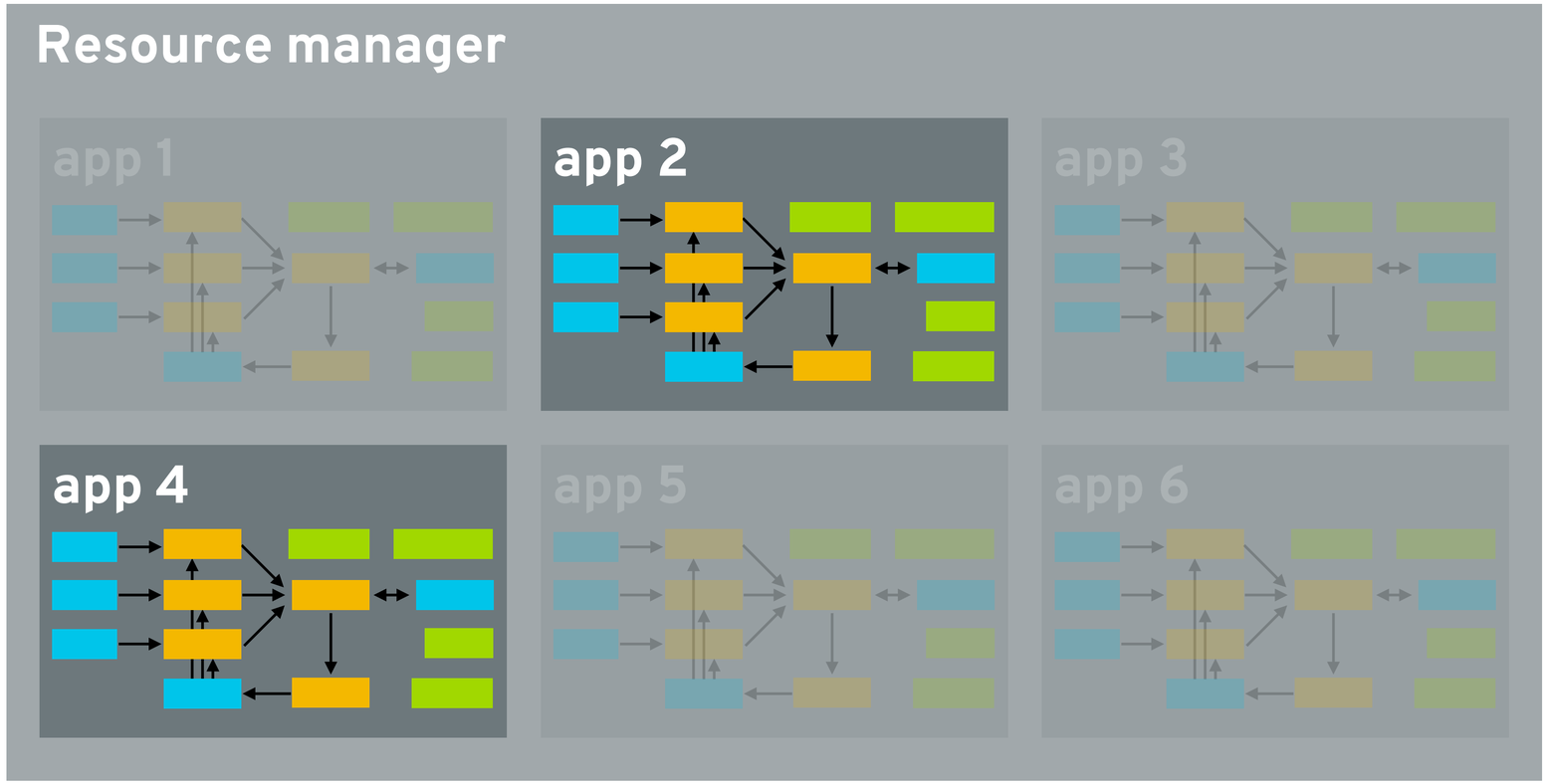
SIDEBAR: THE MONOLITHIC SPARK ANTIPATTERN



ONE CLUSTER PER APPLICATION

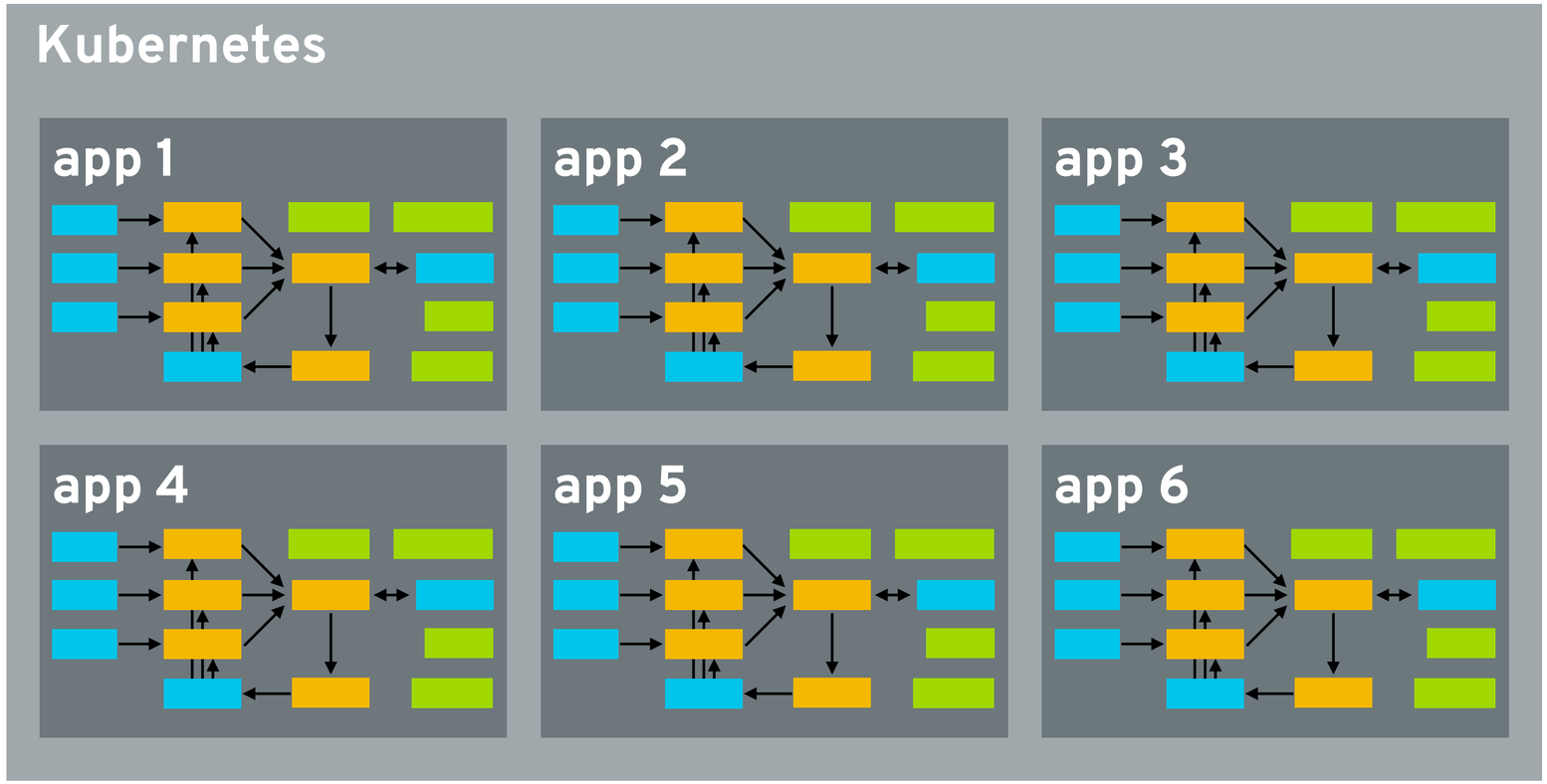


ONE CLUSTER PER APPLICATION



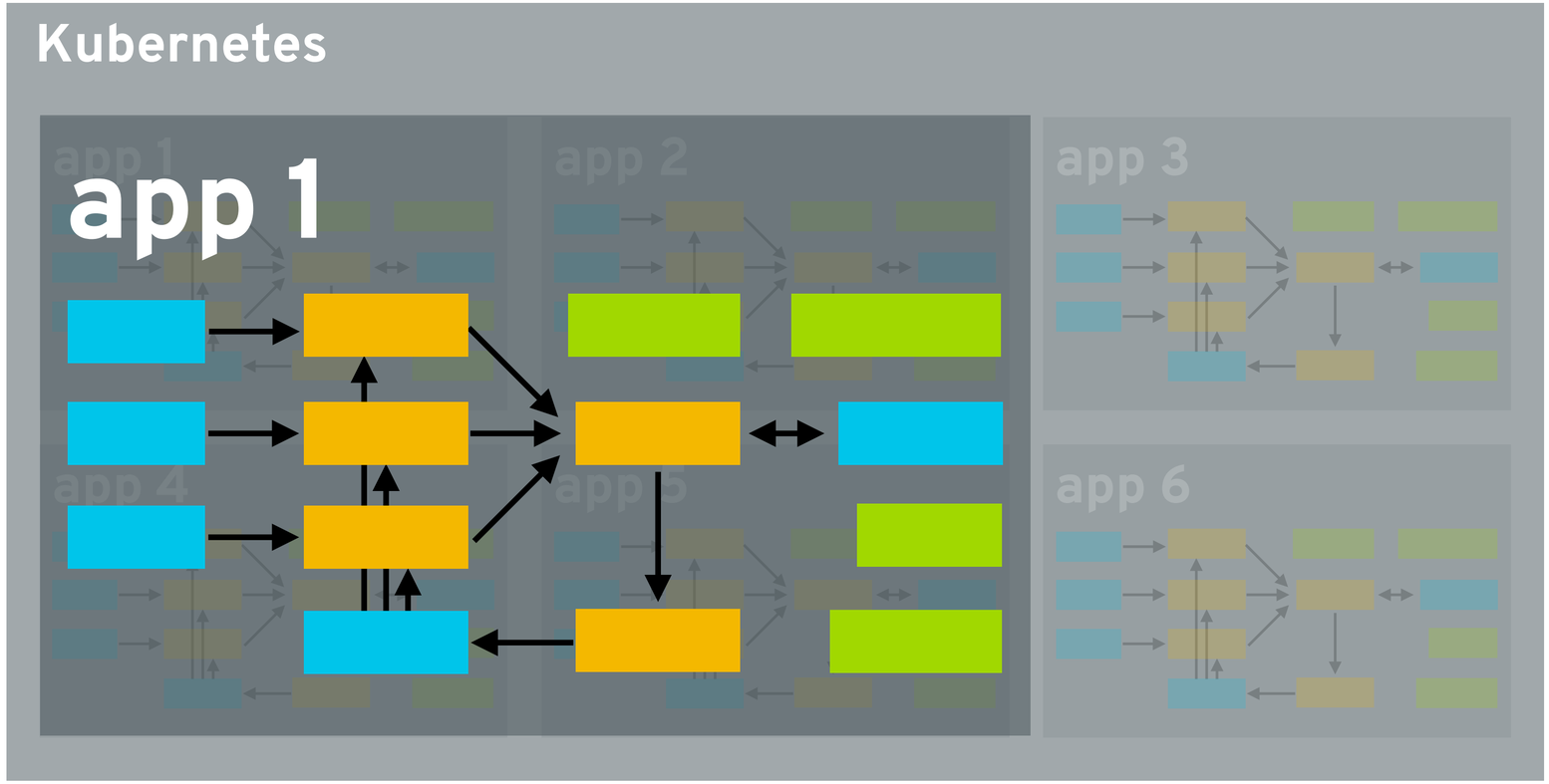
PRACTICALITIES AND POTENTIAL PITFALLS

SCHEDULING


Kubernetes
app 1
app 2
app 3
app 4
app 5
app 6

Object stores
Databases

SCHEDULING



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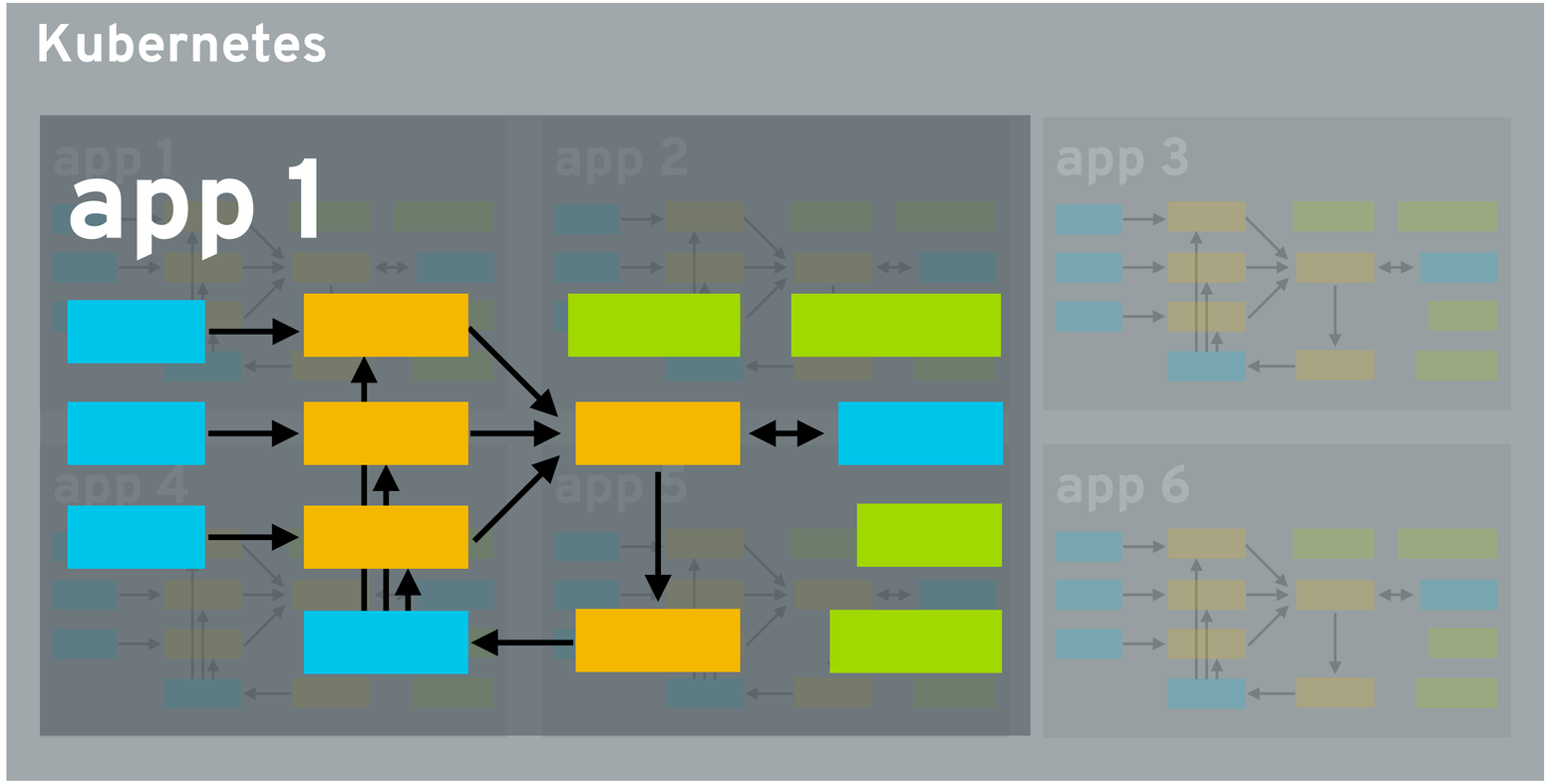

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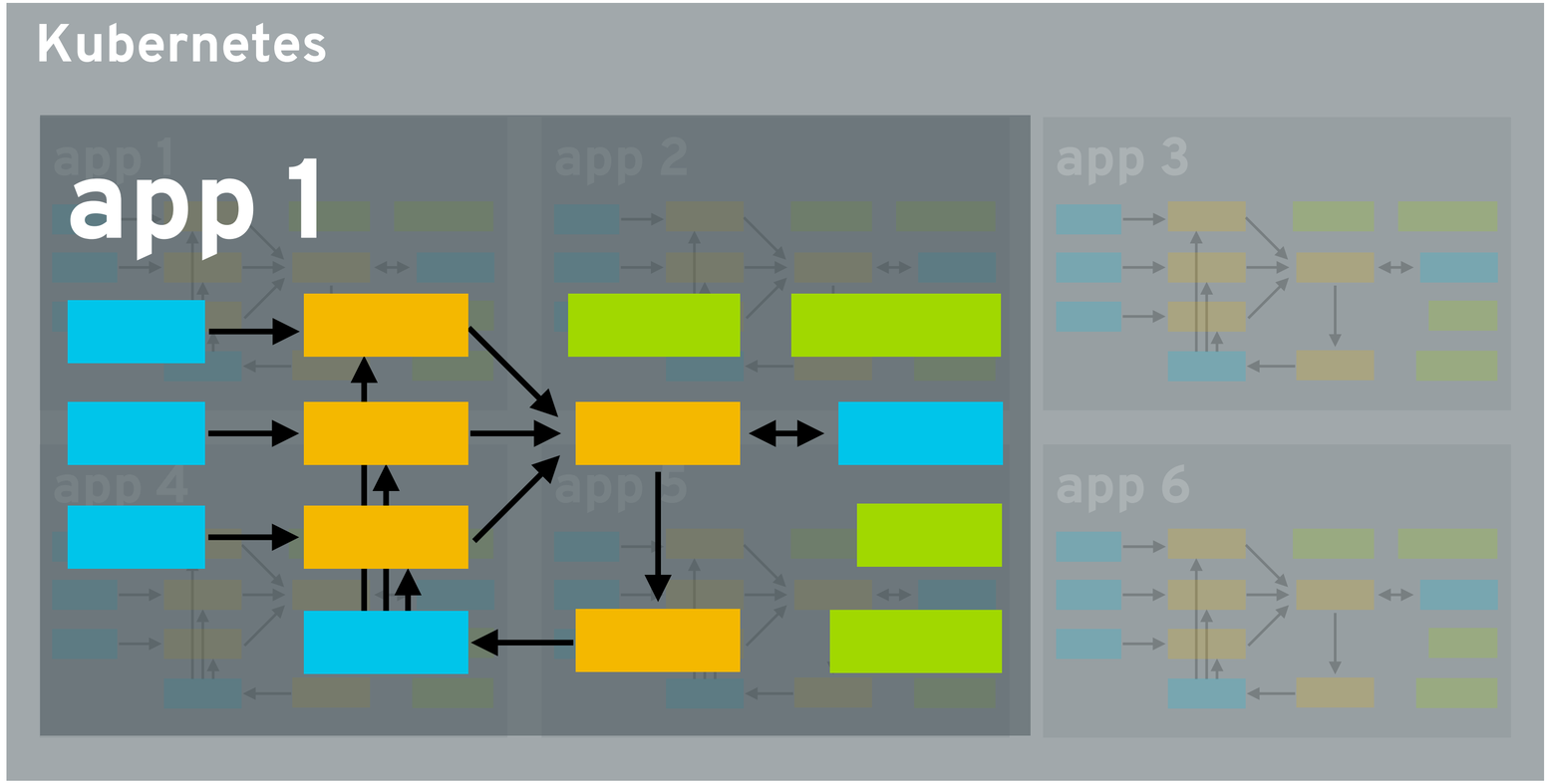

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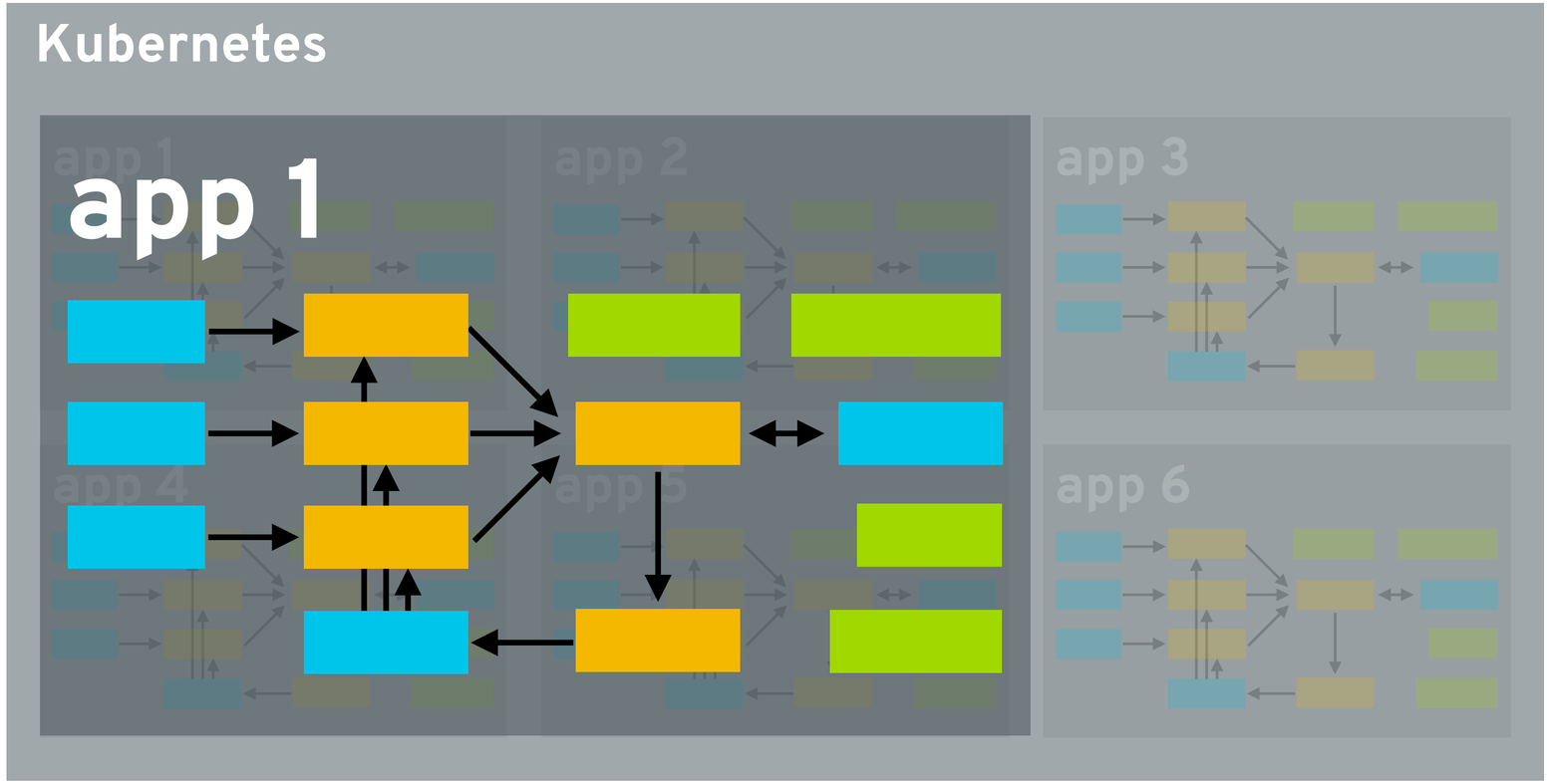

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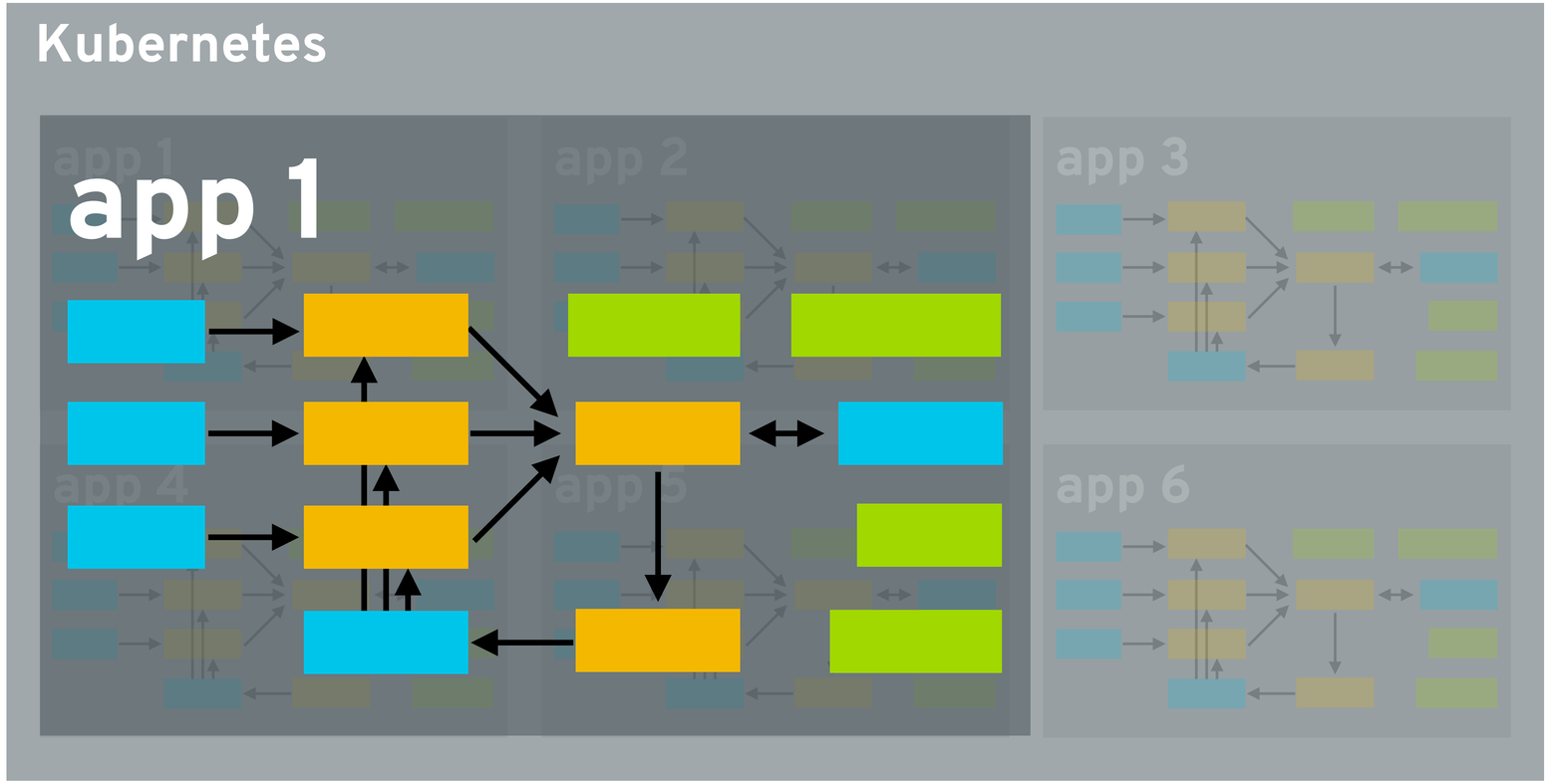
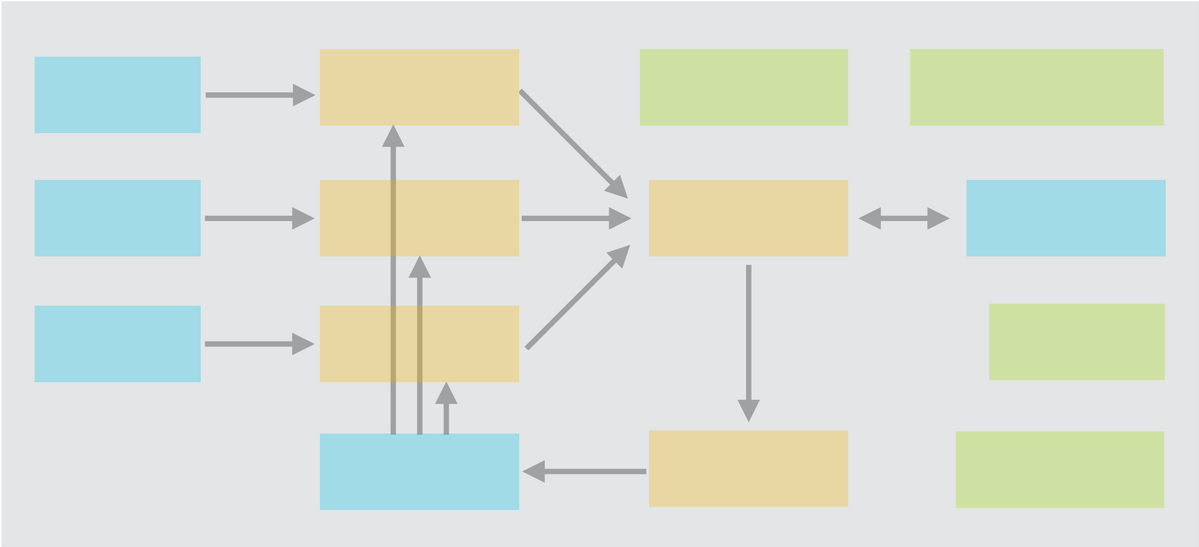


SECURITY



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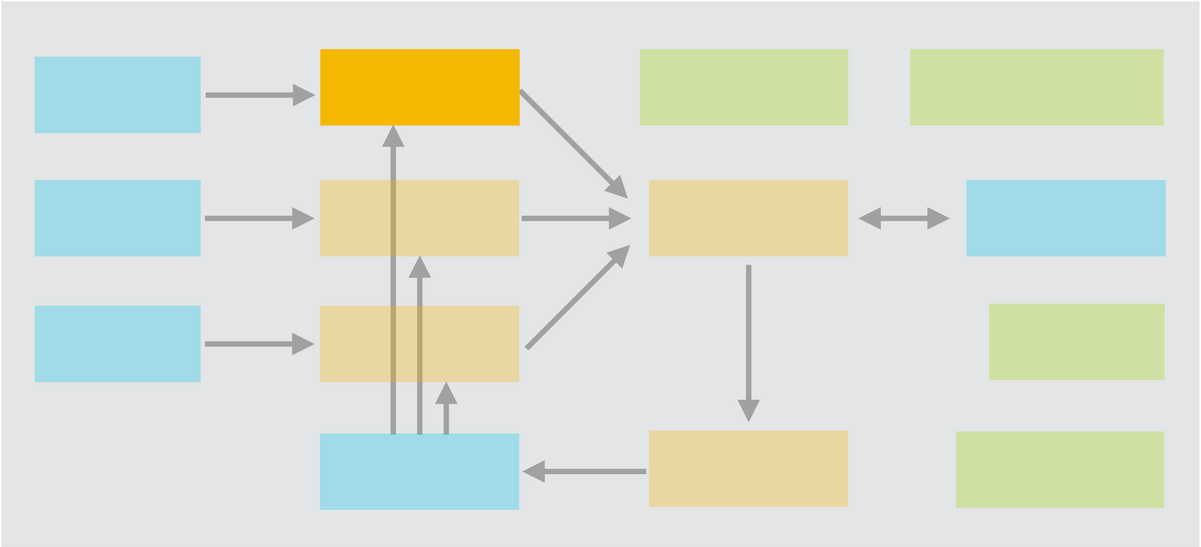

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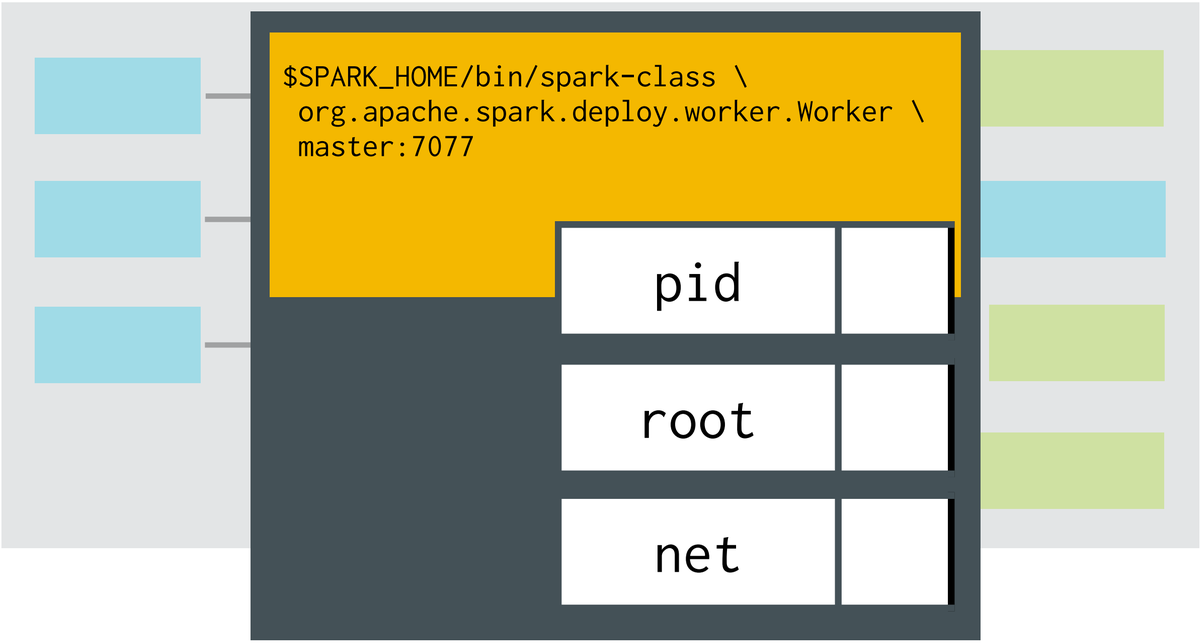
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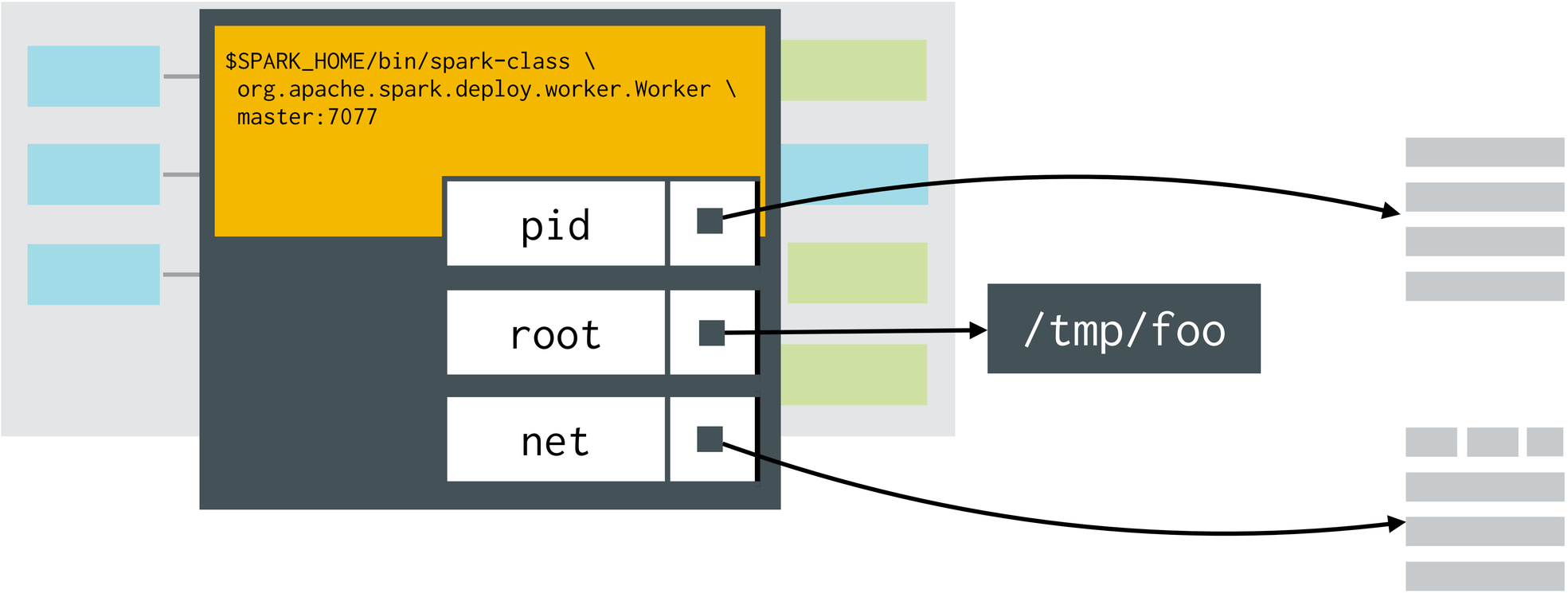
org.apache.spark.deploy.worker.Worker \
master:7077
pid
root
net

SECURITY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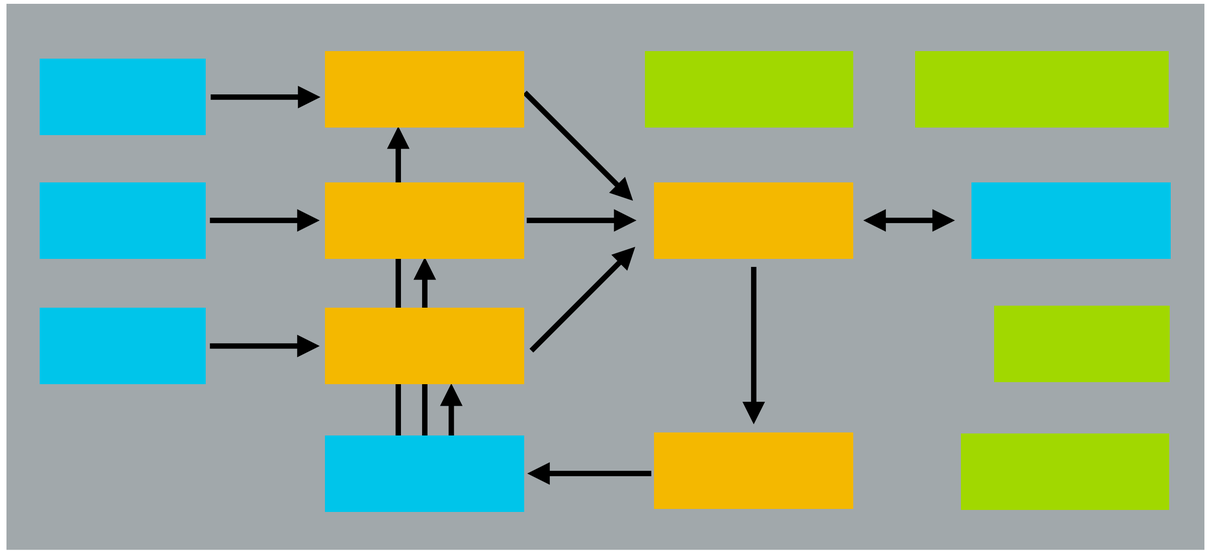
org.apache.spark.deploy.worker.Worker \
master:7077
pid
root
net
/tmp/foo

SECURITY



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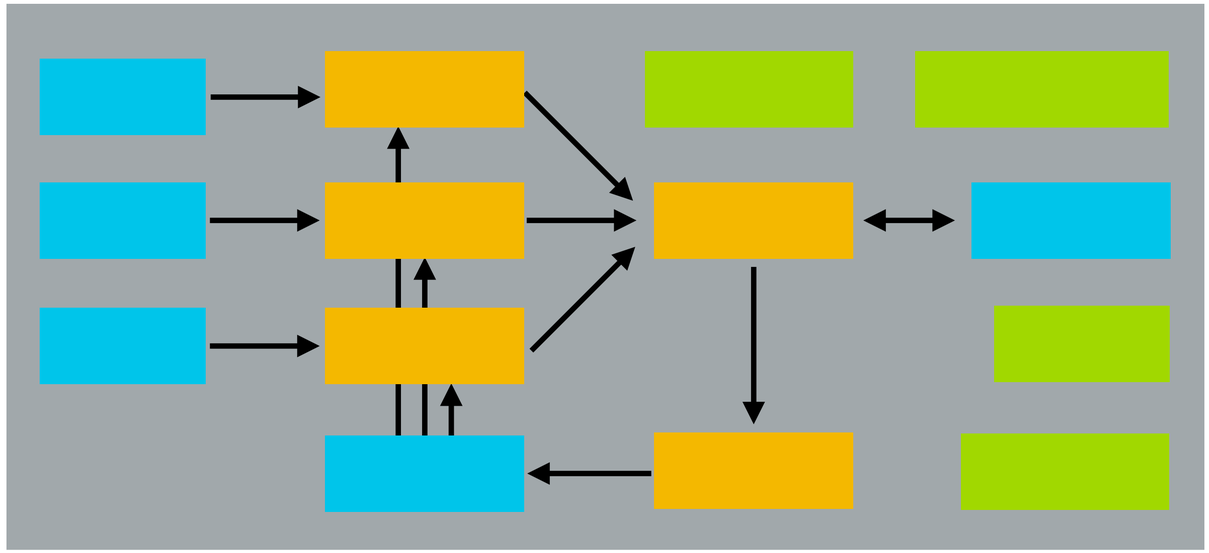
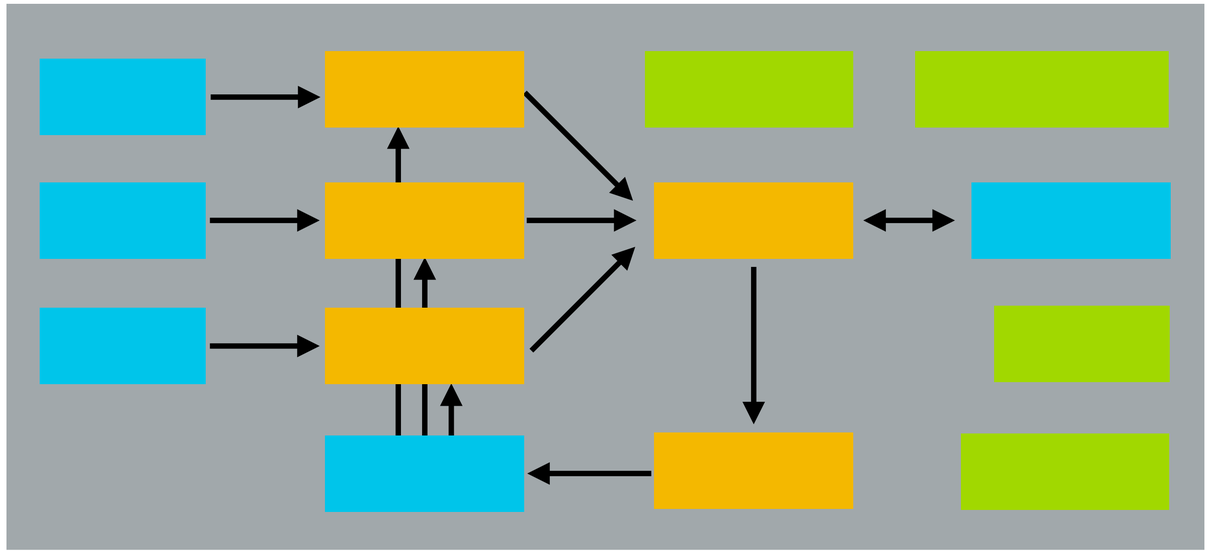


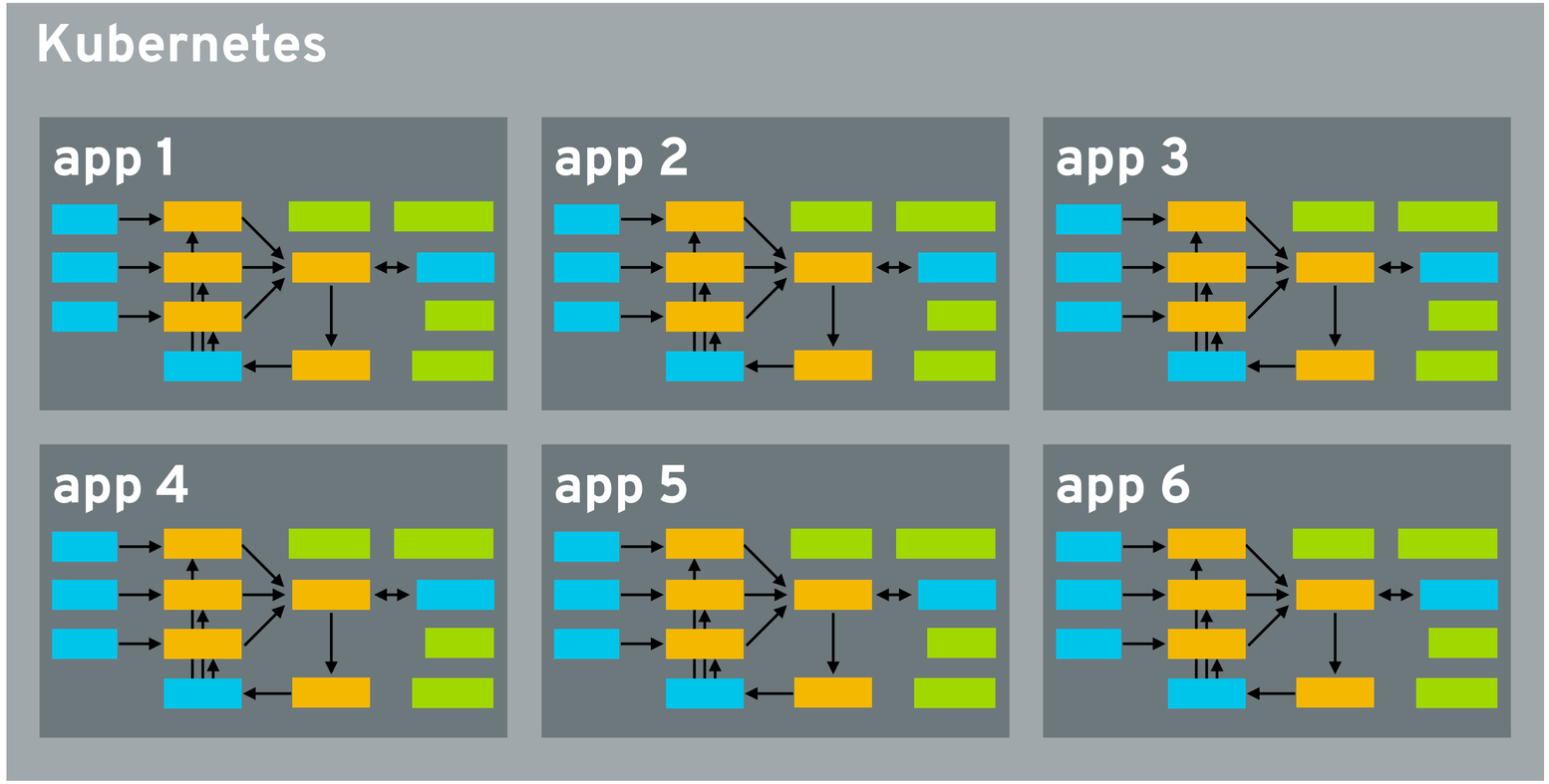
k8s namespace

SECURITY

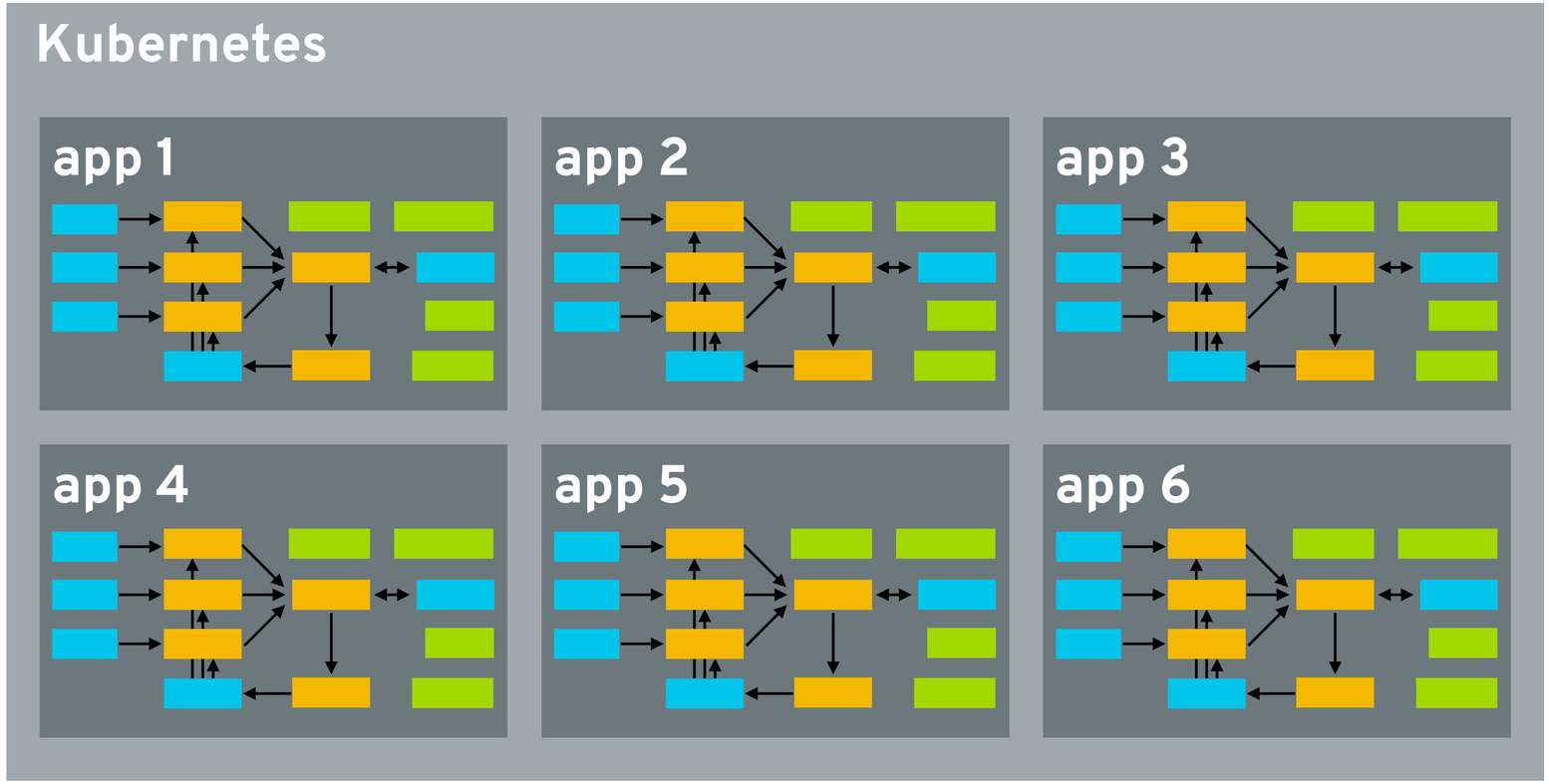


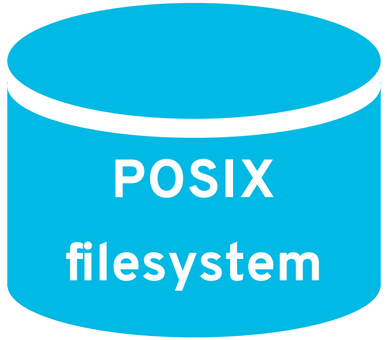
k8s namespace*

STORAGE

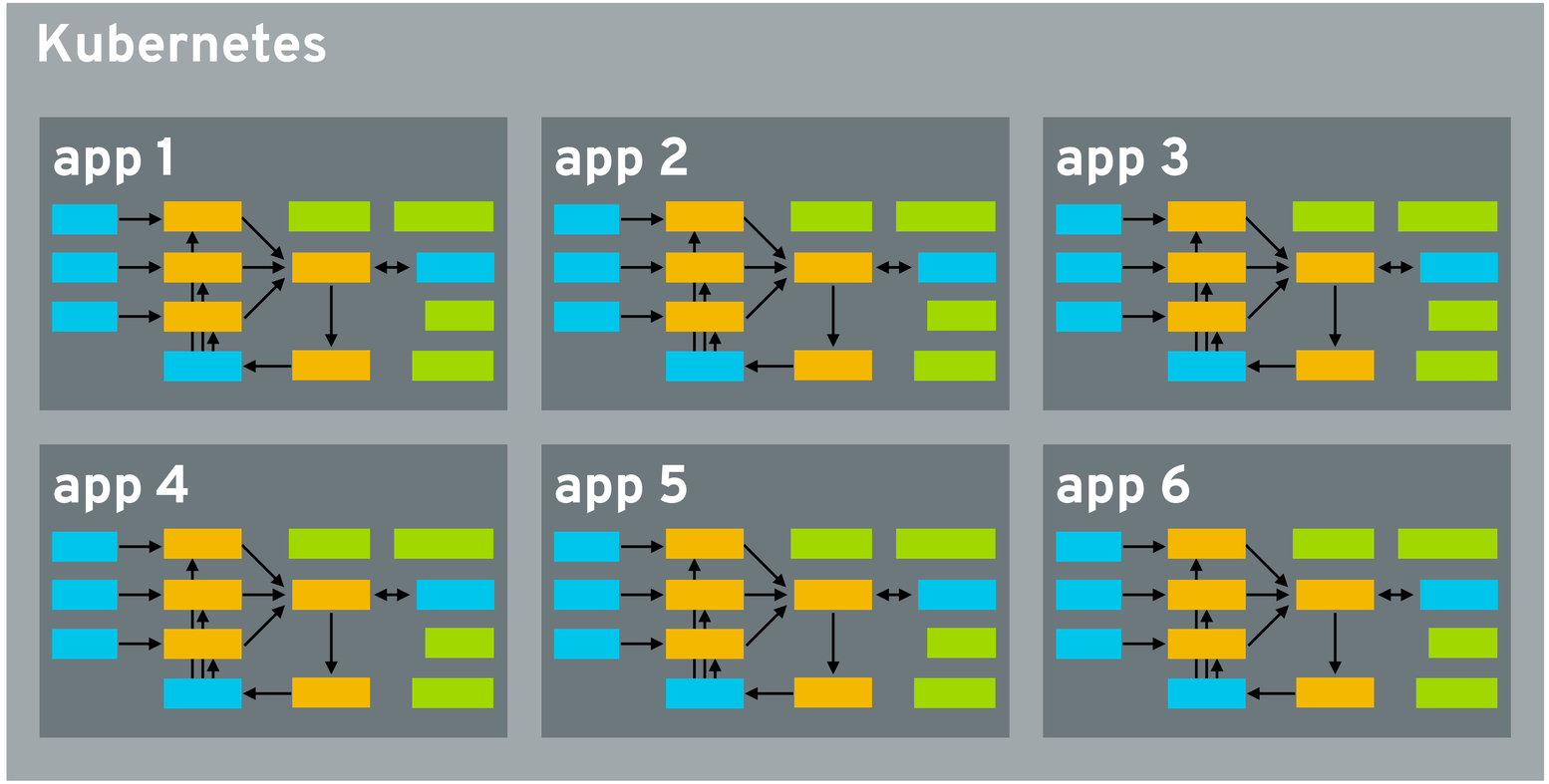


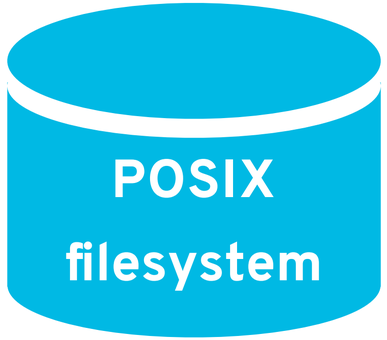
STORAGE





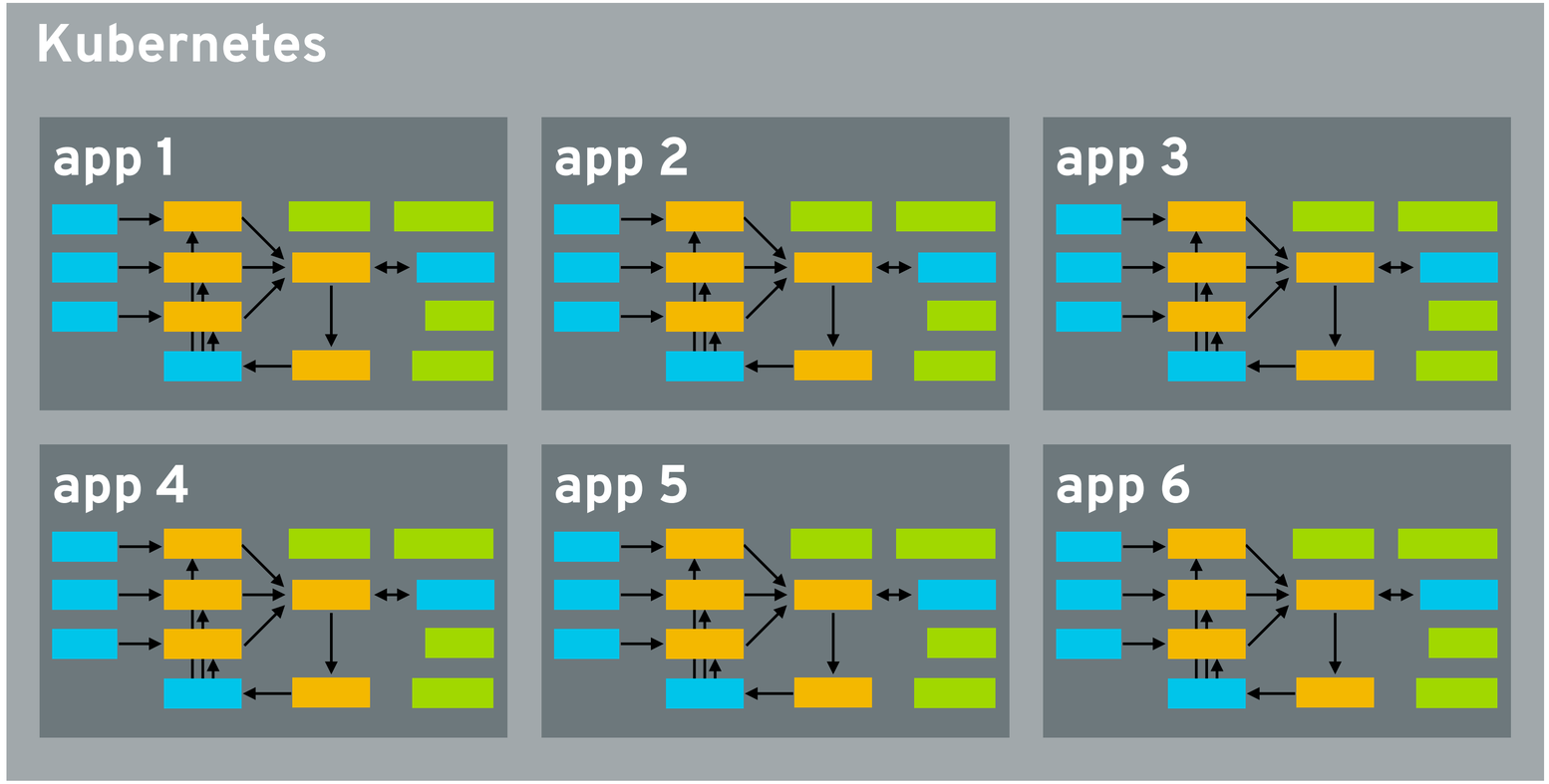
STORAGE

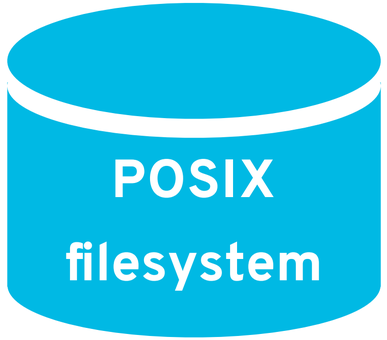




- ✓ familiar interface
- ✓ interoperability with other programs

STORAGE





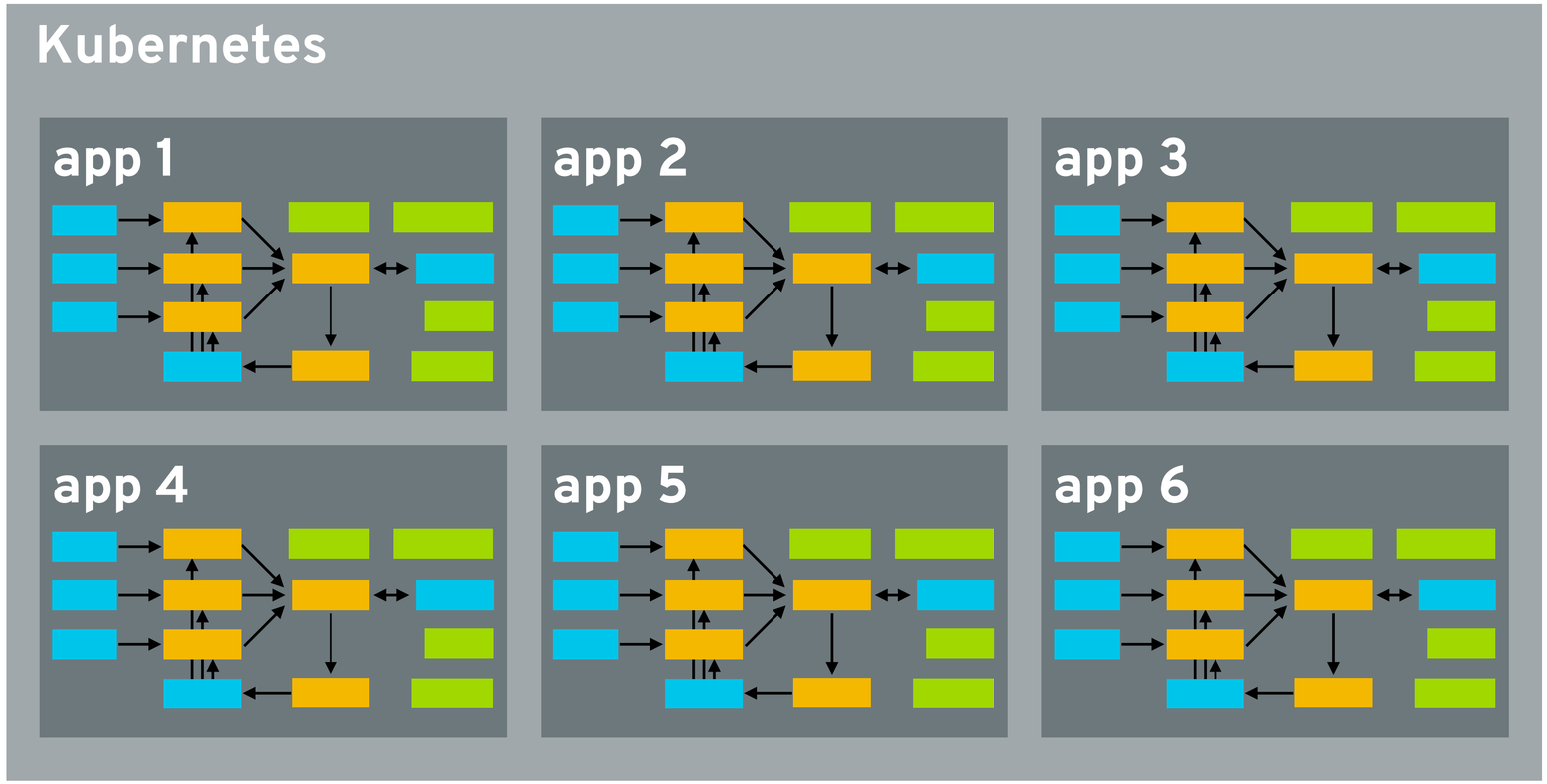
✓ familiar interface

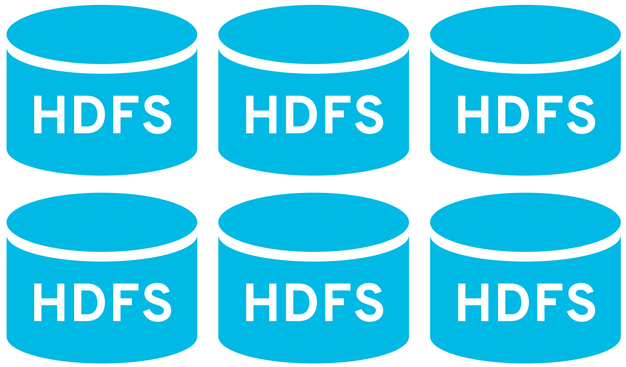
✓ interoperability with other programs

✗ unnecessary semantic guarantees

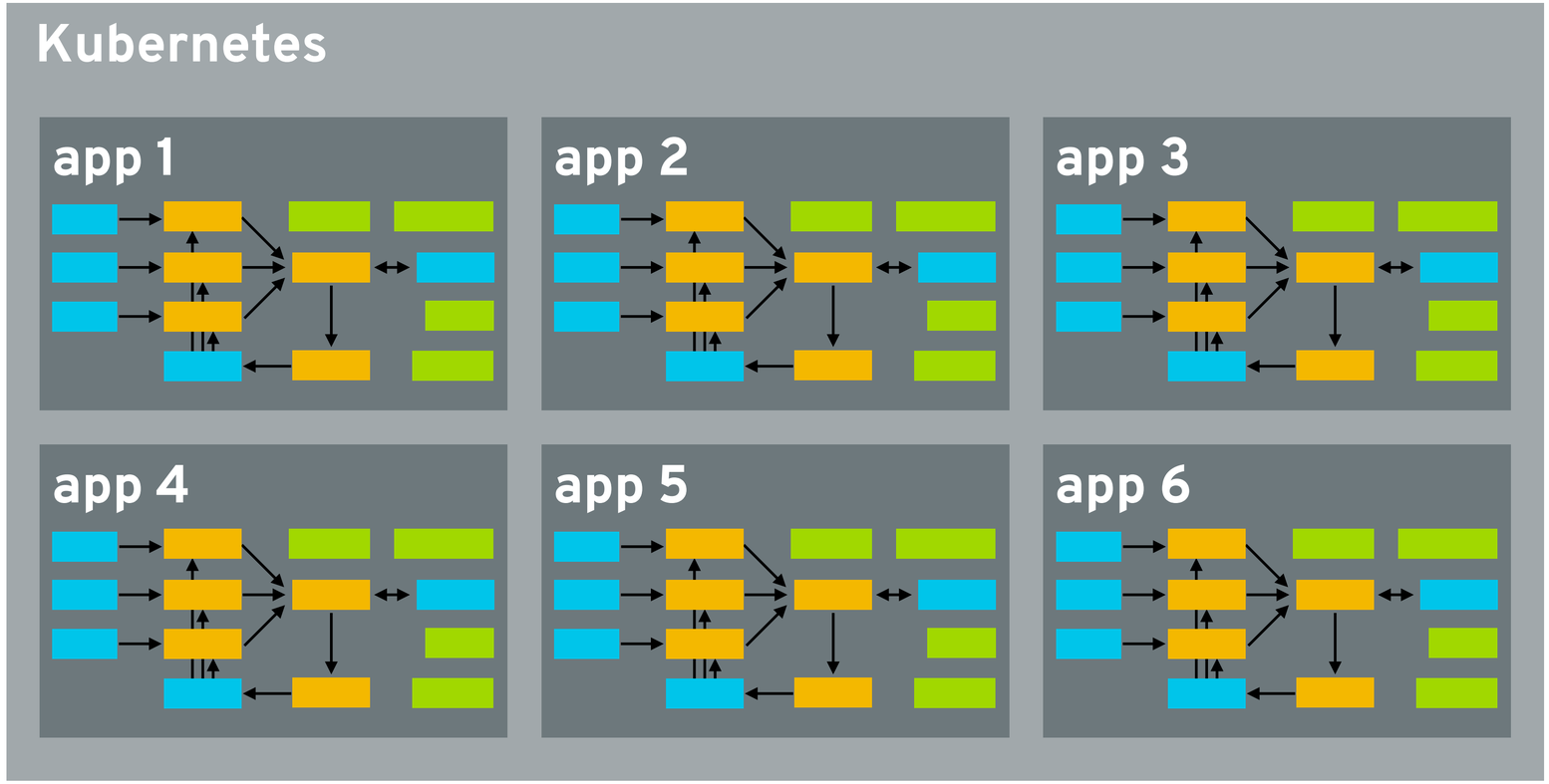
✗ difficult to manage

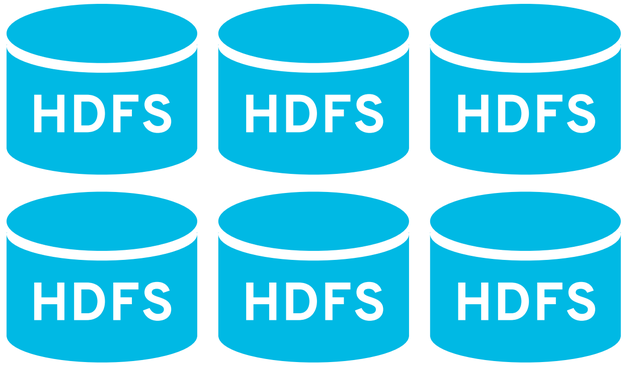
STORAGE


Kubernetes
app 1
app 2
app 3
app 4
app 5
app 6


HDFS
HDFS
HDFS
HDFS
HDFS
HDFS

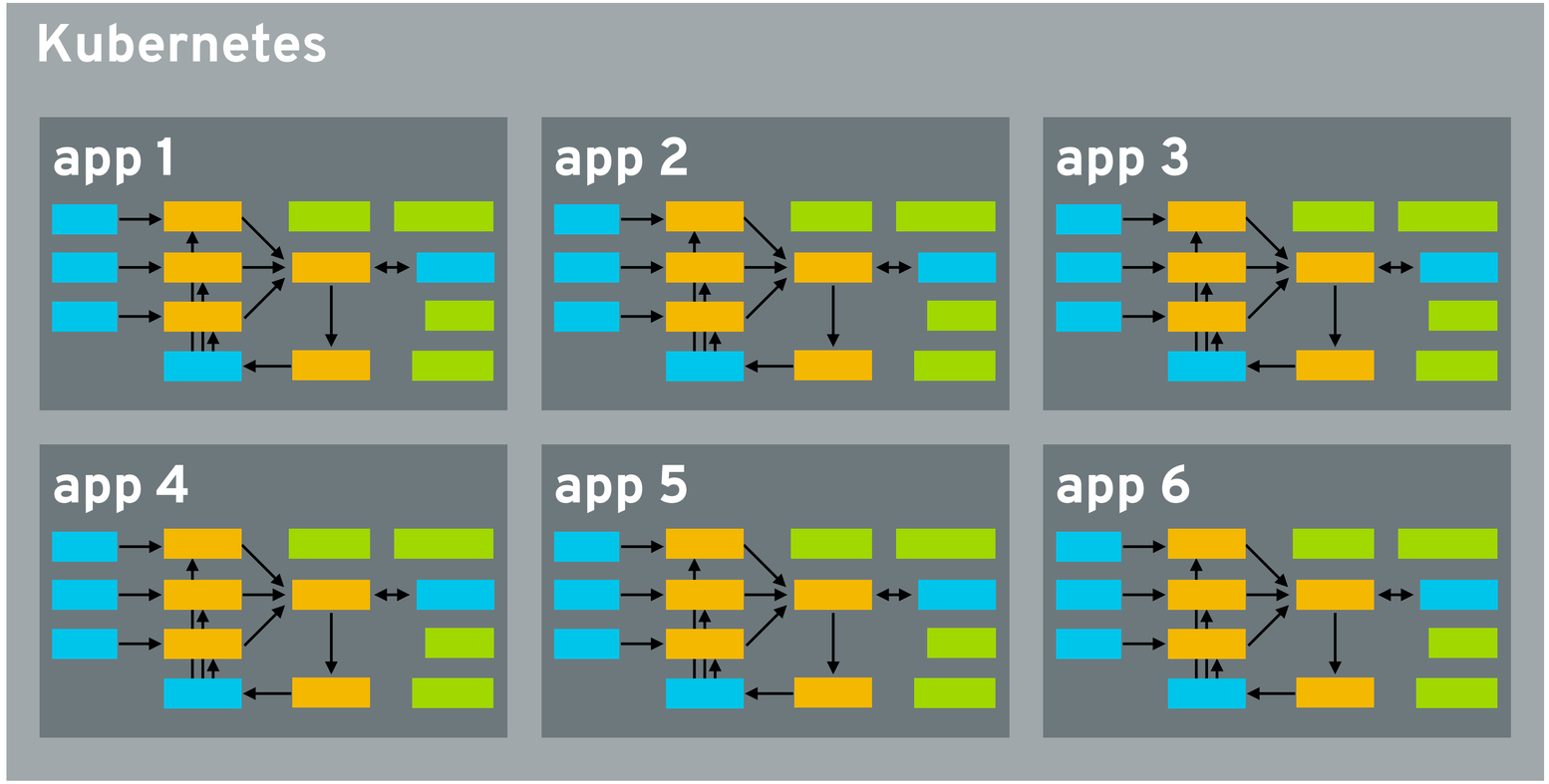
STORAGE

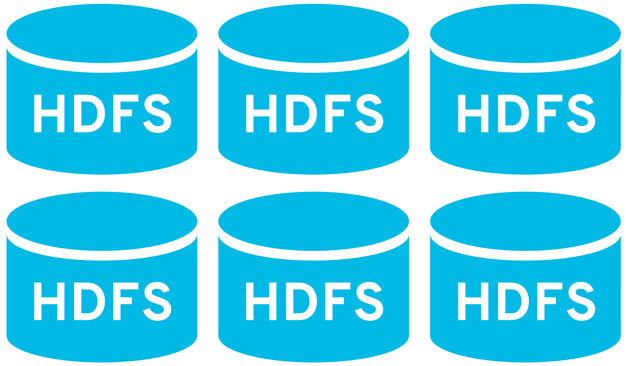




✓ support for legacy Hadoop installations

STORAGE





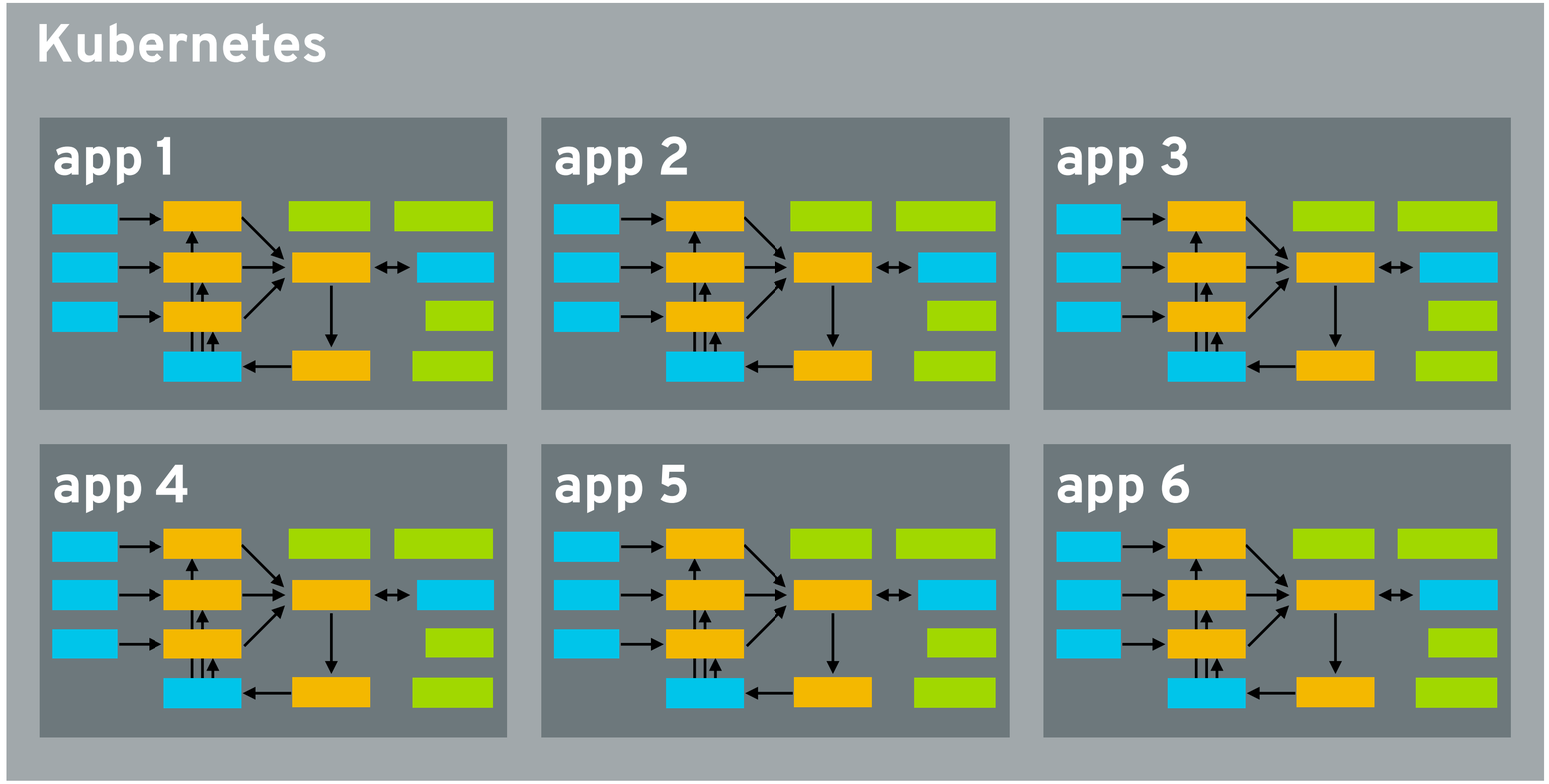
✓ support for legacy Hadoop installations

✗ inelastic

✗ stateful

✗ can't collocate compute and data

STORAGE



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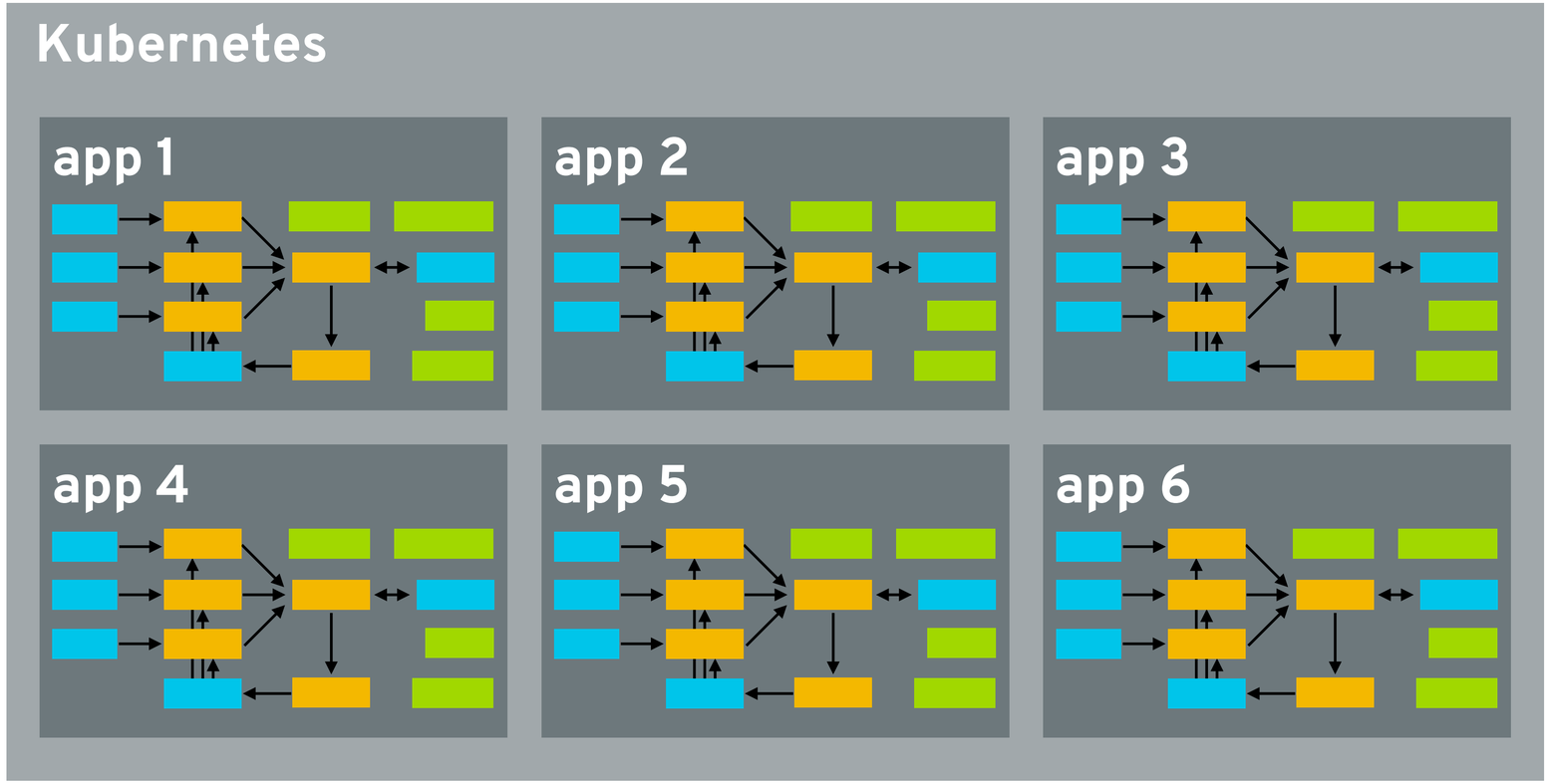
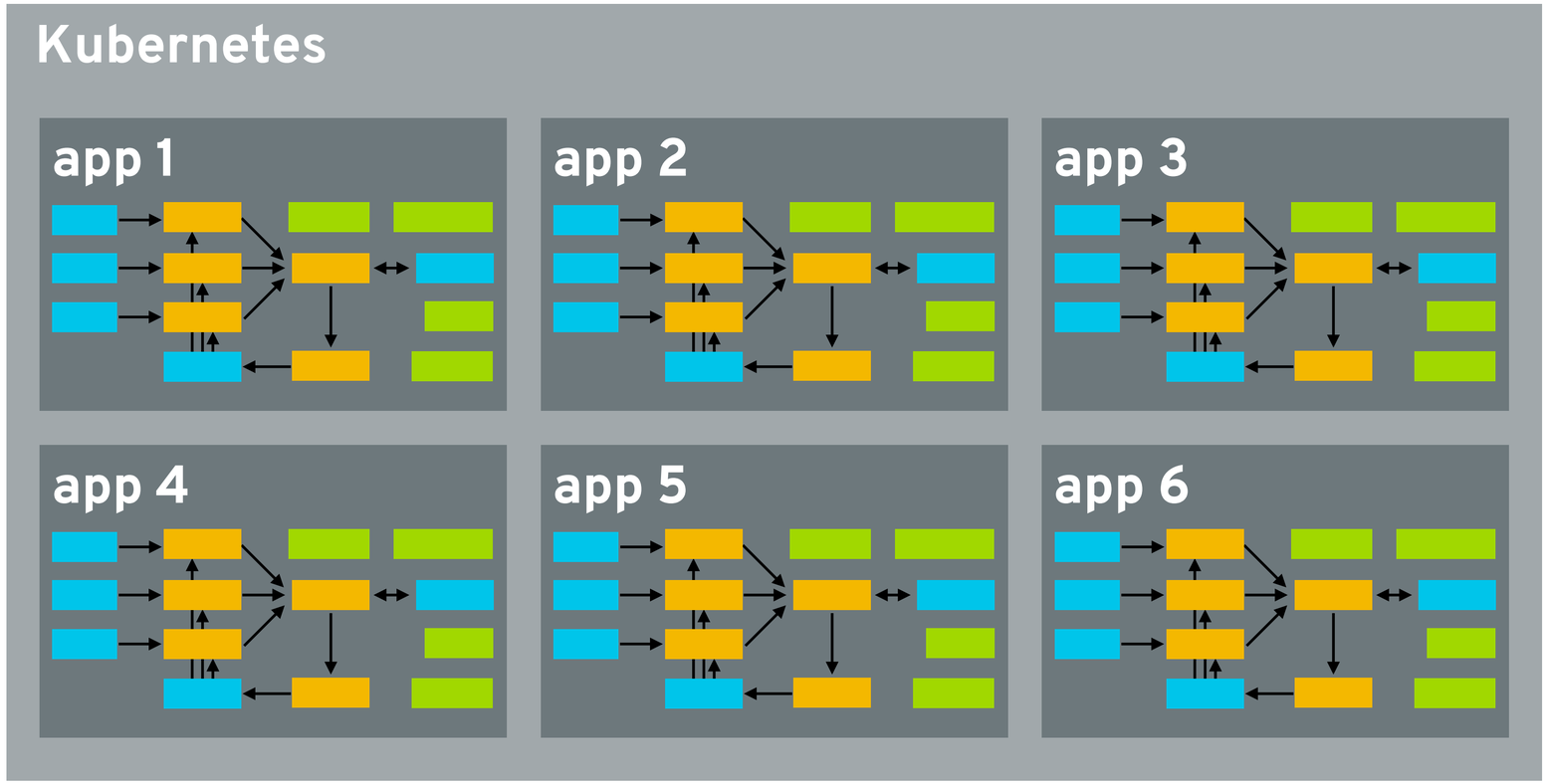


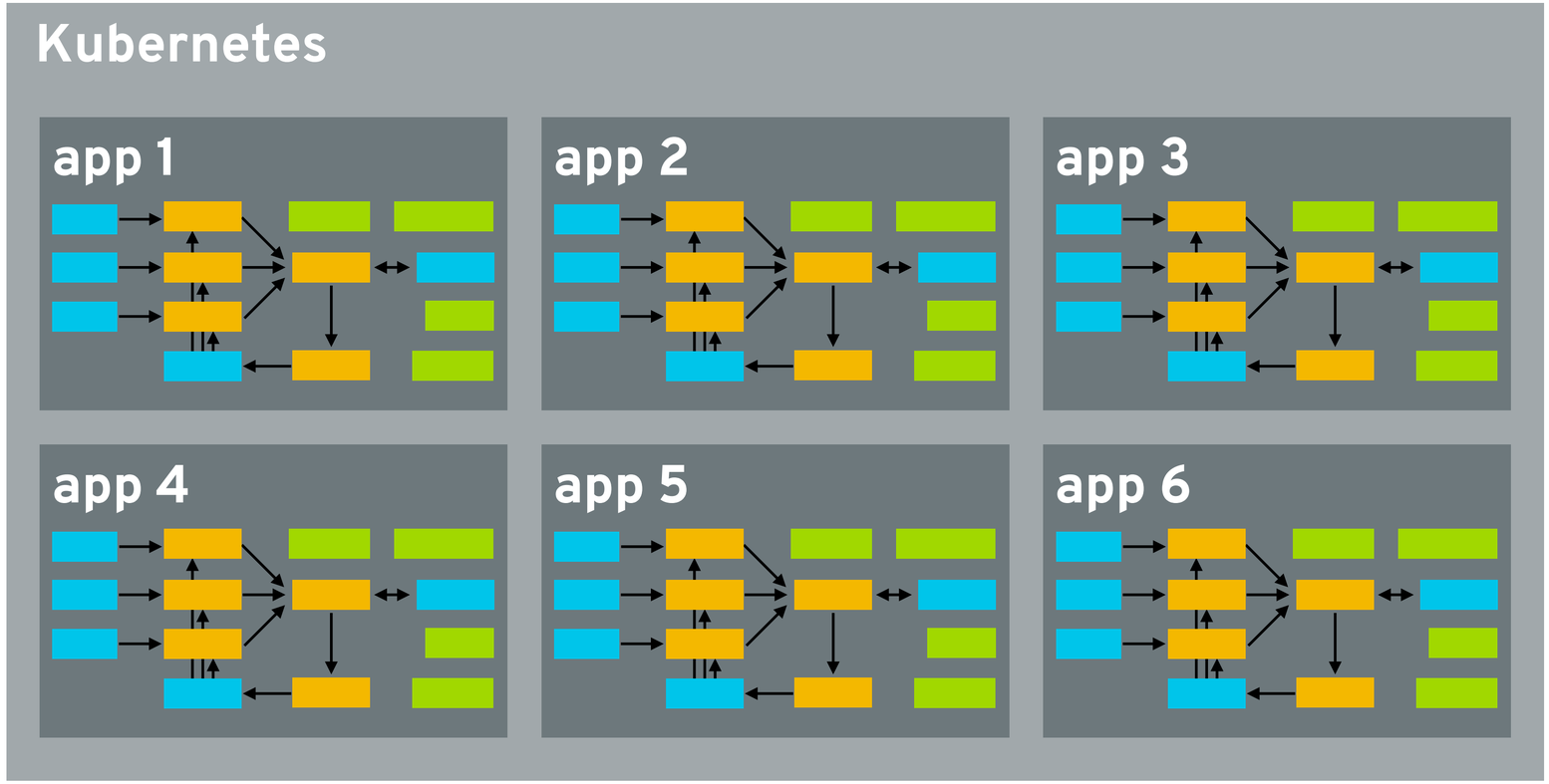
- ✓ interoperability
- ✓ fine-grained AC
- ✓ many implementations

STORAGE



- ✓ interoperability
- ✓ fine-grained AC
- ✓ many implementations
- ✗ consistency model
- ✗ performance (?)

STORAGE

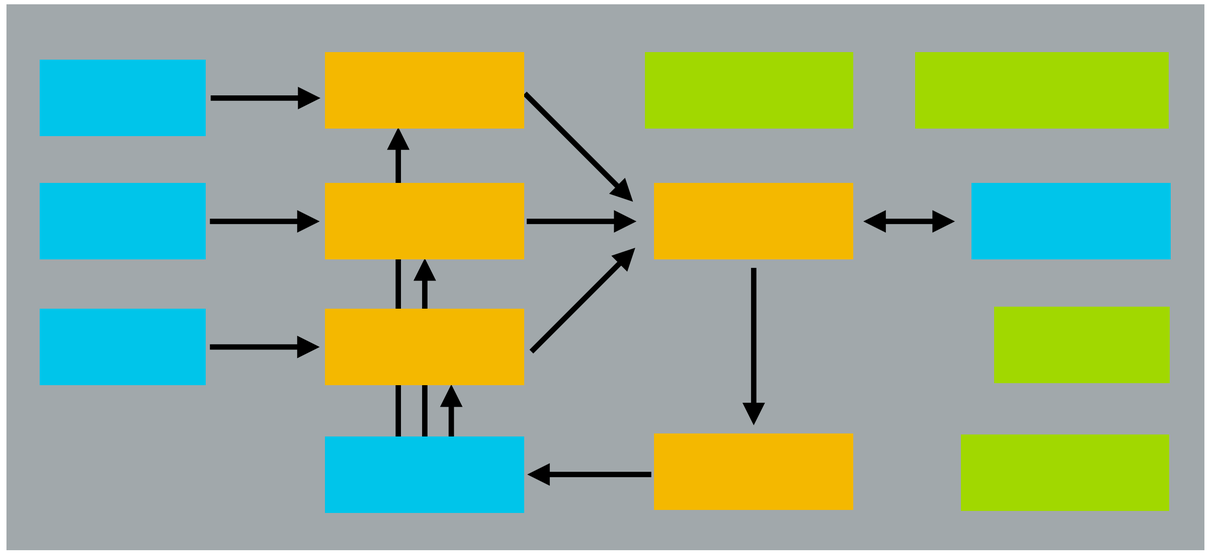


- ✓ interoperability
- ✓ fine-grained AC
- ✓ many implementations
- ✗ consistency model
- ✗ performance

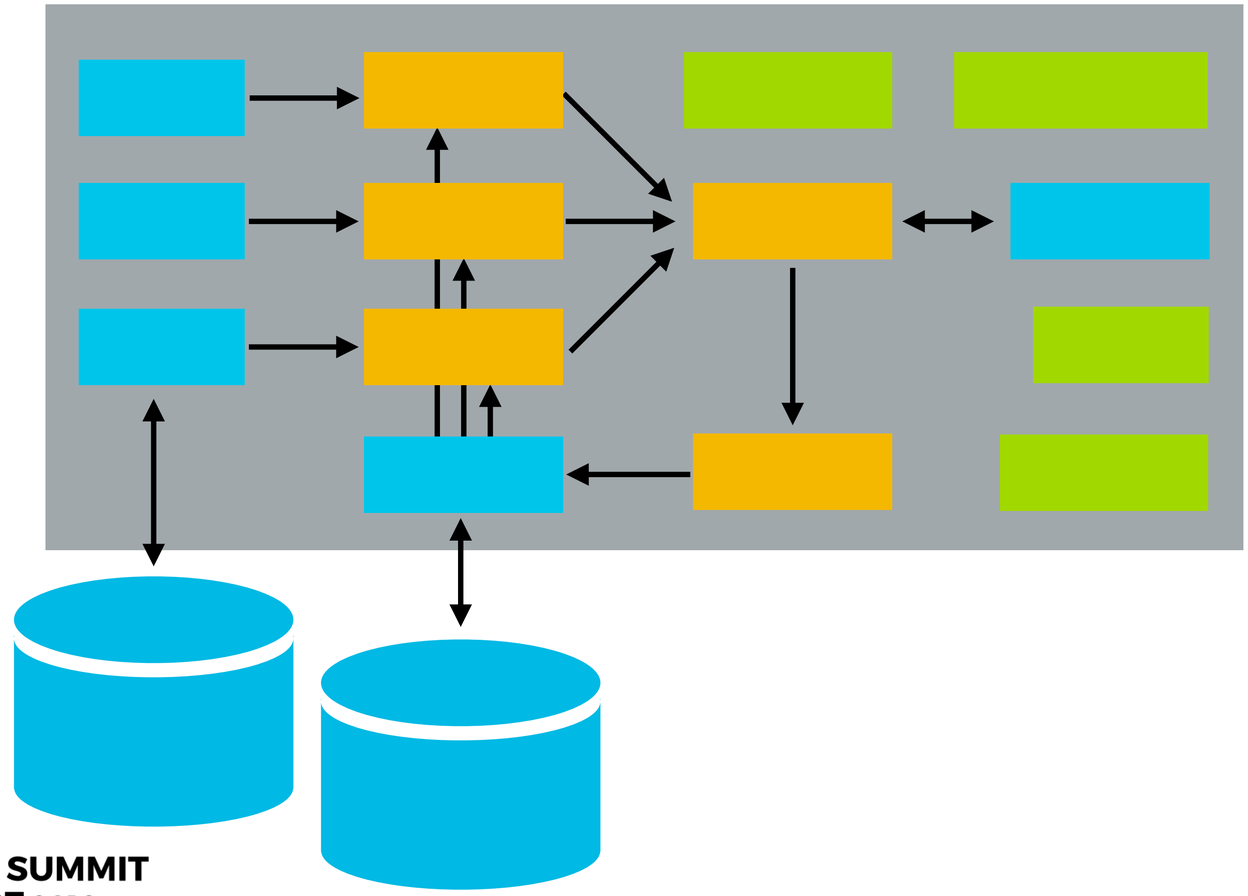
“…in a cloud native architecture, the benefit of HDFS is actually very small and that is why many cloud-first organizations no longer run HDFS, or only run it as a caching layer for S3.”

—Reynold Xin on Quora (http://qr.ae/TAF4cN)

NETWORKING



NETWORKING



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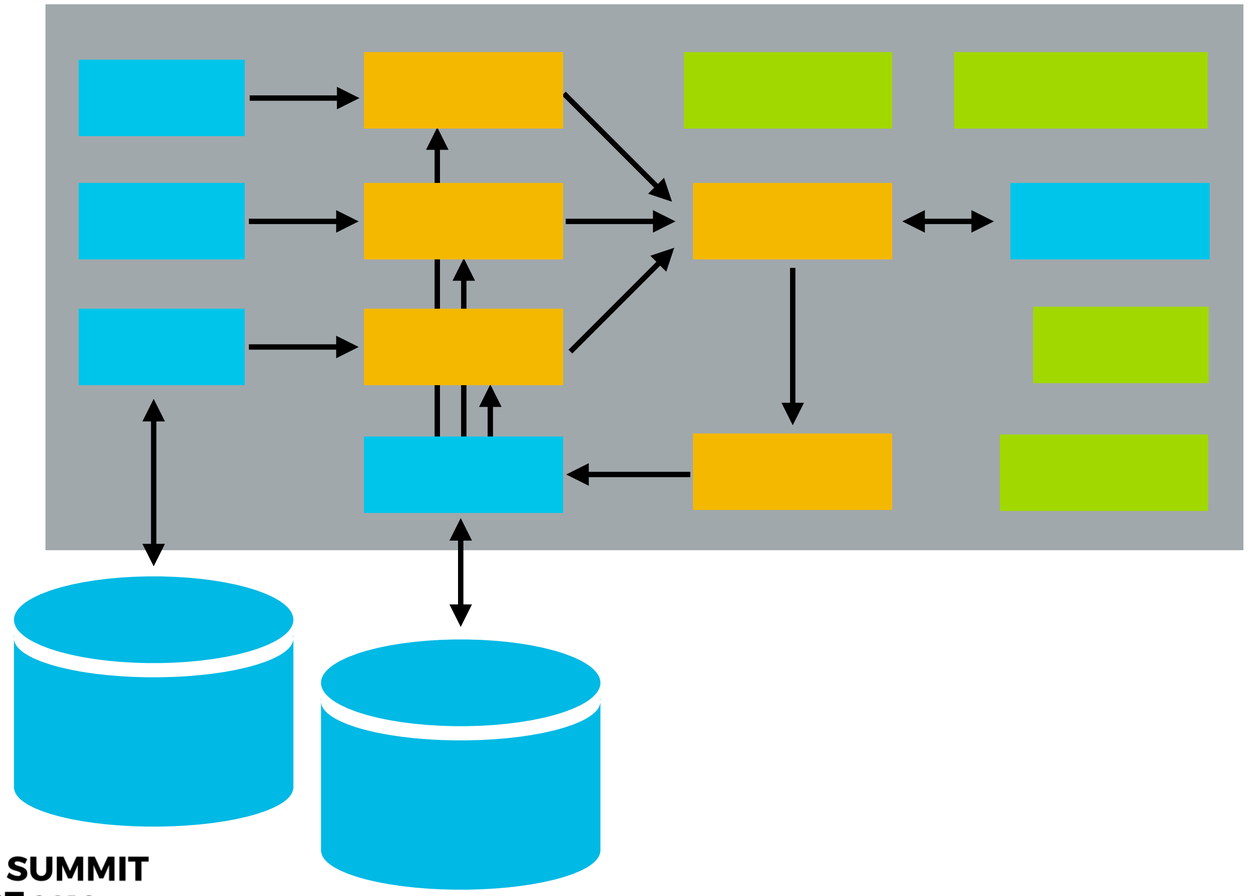

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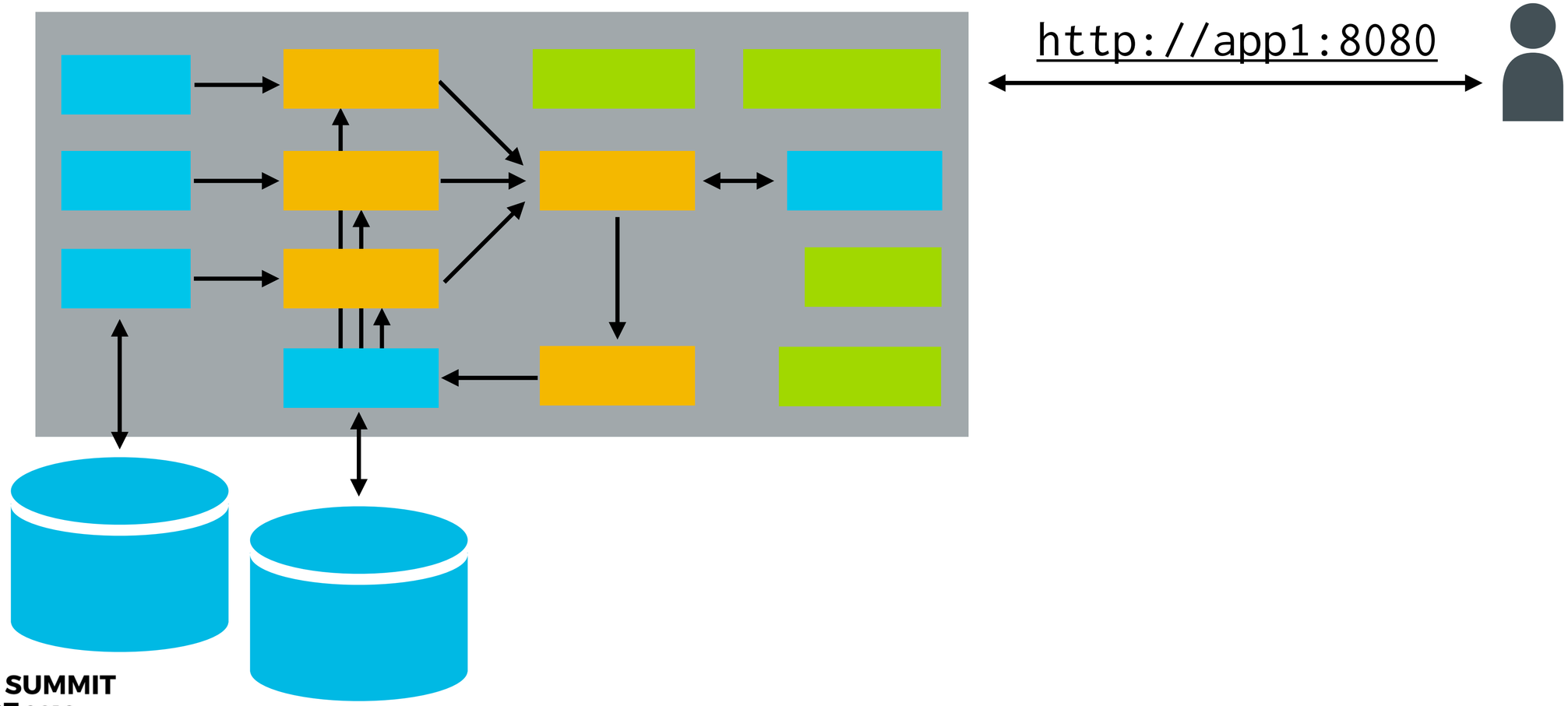
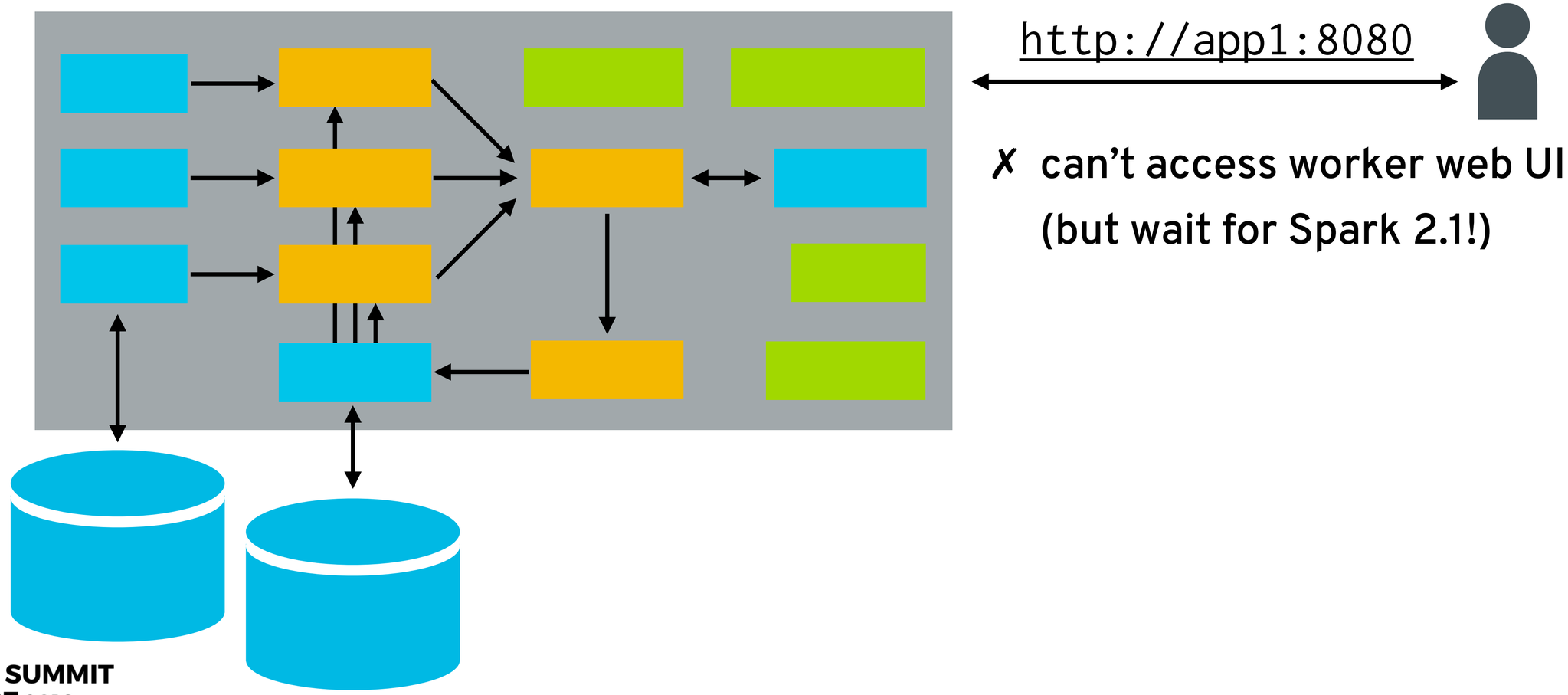


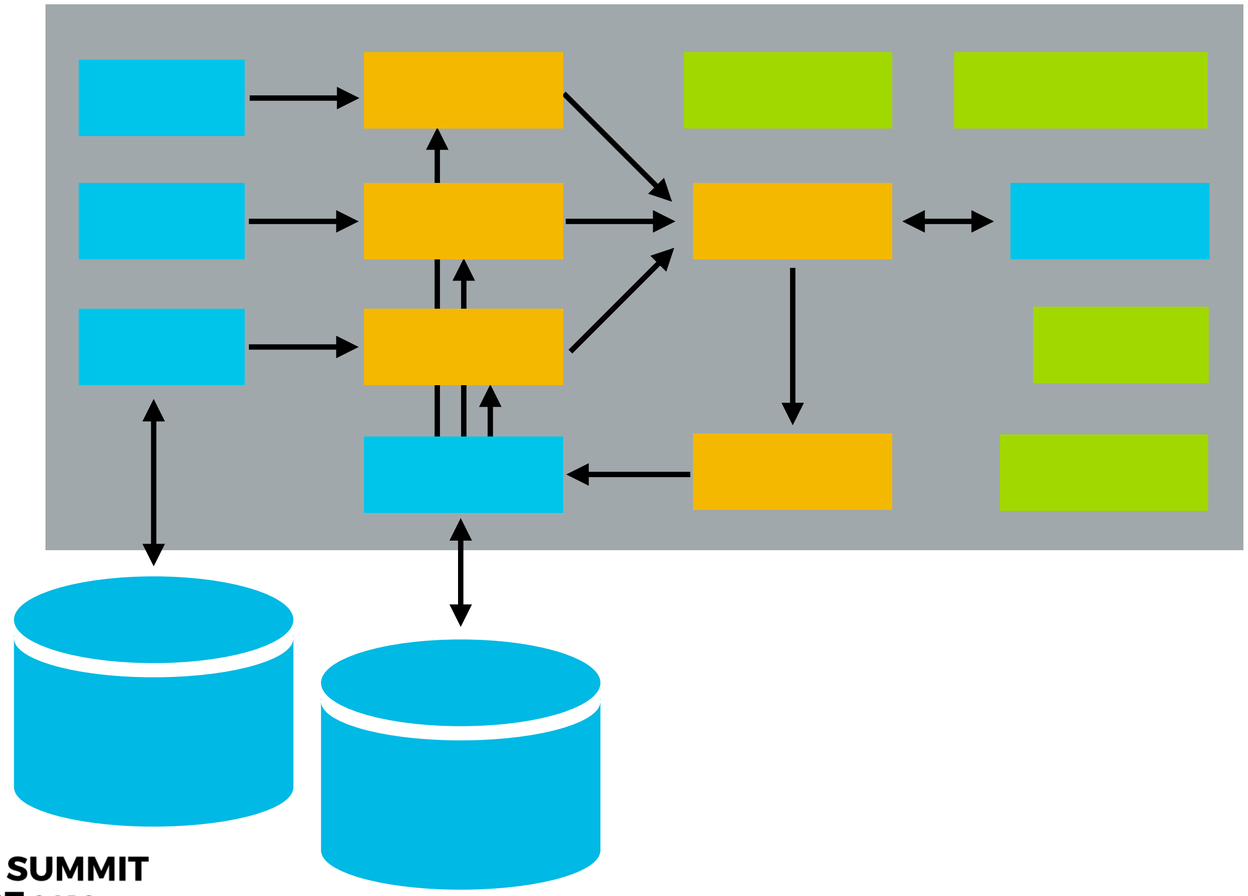
NETWORKING



✗ can't access worker web UI

(but wait for Spark 2.1!)

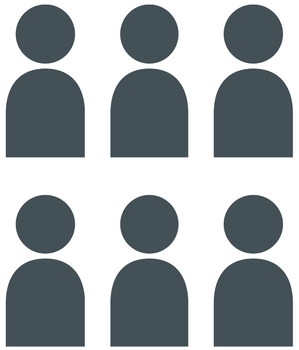
NETWORKING



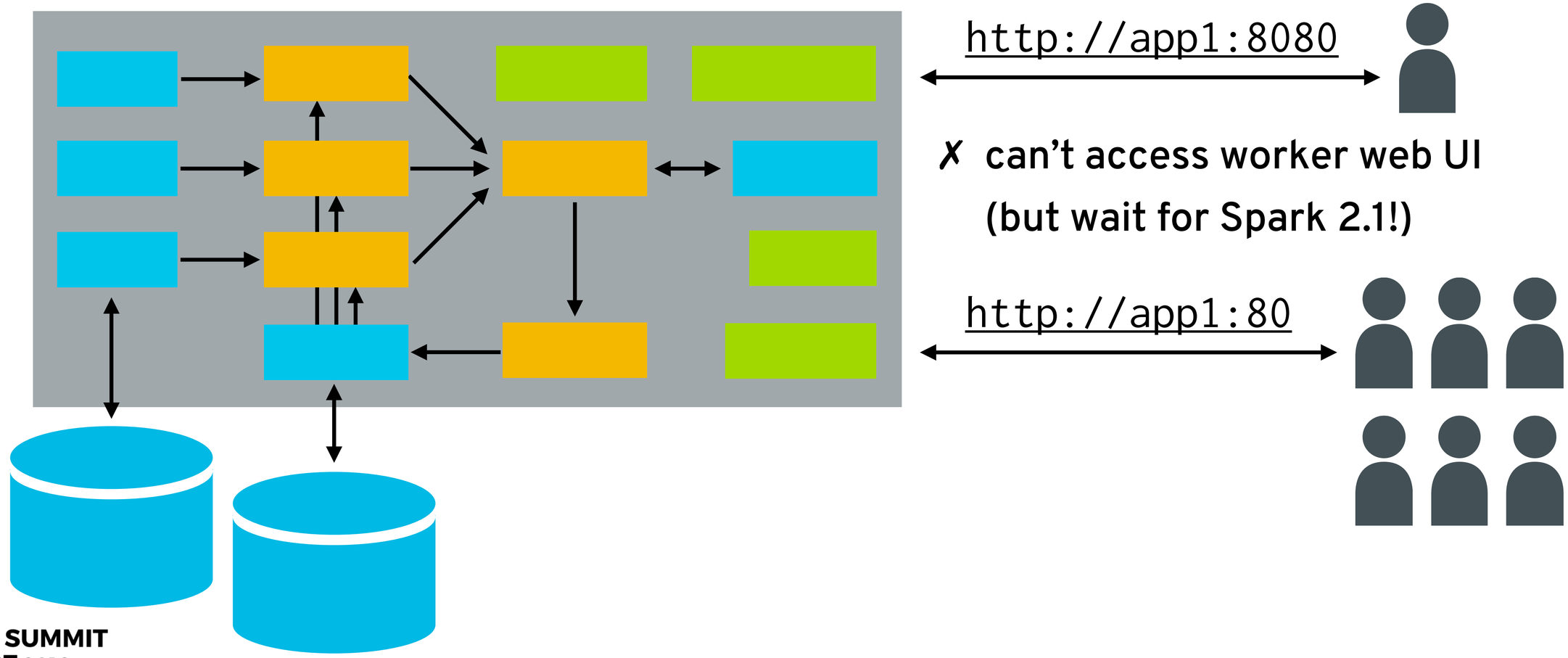
http://app1:8080

✗ can't access worker web UI

(but wait for Spark 2.1!)



NETWORKING


http://app1:8080
✗ can't access worker web UI
(but wait for Spark 2.1!)
http://app1:80

NEXT STEPS: FUTURE WORK & PLAYING ALONG AT HOME

NEXT STEPS

Further performance evaluation

Better developer experience

Improved scheduling of Spark tasks on Kubernetes

TRY IT OUT YOURSELF

Kubernetes standalone Spark example:
https://github.com/kubernetes/kubernetes/tree/master/examples/spark

Enabling Spark on OpenShift: https://github.com/radanalyticsio

Native Spark on Kubernetes proposal:
https://github.com/kubernetes/kubernetes/issues/34377

THANKS!

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https://chapeau.freevariable.com

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EUROPE 2016